Grade 9 Mathematics

Solved Exercises - English Medium - Balochistan Textbook Board (BTBB)

Units solved in this PDF (every exercise section of every unit, in book order):

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Unit 2: Logarithms

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Unit 2: Logarithms (continued: Exercises 2.4-2.5 and Review Exercise 2)

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Unit 10: Practical Geometry of Triangles

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Unit 11: Basic Statistics and Probability (continued: Exercise 11.2 Q6-7 and Exercise 11.3)

Unit 11: Basic Statistics and Probability (continued: Exercises 11.4-11.5 and Review Exercise 11)

Unit 1: Real Numbers (continued: Exercises 1.1-1.2, recovered by OCR cross-check)

Source: BTBB Grade 9 Mathematics textbook file, 300 pages, 95,144,574 bytes. Medium confirmed from the book itself: cover, Contents (PDF p.5) and all units are in English; printed page = PDF page. The book is effectively scanned (fragmentary text layer), so exercises were read from 150-DPI rendered pages. Worked examples in the lessons are NOT exercises and are not reproduced as questions; only the book's own exercise sections are solved, with every solution shown in maximum explicit steps. Book quirks noted honestly (answers follow the book's data unless a correction is stated; every call is also flagged in the answer itself; see the manifest for full detail). UNIT 1: REVIEW EXERCISE 1 Q1(v) prints options (a) and (b) IDENTICALLY as "ac > bc"; (a) is ticked. Exercises 1.1 (p.16) and 1.2 (pp.22-23) were recovered by an OCR cross-check after an earlier pass had recorded Unit 1 as Review-only. Exercise 1.2 Q8(ii) as printed is FALSE: its left side equals 25 x 3^(-1/6) = about 20.82, not 25/3. UNIT 2: Ex 2.3 Q1 prints (vi) twice (9650 in the (iv) place; 3465.25 as the second (vi)); Ex 2.3 Q3 numbering jumps from (ix) to (xii); Ex 2.4 Q1(ix) prints no log symbol (answered as log of the cube root); Ex 2.5 Q4's M is negative because the reference intensity exceeds the measured one; Ex 2.5 Q8 follows the book's own Example 2.27 convention; Ex 2.5 Q9 counts the whole purchase against cash received. UNIT 3: the Unit 3 opener Venn mislabels the third circle "B" for "C"; Ex 3.2 Q8 is answered CORRECTLY (exactly-one = 90%) although the book's worked examples print the union value as the "only one" answer; Ex 3.3 Q1 relations are examples (any valid subset is acceptable); Ex 3.3 Q6 takes x as integers. UNIT 4: Ex 4.2 Q3(vi) duplicates Q3(i); Ex 4.2 Q6(i) x^3+81 needs cube roots (no rational factorisation); Ex 4.6 Q4 prints "3x^2 + 9x^2 - 84x" (combined as printed, verified by crop); Ex 4.7 Q1(vi) and (vii) reduce to the same root; Ex 4.7 Q4(c) reads x as the value making the adjustment vanish (x = 3). UNIT 5: Ex 5.2 Q1(vi) has no real solution (extraneous root); Ex 5.2 Q2(ii) rejects x = -23/8 (right side negative); Review Ex 5 Q1(iii) ticked (a) for positive x and y only; Review Ex 5 Q3(iv) has no solution (denominator zero at the formal root). UNIT 6: Ex 6.4 Q15/Q16 assume acute angles; Ex 6.5 Q5 orientation is not fixed by the printed data (primary answer plus mirror alternative); Ex 6.5 Q6 gives both possible bearings; Review Ex 6 Q5 asks for an angle already given (computable quantities given instead). UNIT 7: Ex 7.1 Q4's third point is printed without a letter; Ex 7.2 Q5(ii) and (iii) have NO solution; Review Ex 7 Q4 as printed is WRONG (slopes 3/4 and 4/3 give product +1, not -1, so the lines are NOT perpendicular). UNIT 8: Ex 8.1 Q8 keeps only the positive-intercept line; Ex 8.3 Q5 takes the obtuse interior angle (180 - acute); Ex 8.2 Q5(iii) "Determinant intercept" read as t-intercept. UNIT 9: p.209

KEY POINT misprints the interior-angle formula as an exterior angle; p.214 table misprints the hexagon exterior angle as 160 degrees (correct: 60). Ex 9.4 Q5(ii) c = 138 degrees (supplement of angle AEB = 42 degrees, verified by crop); Ex 9.4 Q11 prints "AB parallel to BE" (read as AB parallel to CD from the arrow marks); Ex 9.5 Q2 as printed states only the panel data with no instruction (SSS construction given); Ex 9.5 Q5 gives no spacing between lights, so no definite number of locations follows; Ex 9.5 Q3/Q4 "by measurement" values are the analytic equivalents. UNIT 10: Ex 10.1 Q2's 60-degree vertical angle with equal sides makes the triangle equilateral; Ex 10.1 Q8 is an SSA case with only one triangle (the second arc intersection falls beyond Y). UNIT 11: Ex 11.2 Q4's frequencies add to 58 although the text says 60 adults (calculations use 58); Ex 11.2 Q7's frequencies add to 136 although the heading says 127 families (calculations use 136); Ex 11.4 Q7(ii) "not blue or red" is read as neither blue nor red (2/5); Ex 11.5 Q1(i) counts classes wholly above 60. Lesson-only notes: Unit 11 Example 11.13 prints 893/13 = 38.7 (correct: 68.69); the Unit 3 De Morgan lesson example is numbered "Example 2.16" (should be 3.16).

Unit 1: Real Numbers

REVIEW EXERCISE 1 (book pp.24-25)

Which of the following number is not a real number?

(a) sqrt(16)

(b) -5

(c) 3.14

(d) sqrt(-9) ✓

Which of the following is an irrational number?

(a) 0.25

(b) sqrt(2) ✓

(c) -3

(d) 1/8

If a = b then a + c = b + c. This property is called;

(a) reflexive property

(b) symmetric property

(c) additive property ✓

(d) transitive property

If a + c > b + c then which one is true?

(a) a > b ✓

(b) a + c < b + c

(c) a + x = b + c

(d) a + c is not equal to b + c

If a < b and c < 0 then;

(a) ac > bc ✓

(b) ac > bc

(c) ac = bc

(d) ac <= bc

If a < b and b < c then a < c. This property is;

(a) reflexive property

(b) trichotomy

(c) transitive property ✓

(d) additive property

In 5th-root(x + y + z), radicand is;

(a) x

(b) y

(c) z

(d) x + y + z ✓

Simplified form of x^5 / x^2 is;

(a) x^2

(b) x^3 ✓

(c) x^4

(d) x^5

Simplified form of (x^(4/5) y^(1/2)) / (x^(-1/5) y^(-1/2)) is;

(a) x / y

(b) xy ✓

(c) x / y^(-1/5)

(d) y^(1/2) / (xy)

Simplified form of sqrt(a/b) x sqrt(b/c) x sqrt(c/a) is;

(a) 0

(b) 1 ✓

(c) sqrt(abc)

(d) 1 / sqrt(abc)

Q2. Using properties of real numbers, verify the following: (i) a/b - c/d = (ad - bc)/(bd).

Step 1: take the common denominator bd. Step 2: a/b = ad/(bd) and c/d = bc/(bd). Step 3: subtract: ad/(bd) - bc/(bd) = (ad - bc)/(bd). Hence verified.

Q2. (ii) (1/5)(3 + 4) = 3/5 + 4/5.

Step 1: left side: (1/5)(3 + 4) = (1/5)(7) = 7/5. Step 2: right side: 3/5 + 4/5 = 7/5. Step 3: both sides equal 7/5, by the distributive property (1/5)(3 + 4) = (1/5)(3) + (1/5)(4). Hence verified.

Q2. (iii) If 9 > 5 then -27 < -15.

Step 1: multiply both sides of 9 > 5 by -3. Step 2: multiplying an inequality by a negative number reverses the sign. Step 3: 9 x (-3) = -27 and 5 x (-3) = -15, so -27 < -15. Hence verified.

Q2. (iv) If 36 < 48 then -6 > -8.

Step 1: divide both sides of 36 < 48 by -6. Step 2: dividing an inequality by a negative number reverses the sign. Step 3: 36 / (-6) = -6 and 48 / (-6) = -8, so -6 > -8. Hence verified.

Q3. Simplify the following: (i) ((625 x^4 y^-8) / (81 x^-8 y^-4))^(1/4).

Step 1: inside the bracket, 625 = 5^4 and 81 = 3^4. Step 2: x^4 / x^-8 = x^(4+8) = x^12. Step 3: y^-8 / y^-4 = y^(-8+4) = y^-4. Step 4: the bracket becomes (5^4 x^12 y^-4) / 3^4. Step 5: take the fourth root: (5^4)^(1/4) = 5, (x^12)^(1/4) = x^3, (y^-4)^(1/4) = y^-1, (3^4)^(1/4) = 3. Step 6: answer = 5 x^3 / (3 y).

Q3. (ii) ((25 x^-4 y^-2) / (9 x^2 y^-6))^(1/2).

Step 1: inside the bracket, 25 = 5^2 and 9 = 3^2. Step 2: x^-4 / x^2 = x^(-4-2) = x^-6. Step 3: y^-2 / y^-6 = y^(-2+6) = y^4. Step 4: the bracket becomes (5^2 x^-6 y^4) / 3^2. Step 5: take the square root: (5^2)^(1/2) = 5, (x^-6)^(1/2) = x^-3, (y^4)^(1/2) = y^2, (3^2)^(1/2) = 3. Step 6: answer = 5 y^2 / (3 x^3).

Q4. Show that: (i) (81^(3/4) x 25^(1/2)) / (125^(2/3) x 27^(1/3)) = 9/5.

Step 1: 81^(3/4) = (3^4)^(3/4) = 3^3 = 27. Step 2: 25^(1/2) = 5. Step 3: numerator = 27 x 5 = 135. Step 4: 125^(2/3) = (5^3)^(2/3) = 5^2 = 25. Step 5: 27^(1/3) = 3. Step 6: denominator = 25 x 3 = 75. Step 7: 135/75 = 9/5. Hence shown.

Q4. (ii) 3rd-root(x^(a+b) / x^(b+c)) x 3rd-root(x^(b+c) / x^(c+a)) x 3rd-root(x^(c+a) / x^(a+b)) = 1.

Step 1: the product of cube roots is the cube root of the product. Step 2: multiply the three fractions: the numerator is x^(a+b) x^(b+c) x^(c+a) and the denominator is x^(b+c) x^(c+a) x^(a+b) — they are identical. Step 3: the product inside the cube root is 1. Step 4: 3rd-root(1) = 1. Hence shown.

Unit 2: Logarithms

EXERCISE 2.1 (book p.30)

Q1. Write the following in scientific notation: (i) 7654000.

Step 1: move the decimal point 6 places to the left: 7654000 = 7.654 x 10^6. Step 2: 1 <= 7.654 < 10, so this is scientific notation. Answer: 7.654 x 10^6.

Q1. (ii) 0.3247.

Step 1: move the decimal point 1 place to the right: 0.3247 = 3.247 x 10^-1. Step 2: 1 <= 3.247 < 10. Answer: 3.247 x 10^-1.

Q1. (iii) 0.000027.

Step 1: move the decimal point 5 places to the right: 0.000027 = 2.7 x 10^-5. Step 2: 1 <= 2.7 < 10. Answer: 2.7 x 10^-5.

Q1. (iv) 567 x 10^-6.

Step 1: 567 = 5.67 x 10^2. Step 2: 567 x 10^-6 = 5.67 x 10^2 x 10^-6 = 5.67 x 10^(2-6). Answer: 5.67 x 10^-4.

Q1. (v) 507.9 x 10^5.

Step 1: 507.9 = 5.079 x 10^2. Step 2: 507.9 x 10^5 = 5.079 x 10^2 x 10^5 = 5.079 x 10^(2+5). Answer: 5.079 x 10^7.

Q1. (vi) 89.99 x 10^-3.

Step 1: 89.99 = 8.999 x 10^1. Step 2: 89.99 x 10^-3 = 8.999 x 10^1 x 10^-3 = 8.999 x 10^(1-3). Answer: 8.999 x 10^-2.

Q2. Write the following in standard form: (i) 1.96 x 10^-5.

Step 1: the power is -5, so move the decimal point 5 places to the left. Step 2: 1.96 x 10^-5 = 0.0000196. Answer: 0.0000196.

Q2. (ii) 4.23 x 10^6.

Step 1: the power is 6, so move the decimal point 6 places to the right. Step 2: 4.23 x 10^6 = 4230000. Answer: 4230000.

Q2. (iii) 1.4 x 10^-7.

Step 1: the power is -7, so move the decimal point 7 places to the left. Step 2: 1.4 x 10^-7 = 0.00000014. Answer: 0.00000014.

Q3. Simplify and write in scientific notation: (i) (4.68 x 10^11) / (2 x 10^-10).

Step 1: divide the coefficients: 4.68 / 2 = 2.34. Step 2: divide the powers: 10^11 / 10^-10 = 10^(11-(-10)) = 10^21. Step 3: combine: 2.34 x 10^21, already in scientific notation. Answer: 2.34 x 10^21.

Q3. (ii) (7.26 x 7 x 10^-7) / (3 x 10^-4).

Step 1: numerator coefficient: 7.26 x 7 = 50.82. Step 2: 50.82 / 3 = 16.94. Step 3: powers: 10^-7 / 10^-4 = 10^(-7-(-4)) = 10^-3. Step 4: 16.94 x 10^-3 = 1.694 x 10^-2. Answer:

1.694 x 10^-2.

Q3. (iii) (9.63 x 10^9) / (6 x 10^-5).

Step 1: 9.63 / 6 = 1.605. Step 2: 10^9 / 10^-5 = 10^(9-(-5)) = 10^14. Answer: 1.605 x 10^14.

Q3. (iv) 4.9 x 10^7 + 4.8 x 10^6.

Step 1: make the powers equal: 4.8 x 10^6 = 0.48 x 10^7. Step 2: add: 4.9 x 10^7 + 0.48 x 10^7 = 5.38 x 10^7. Answer: 5.38 x 10^7.

Q3. (v) 6.3 x 10^4 x 4.6 x 10^5.

Step 1: multiply the coefficients: 6.3 x 4.6 = 28.98. Step 2: multiply the powers: 10^4 x 10^5 = 10^(4+5) = 10^9. Step 3: 28.98 x 10^9 = 2.898 x 10^10. Answer: 2.898 x 10^10.

Q4. Write in scientific notation the mass of earth 6 x 10^24 kg after converting it into gram.

Step 1: 1 kg = 1000 g = 10^3 g. Step 2: 6 x 10^24 kg = 6 x 10^24 x 10^3 g = 6 x 10^27 g. Answer: 6 x 10^27 g.

Q5. Write in scientific notation the speed of sound 1236 km/h after converting it into cm/h.

Step 1: 1 km = 100000 cm = 10^5 cm. Step 2: 1236 km/h = 1236 x 10^5 cm/h = 123600000 cm/h. Step 3: 123600000 = 1.236 x 10^8. Answer: 1.236 x 10^8 cm/h.

Q6. Convert the 5.4 x 10^6 cm into meter and write in standard form.

Step 1: 1 m = 100 cm, so divide by 100: 5.4 x 10^6 / 10^2 = 5.4 x 10^4 m. Step 2: standard form: 5.4 x 10^4 = 54000. Answer: 54000 m.

EXERCISE 2.2 (book pp.31-32)

Q1. Write the following in the logarithmic form: (i) 7^2 = 49.

Step 1: use a^x = y if and only if x = log_a y. Step 2: here a = 7, x = 2, y = 49. Answer: log_7 49 = 2.

Q1. (ii) 2^8 = 256.

Step 1: a = 2, x = 8, y = 256. Step 2: x = log_a y. Answer: log_2 256 = 8.

Q1. (iii) z^(4/5) = 1.

Step 1: a = z, x = 4/5, y = 1. Step 2: x = log_a y. Answer: log_z 1 = 4/5.

Q1. (iv) 5^-3 = 1/125.

Step 1: a = 5, x = -3, y = 1/125. Step 2: x = log_a y. Answer: log_5 (1/125) = -3.

Q1. (v) 3^3 = 27.

Step 1: a = 3, x = 3, y = 27. Step 2: x = log_a y. Answer: log_3 27 = 3.

Q2. Write the following in the exponential form: (i) log_8 1 = 0.

Step 1: use x = log_a y if and only if a^x = y. Step 2: a = 8, x = 0, y = 1. Answer: 8^0 = 1.

Q2. (ii) log_6 x = 5.

Step 1: a = 6, x = 5, y = x. Step 2: a^x = y. Answer: 6^5 = x.

Q2. (iii) log_4 64 = 3.

Step 1: a = 4, x = 3, y = 64. Step 2: a^x = y. Answer: 4^3 = 64.

Q2. (iv) log_7 y = 2.

Step 1: a = 7, x = 2, y = y. Step 2: a^x = y. Answer: 7^2 = y.

Q2. (v) log_5 4 = x.

Step 1: a = 5, x = x, y = 4. Step 2: a^x = y. Answer: 5^x = 4.

Q3. Find the value of x from the following statements: (i) log_3 x = 5.

Step 1: exponential form: 3^5 = x. Step 2: 3^5 = 243. Answer: x = 243.

Q3. (ii) log_4 x = 3.

Step 1: exponential form: 4^3 = x. Step 2: 4^3 = 64. Answer: x = 64.

Q3. (iii) log_25 x = -1/2.

Step 1: exponential form: 25^(-1/2) = x. Step 2: 25^(-1/2) = 1 / sqrt(25) = 1/5. Answer: x = 1/5.

Q3. (iv) log_x 4 = 2.

Step 1: exponential form: x^2 = 4. Step 2: x = 2 (base of a logarithm is positive). Answer: x = 2.

Q3. (v) log_x 625 = 4.

Step 1: exponential form: x^4 = 625. Step 2: 625 = 5^4, so x = 5. Answer: x = 5.

Q3. (vi) log_x 27 = 3.

Step 1: exponential form: x^3 = 27. Step 2: 27 = 3^3, so x = 3. Answer: x = 3.

Q3. (vii) log_4 64 = x.

Step 1: exponential form: 4^x = 64. Step 2: 64 = 4^3, so x = 3. Answer: x = 3.

Q3. (viii) log_64 8 = x/2.

Step 1: exponential form: 64^(x/2) = 8. Step 2: 64 = 8^2, so (8^2)^(x/2) = 8^x = 8^1.

Step 3: x = 1. Answer: x = 1.

Unit 2: Logarithms (continued: Exercises 2.3-2.5 and Review Exercise 2)

EXERCISE 2.3 (book p.34)

Q1. Find the characteristic of the logarithm of the following numbers: (i) 6.

Step 1: 6 has 1 digit. Step 2: characteristic = number of digits - 1 = 1 - 1 = 0. Answer: 0.

Q1. (ii) 84.

Step 1: 84 has 2 digits. Step 2: characteristic = 2 - 1 = 1. Answer: 1.

Q1. (iii) 561.

Step 1: 561 has 3 digits. Step 2: characteristic = 3 - 1 = 2. Answer: 2.

Q1. (iv) 9650. (Printed as (vi) in the book; see the note in the answer.)

Step 1: 9650 has 4 digits. Step 2: characteristic = 4 - 1 = 3. Answer: 3. Note: the book prints (vi) twice in Q1; the first (vi), 9650, stands in the (iv) place.

Q1. (v) 67.24.

Step 1: 67.24 has 2 digits before the decimal point. Step 2: characteristic = 2 - 1 = 1. Answer: 1.

Q1. (vi) 3465.25. (The second (vi) printed in the book.)

Step 1: 3465.25 has 4 digits before the decimal point. Step 2: characteristic = 4 - 1 = 3. Answer: 3.

Q1. (vii) 0.002.

Step 1: 0.002 is less than 1; its first significant digit is the 3rd decimal place. Step 2: characteristic = -3, written bar3. Answer: bar3.

Q1. (viii) 1600.

Step 1: 1600 has 4 digits. Step 2: characteristic = 4 - 1 = 3. Answer: 3.

Q1. (ix) 406.3.

Step 1: 406.3 has 3 digits before the decimal point. Step 2: characteristic = 3 - 1 = 2. Answer: 2.

Q1. (x) 24.36.

Step 1: 24.36 has 2 digits before the decimal point. Step 2: characteristic = 2 - 1 = 1. Answer: 1.

Q2. Find the logarithm of the following numbers: (i) 8.

Step 1: characteristic is 0 (1 digit). Step 2: mantissa of 8 is .9031. Answer: log 8 = 0.9031.

Q2. (ii) 92.

Step 1: characteristic is 1 (2 digits). Step 2: mantissa of 92 is .9638. Answer: log 92 = 1.9638.

Q2. (iii) 123.

Step 1: characteristic is 2 (3 digits). Step 2: mantissa of 123 is .0899. Answer: log 123 = 2.0899.

Q2. (iv) 4546.

Step 1: characteristic is 3 (4 digits). Step 2: mantissa of 4546 is .6576. Answer: log 4546 = 3.6576.

Q2. (v) 4000.

Step 1: characteristic is 3 (4 digits). Step 2: mantissa of 4000 is .6021. Answer: log 4000 = 3.6021.

Q2. (vi) 6005.

Step 1: characteristic is 3 (4 digits). Step 2: mantissa of 6005 is .7785. Answer: log 6005 = 3.7785.

Q2. (vii) 9.625.

Step 1: characteristic is 0 (1 digit before the point). Step 2: mantissa of 9.625 is .9834. Answer: log 9.625 = 0.9834.

Q2. (viii) 165.5.

Step 1: characteristic is 2 (3 digits before the point). Step 2: mantissa of 165.5 is .2188. Answer: log 165.5 = 2.2188.

Q2. (ix) 0.224.

Step 1: 0.224 is less than 1, so the characteristic is bar1 (negative) and the mantissa stays positive. Step 2: mantissa of 224 is .3502. Answer: log 0.224 = bar1.3502.

Q2. (x) 1.645.

Step 1: characteristic is 0. Step 2: mantissa of 1.645 is .2162. Answer: log 1.645 = 0.2162.

Q2. (xi) 0.00257.

Step 1: characteristic is bar3 (first significant digit in the 3rd decimal place). Step 2: mantissa of 257 is .4099. Answer: log 0.00257 = bar3.4099.

Q2. (xii) 0.0145.

Step 1: characteristic is bar2. Step 2: mantissa of 145 is .1614. Answer: log 0.0145 = bar2.1614.

Q2. (xiii) 0.0067.

Step 1: characteristic is bar3. Step 2: mantissa of 67 is .8261. Answer: log 0.0067 = bar3.8261.

Q2. (xiv) 3.85.

Step 1: characteristic is 0. Step 2: mantissa of 3.85 is .5855. Answer: log 3.85 = 0.5855.

Q2. (xv) 0.00136.

Step 1: characteristic is bar3. Step 2: mantissa of 136 is .1335. Answer: log 0.00136 = bar3.1335.

Q3. Find the value of x from the following statements: (i) log x = 0.3642.

Step 1: characteristic 0, mantissa .3642. Step 2: x = antilog(0.3642) = 2.3131. Answer: x =

2.3131.

Q3. (ii) log x = 1.9654.

Step 1: characteristic 1, mantissa .9654. Step 2: x = antilog(1.9654) = 92.3422. Answer: x = 92.3422.

Q3. (iii) log x = 2.5426.

Step 1: characteristic 2, mantissa .5426. Step 2: x = antilog(2.5426) = 348.8189. Answer: x = 348.8189.

Q3. (iv) log x = 3.1942.

Step 1: characteristic 3, mantissa .1942. Step 2: x = antilog(3.1942) = 1563.8677. Answer: x = 1563.8677.

Q3. (v) log x = bar1.2314.

Step 1: bar1.2314 = -1 + 0.2314 = -0.7686. Step 2: x = antilog(-0.7686) = 0.1704. Answer: x = 0.1704.

Q3. (vi) log x = bar2.2125.

Step 1: bar2.2125 = -2 + 0.2125 = -1.7875. Step 2: x = antilog(-1.7875) = 0.0163. Answer: x = 0.0163.

Q3. (vii) log x = 0.0034.

Step 1: characteristic 0, mantissa .0034. Step 2: x = antilog(0.0034) = 1.0079. Answer: x = 1.0079.

Q3. (viii) log x = bar2.3214.

Step 1: bar2.3214 = -2 + 0.3214 = -1.6786. Step 2: x = antilog(-1.6786) = 0.0210. Answer: x = 0.0210.

Q3. (ix) log x = 3.1424.

Step 1: characteristic 3, mantissa .1424. Step 2: x = antilog(3.1424) = 1388.0337. Answer: x = 1388.0337.

Q3. (xii) log x = bar1.6942. (Printed as (xii); parts (x) and (xi) are not printed in the book.)

Step 1: bar1.6942 = -1 + 0.6942 = -0.3058. Step 2: x = antilog(-0.3058) = 0.4945. Answer: x = 0.4945.

Unit 2: Logarithms (continued: Exercises 2.4-2.5 and Review Exercise 2)

EXERCISE 2.4 (book pp.37-38)

Q1. Expand the following by using laws of logarithms: (i) log(451.5 x 3.54).

Step 1: use the product law log(mn) = log m + log n. Answer: log 451.5 + log 3.54.

Q1. (ii) log(2.67 x 3.45).

Step 1: use the product law log(mn) = log m + log n. Answer: log 2.67 + log 3.45.

Q1. (iii) log(35.7 / 25.57).

Step 1: use the quotient law log(m/n) = log m - log n. Answer: log 35.7 - log 25.57.

Q1. (iv) log((8 x 3) / 7).

Step 1: product law for the numerator: log(8 x 3) = log 8 + log 3. Step 2: quotient law: subtract log 7. Answer: log 8 + log 3 - log 7.

Q1. (v) log((56.25 x 4.2) / 265.46).

Step 1: numerator: log 56.25 + log 4.2. Step 2: quotient law: subtract log 265.46. Answer: log 56.25 + log 4.2 - log 265.46.

Q1. (vi) log(215.6)^(1/2).

Step 1: use the power law log(m^n) = n log m with n = 1/2. Answer: (1/2) log 215.6.

Q1. (vii) log(3rd-root(451.6 / 384)).

Step 1: 3rd-root(451.6 / 384) = (451.6 / 384)^(1/3). Step 2: power law gives (1/3) log(451.6 / 384). Step 3: quotient law gives (1/3)(log 451.6 - log 384). Answer: (1/3)(log 451.6 - log 384).

Q1. (viii) log(sqrt((7.6 x 4.8) / 2.5)).

Step 1: sqrt((7.6 x 4.8) / 2.5) = ((7.6 x 4.8) / 2.5)^(1/2). Step 2: power law gives (1/2) log((7.6 x 4.8) / 2.5). Step 3: product and quotient laws give (1/2)(log 7.6 + log 4.8 - log 2.5). Answer: (1/2)(log 7.6 + log 4.8 - log 2.5).

Q1. (ix) log(3rd-root(384.5)). (Note: part (ix) prints without an explicit log symbol; it is answered as the log of the cube root, like parts (vii) and (x).)

Step 1: 3rd-root(384.5) = (384.5)^(1/3). Step 2: power law gives (1/3) log 384.5. Answer: (1/3) log 384.5.

Q1. (x) log(3rd-root(384.5) / sqrt(251.7)).

Step 1: quotient law: log(3rd-root(384.5)) - log(sqrt(251.7)). Step 2: power law on each: (1/3) log 384.5 - (1/2) log 251.7. Answer: (1/3) log 384.5 - (1/2) log 251.7.

Q1. (xi) log(36.25 / (31.5 x 41.6)).

Step 1: quotient law: log 36.25 - log(31.5 x 41.6). Step 2: product law: log 36.25 - log 31.5 - log 41.6. Answer: log 36.25 - log 31.5 - log 41.6.

Q1. (xii) log(1 / (69.6 x 41.96)).

Step 1: quotient law: log 1 - log(69.6 x 41.96). Step 2: log 1 = 0 and product law: 0 - log 69.6 - log 41.96. Answer: -log 69.6 - log 41.96.

Q2. Express the following in the form of a single logarithm: (i) log x + log y + log z.

Step 1: product law backwards: log x + log y = log(xy). Step 2: add log z: log(xy) + log z = log(xyz). Answer: log(xyz).

Q2. (ii) log 35.75 - log 21.7.

Step 1: quotient law backwards: log m - log n = log(m/n). Answer: log(35.75 / 21.7).

Q2. (iii) (1/6) log 31.6.

Step 1: power law backwards: n log m = log(m^n) with n = 1/6. Answer: log(31.6^(1/6)).

Q2. (iv) (1/2) log a + (1/2) log b + (1/2) log c.

Step 1: power law on each: log(a^(1/2)) + log(b^(1/2)) + log(c^(1/2)). Step 2: product law: log(a^(1/2) b^(1/2) c^(1/2)) = log(sqrt(abc)). Answer: log(sqrt(abc)).

Q2. (v) log 61.5 + log 5 - (1/3) log 4.

Step 1: product law: log 61.5 + log 5 = log(61.5 x 5). Step 2: power law: (1/3) log 4 = log(4^(1/3)). Step 3: quotient law: log((61.5 x 5) / 4^(1/3)). Answer: log((61.5 x 5) / 4^(1/3)).

Q2. (vi) (1/7) log 81.

Step 1: power law backwards with n = 1/7. Answer: log(81^(1/7)).

Q3. If log 3 = 0.4771, log 5 = 0.6990 and log 7 = 0.8451 then find the values of the following without using tables: (i) log 21.

Step 1: 21 = 3 x 7. Step 2: log 21 = log 3 + log 7 = 0.4771 + 0.8451. Answer: 1.3222.

Q3. (ii) log 81.

Step 1: 81 = 3^4. Step 2: log 81 = 4 log 3 = 4 x 0.4771. Answer: 1.9084.

Q3. (iii) log sqrt(35).

Step 1: sqrt(35) = 35^(1/2) and 35 = 5 x 7. Step 2: log sqrt(35) = (1/2)(log 5 + log 7) = (1/2)(0.6990 + 0.8451) = (1/2)(1.5441). Answer: 0.7721.

Q3. (iv) log 75.

Step 1: 75 = 3 x 5^2. Step 2: log 75 = log 3 + 2 log 5 = 0.4771 + 2 x 0.6990 = 0.4771 + 1.3980. Answer: 1.8751.

Q3. (v) log(3rd-root(5) / 3).

Step 1: log(3rd-root(5) / 3) = log(3rd-root(5)) - log 3 = (1/3) log 5 - log 3. Step 2: (1/3)(0.6990) - 0.4771 = 0.2330 - 0.4771. Answer: -0.2441.

Q3. (vi) log sqrt(4 1/5).

Step 1: 4 1/5 = 21/5, so sqrt(21/5) = (21/5)^(1/2). Step 2: log = (1/2)(log 21 - log 5) = (1/2)(1.3222 - 0.6990) = (1/2)(0.6232). Answer: 0.3116.

Q3. (vii) log sqrt(147).

Step 1: 147 = 3 x 7^2, so log sqrt(147) = (1/2)(log 3 + 2 log 7). Step 2: (1/2)(0.4771 + 2 x 0.8451) = (1/2)(0.4771 + 1.6902) = (1/2)(2.1673). Answer: 1.0837.

Q3. (viii) log 0.105.

Step 1: 0.105 = 105 / 1000 = (3 x 5 x 7) / 10^3. Step 2: log 0.105 = log 3 + log 5 + log 7 - 3 log 10 = 0.4771 + 0.6990 + 0.8451 - 3 = 2.0212 - 3. Answer: -0.9788.

Q4. Show that: (i) log_b a = (log_d a) / (log_d b).

Step 1: let log_b a = x, so b^x = a. Step 2: take log_d of both sides: log_d(b^x) = log_d a. Step 3: power law: x log_d b = log_d a. Step 4: x = (log_d a) / (log_d b). Hence log_b a = (log_d a) / (log_d b).

Q4. (ii) log_a m = log_c m x log_a c.

Step 1: by part (i) with base c: log_a m = (log_c m) / (log_c a). Step 2: also by part (i): log_c a = 1 / (log_a c), since log_a c = (log_c c) / (log_c a) = 1 / (log_c a). Step 3: substitute: log_a m = log_c m x log_a c. Hence shown.

EXERCISE 2.5 (book pp.43-44)

Q1. Find the number of digits in: (i) 2^9.

Step 1: 2^9 = 512. Step 2: 512 has 3 digits. Answer: 3 digits.

Q1. (ii) 9^5.

Step 1: 9^5 = 59049. Step 2: 59049 has 5 digits. Answer: 5 digits.

Q1. (iii) 7^6.

Step 1: 7^6 = 117649. Step 2: 117649 has 6 digits. Answer: 6 digits.

Q1. (iv) 9^33.

Step 1: let x = 9^33, so log x = 33 log 9 = 33 x 0.9542 = 31.4886. Step 2: number of digits = characteristic + 1 = 31 + 1 = 32. Answer: 32 digits.

Q2. Evaluate the following with the help of logarithms: (i) 68.95 x 4.365^2.

Step 1: let x = 68.95 x 4.365^2. Step 2: log x = log 68.95 + 2 log 4.365 = 1.8385 + 2 x 0.6400 = 3.1185. Step 3: x = antilog(3.1185) = 1313.72. Answer: 1313.72.

Q2. (ii) 5946 / 3815.

Step 1: let x = 5946 / 3815. Step 2: log x = log 5946 - log 3815 = 0.1927. Step 3: x = antilog(0.1927) = 1.5586. Answer: 1.5586.

Q2. (iii) 30.67 / 19.27.

Step 1: let x = 30.67 / 19.27. Step 2: log x = log 30.67 - log 19.27 = 0.2018. Step 3: x = antilog(0.2018) = 1.5916. Answer: 1.5916.

Q2. (iv) 591.6 / 382.6.

Step 1: let x = 591.6 / 382.6. Step 2: log x = log 591.6 - log 382.6 = 0.1893. Step 3: x = antilog(0.1893) = 1.5463. Answer: 1.5463.

Q2. (v) (8.67)^4.

Step 1: let x = (8.67)^4. Step 2: log x = 4 log 8.67 = 3.7521. Step 3: x = antilog(3.7521) =

5650.36. Answer: 5650.36.

Q2. (vi) (2.69 x 3.561) / 48.7.

Step 1: let x = (2.69 x 3.561) / 48.7. Step 2: log x = log 2.69 + log 3.561 - log 48.7 = -0.7062. Step 3: x = antilog(-0.7062) = 0.1967. Answer: 0.1967.

Q2. (vii) 6th-root(32.6).

Step 1: let x = 32.6^(1/6). Step 2: log x = (1/6) log 32.6 = 0.2522. Step 3: x = antilog(0.2522) = 1.7873. Answer: 1.7873.

Q2. (viii) 3rd-root(48.54).

Step 1: let x = 48.54^(1/3). Step 2: log x = (1/3) log 48.54 = 0.5620. Step 3: x = antilog(0.5620) = 3.6478. Answer: 3.6478.

Q2. (ix) (167)^(1/7).

Step 1: let x = 167^(1/7). Step 2: log x = (1/7) log 167 = 0.3175. Step 3: x = antilog(0.3175) = 2.0775. Answer: 2.0775.

Q2. (x) ((8.62)^2 x (1.6157)^3) / 51.6.

Step 1: let x = ((8.62)^2 x (1.6157)^3) / 51.6. Step 2: log x = 2 log 8.62 + 3 log 1.6157 - log 51.6 = 0.7834. Step 3: x = antilog(0.7834) = 6.0736. Answer: 6.0736.

Q2. (xi) (3rd-root(37.54) x (38.4)^(1/2)) / 5th-root(31.5).

Step 1: let x = (3rd-root(37.54) x (38.4)^(1/2)) / 5th-root(31.5). Step 2: log x = (1/3) log 37.54 + (1/2) log 38.4 - (1/5) log 31.5 = 1.0173. Step 3: x = antilog(1.0173) = 10.4072. Answer: 10.4072.

Q2. (xii) 3rd-root(279.6) / 5th-root(324.5).

Step 1: let x = 3rd-root(279.6) / 5th-root(324.5). Step 2: log x = (1/3) log 279.6 - (1/5) log 324.5 = 0.3133. Step 3: x = antilog(0.3133) = 2.0572. Answer: 2.0572.

Q2. (xiii) sqrt(5 1/3) / 3rd-root(28).

Step 1: 5 1/3 = 16/3, so let x = sqrt(16/3) / 3rd-root(28). Step 2: log x = (1/2) log(16/3) - (1/3) log 28 = -0.1189. Step 3: x = antilog(-0.1189) = 0.7605. Answer: 0.7605.

Q2. (xiv) 11th-root(781.6).

Step 1: let x = 781.6^(1/11). Step 2: log x = (1/11) log 781.6 = 0.2630. Step 3: x = antilog(0.2630) = 1.8323. Answer: 1.8323.

Q2. (xv) (3rd-root(24.58) x 6th-root(40.9)) / sqrt(83.85).

Step 1: let x = (3rd-root(24.58) x 6th-root(40.9)) / sqrt(83.85). Step 2: log x = (1/3) log 24.58 + (1/6) log 40.9 - (1/2) log 83.85 = -0.2296. Step 3: x = antilog(-0.2296) = 0.5894. Answer: 0.5894.

Q3. A person invests Rs.4,50,000 in a saving account that offers 4.5% annual compound interest. Using the formula A = p(1+i)^n, calculate the total amount A in the account after 5 years using logarithms.

Step 1: p = 450000, i = 0.045, n = 5, so A = 450000 x (1.045)^5. Step 2: log A = log 450000 + 5 log 1.045 = 5.6532 + 5 x 0.0191 = 5.7487. Step 3: A = antilog(5.7487) =

560782 (approx). Answer: Rs.5,60,782.

Q4. During an earth quake, the intensity of seismic waves is measured at 4.5 cm. Scientists use the formula M = log(I / I_o) to calculate the magnitude of the earthquake, where I_o is the standard reference intensity, given as 2.57 x 10^2. Find the magnitude of the earthquake.

Step 1: M = log(4.5 / (2.57 x 10^2)) = log(4.5 / 257) = log(0.01751). Step 2: log(0.01751) = -1.7567. Answer: M = -1.7567. Note: the magnitude comes out negative because the given reference intensity (257) is larger than the measured intensity (4.5); the answer follows the book's data exactly.

Q5. The model for growth and decay is represented by the formula y = c e^(kt), where t is the time, c is the original cost, y is the amount after time t, k is the constant and e is approximately 2.718. Find: (i) y if c = Rs.4,80,000, t = 4, k = 1.59.

Step 1: y = 480000 x e^(1.59 x 4) = 480000 x e^6.36. Step 2: e^6.36 = 578.26 (with e = 2.718). Step 3: y = 480000 x 578.26 = 277,558,251 (approx). Answer: Rs.27,75,58,251 (approx).

Q5. (ii) y if c = Rs.7,50,000, t = 3, k = -1.72.

Step 1: y = 750000 x e^(-1.72 x 3) = 750000 x e^-5.16. Step 2: e^-5.16 = 0.005742. Step 3: y = 750000 x 0.005742 = 4306.27. Answer: Rs.4306 (approx).

Q6. The temperature of a room is 22 degrees C. Given that the freezing point of water is 0 degrees C, determine the equivalent temperature in Fahrenheit.

Step 1: use F = (C x 1.8) + 32. Step 2: F = (22 x 1.8) + 32 = 39.6 + 32. Answer: 71.6 degrees F.

Q7. A saving account in a bank has a balance of Rs.60,000. The bank pays an annual interest rate of 8%, compounded quarterly. Calculate the total balance in the account at the end of one year.

Step 1: A = P(1 + r/n)^(nt) with P = 60000, r = 0.08, n = 4, t = 1. Step 2: A = 60000 x (1 + 0.08/4)^4 = 60000 x (1.02)^4. Step 3: (1.02)^4 = 1.08243216, so A = 60000 x 1.08243216 = 64945.93. Answer: Rs.64,946 (approx).

Q8. A person has a monthly salary of Rs.32,500 and also receives an annual bonus of Rs.5,500. Find the person's total income for the current year. Then, calculate the percentage increase in income compared to the previous year, when the total income (including bonus) was Rs.7,27,575.

Step 1: total income this year = 32500 x 12 + 5500 = 390000 + 5500 = Rs.3,95,500. Step 2: following the book's own worked Example 2.27, percentage = ((previous - current) / current) x 100 = ((727575 - 395500) / 395500) x 100 = (332075 / 395500) x 100 = 83.96%. Answer: total income Rs.3,95,500; 83.96% by the book's Example 2.27 convention (note: the previous year's income is larger than this year's, so read as the previous year

exceeding the current year by 83.96% of the current year's income).

Q9. A shopkeeper purchases 200 kg of rice at a rate of Rs.18 per kg. He then sells 150 kg of the rice at Rs.20 per kg. Determine whether the shopkeeper makes a profit or a loss and calculate the corresponding percentage of profit or loss.

Step 1: total cost price = 200 x 18 = Rs.3600. Step 2: total selling price received = 150 x 20 = Rs.3000. Step 3: selling price < cost price, so it is a loss of 3600 - 3000 = Rs.600. Step 4: loss percentage = (600 / 3600) x 100 = 16.67%. Answer: loss of Rs.600, i.e. 16.67%. Note: this counts the whole purchase against the cash received; the unsold 50 kg (cost Rs.900) is not counted as sold.

REVIEW EXERCISE 2 (book pp.44-45)

Q1. Choose the correct answer: (i) If log_10 x = 4 then x is:

(a) 500

(b) 100

(c) 1000

(d) 10000 ✓

Q1. (ii) The characteristic of log 34.67 is:

(a) 0

(b) 1 ✓

(c) 2

(d) 4

Q1. (iii) The base of common logarithm is:

(a) 5

(b) 10 ✓

(c) e

(d) 100

Q1. (iv) log abc is:

(a) log a log b log c

(b) log a + log b + log c ✓

(c) log(a/c)

(d) log a - log b - log c

Q1. (v) Scientific notation of 0.00675 is:

(a) 6.75 x 10^-3 ✓

(b) 6.75 x 10^3

(c) 0.675 x 10^-2

(d) 67.5 x 10^-4

Q2. Using logarithm, evaluate the following: (i) 3rd-root(45.8) / sqrt(32).

Step 1: let x = 3rd-root(45.8) / sqrt(32). Step 2: log x = (1/3) log 45.8 - (1/2) log 32 = -0.1989. Step 3: x = antilog(-0.1989) = 0.6325. Answer: 0.6325.

Q2. (ii) (3.98)^(2/3).

Step 1: let x = (3.98)^(2/3). Step 2: log x = (2/3) log 3.98 = 0.3999. Step 3: x = antilog(0.3999) = 2.5114. Answer: 2.5114.

Q3. A scientist is studying the growth of bacteria in a lab. The number of bacteria after t hours is modeled by the formula N = N_o e^(kt), where N_o = 5,000 (the initial number of bacteria), N is the population after time t, k is the growth constant and e is approximately 2.718. After 6 hours, the bacteria population grows to 20,000. Use logarithms to find the value of k.

Step 1: 20000 = 5000 x e^(6k), so e^(6k) = 20000 / 5000 = 4. Step 2: take natural logs: 6k = ln 4 = 1.3863. Step 3: k = 1.3863 / 6 = 0.2310. Answer: k = 0.2310.

Q4. Sara bought 5/2 meters of cloth to stitch a dress. The price of the cloth is Rs.240 per meter. She paid the shopkeeper and later sold the stitched dress for Rs.1,800. Find the total cost of the cloth, the profit she made and the profit percentage.

Step 1: total cost = (5/2) x 240 = Rs.600. Step 2: profit = selling price - cost = 1800 - 600 = Rs.1200. Step 3: profit percentage = (1200 / 600) x 100 = 200%. Answer: cost Rs.600, profit Rs.1200, profit percentage 200%.

Unit 3: Sets and Functions

EXERCISE 3.1 (book p.54)

Q1. If C = {2,5,7,9}, D = {9,12,13,15}, then find: (i) C union D.

Step 1: union collects every element of C or D, each written once. Step 2: C union D = {2,5,7,9,12,13,15}. Answer: {2,5,7,9,12,13,15}.

Q1. (ii) C intersection D.

Step 1: intersection keeps only elements common to both sets. Step 2: the only common element is 9. Answer: {9}.

Q1. (iii) D \ C.

Step 1: D \ C keeps elements of D that are not in C. Step 2: remove 9 from D. Answer: {12,13,15}.

Q1. (iv) C \ D.

Step 1: C \ D keeps elements of C that are not in D. Step 2: remove 9 from C. Answer: {2,5,7}.

Q2. If A = set of first ten positive odd numbers, B = set of first 12 natural numbers. Find the following and draw their Venn diagrams: (i) A union B.

Step 1: A = {1,3,5,7,9,11,13,15,17,19} and B = {1,2,3,4,5,6,7,8,9,10,11,12}. Step 2: union = {1,2,3,4,5,6,7,8,9,10,11,12,13,15,17,19}. Step 3: Venn diagram: two overlapping circles; the overlap holds {1,3,5,7,9,11}, the A-only part holds {13,15,17,19}, the B-only part holds {2,4,6,8,10,12}. Answer: {1,2,3,4,5,6,7,8,9,10,11,12,13,15,17,19}.

Q2. (ii) A - B.

Step 1: A - B keeps elements of A not in B. Step 2: remove {1,3,5,7,9,11} from A. Answer: {13,15,17,19}. Venn diagram: only the A-only part of the circles is shaded.

Q2. (iii) A intersection B.

Step 1: intersection keeps elements common to A and B. Step 2: common elements are 1,3,5,7,9,11. Answer: {1,3,5,7,9,11}. Venn diagram: only the overlapping lens is shaded.

Q2. (iv) B - A.

Step 1: B - A keeps elements of B not in A. Step 2: remove {1,3,5,7,9,11} from B. Answer: {2,4,6,8,10,12}. Venn diagram: only the B-only part is shaded.

Q3. If U = {x | x in N, 5 <= x <= 20}, E = {x | x in O, 5 <= x <= 19}, F = {x | x in E, 6 <= x <= 20}. Find: (i) E \ F.

Step 1: U = {5,6,...,20}; E = odd numbers from 5 to 19 = {5,7,9,11,13,15,17,19}; F = members of E from 6 to 20 = {7,9,11,13,15,17,19}. Step 2: E \ F removes F from E, leaving 5. Answer: {5}.

Q3. (ii) F \ E.

Step 1: every member of F is already in E. Step 2: nothing remains. Answer: {} (the empty

set).

Q3. (iii) E^c (complement of E in U).

Step 1: E^c = U - E. Step 2: remove the odd numbers 5..19 from {5,...,20}, leaving the even numbers. Answer: {6,8,10,12,14,16,18,20}.

Q3. (iv) F^c (complement of F in U).

Step 1: F^c = U - F. Step 2: remove {7,9,11,13,15,17,19} from U. Answer: {5,6,8,10,12,14,16,18,20}.

Q3. (v) (E union F)^c.

Step 1: E union F = E = {5,7,9,11,13,15,17,19} since F is inside E. Step 2: complement in U = U - E. Answer: {6,8,10,12,14,16,18,20}.

Q3. (vi) (E intersection F)^c.

Step 1: E intersection F = F = {7,9,11,13,15,17,19}. Step 2: complement in U = U - F. Answer: {5,6,8,10,12,14,16,18,20}.

Q3. (vii) E^c union F^c.

Step 1: E^c = {6,8,10,12,14,16,18,20} and F^c = {5,6,8,10,12,14,16,18,20}. Step 2: union = {5,6,8,10,12,14,16,18,20}. Answer: {5,6,8,10,12,14,16,18,20} (this is De Morgan's law: E^c union F^c = (E intersection F)^c).

Q3. (viii) E^c intersection F^c.

Step 1: E^c = {6,8,10,12,14,16,18,20} and F^c = {5,6,8,10,12,14,16,18,20}. Step 2: common elements = {6,8,10,12,14,16,18,20}. Answer: {6,8,10,12,14,16,18,20} (De Morgan's law: E^c intersection F^c = (E union F)^c).

EXERCISE 3.2 (book pp.65-67)

Q1. If A = {1,2,3,4,5}, B = {4,5,6,7,8}, C = {7,8,9,10,11} then find: (i) A union (B union C).

Step 1: B union C = {4,5,6,7,8,9,10,11}. Step 2: A union that = {1,2,3,4,5,6,7,8,9,10,11}. Answer: {1,2,3,4,5,6,7,8,9,10,11}.

Q1. (ii) (A union B) union C.

Step 1: A union B = {1,2,3,4,5,6,7,8}. Step 2: union with C = {1,2,3,4,5,6,7,8,9,10,11}. Answer: {1,2,3,4,5,6,7,8,9,10,11} (equal to part (i): associative law for union).

Q1. (iii) A intersection (B intersection C).

Step 1: B intersection C = {7,8}. Step 2: A intersection {7,8} = {} since A has neither 7 nor 8. Answer: {} (the empty set).

Q1. (iv) (A intersection B) intersection C.

Step 1: A intersection B = {4,5}. Step 2: {4,5} intersection C = {} since C has neither 4 nor 5. Answer: {} (equal to part (iii): associative law for intersection).

Q1. (v) A union (B intersection C).

Step 1: B intersection C = {7,8}. Step 2: A union {7,8} = {1,2,3,4,5,7,8}. Answer:

{1,2,3,4,5,7,8}.

Q1. (vi) A intersection (B union C).

Step 1: B union C = {4,5,6,7,8,9,10,11}. Step 2: A intersection that = {4,5}. Answer: {4,5}.

Q1. (vii) (A union B) intersection (A union C).

Step 1: A union B = {1,2,3,4,5,6,7,8}; A union C = {1,2,3,4,5,7,8,9,10,11}. Step 2: common elements = {1,2,3,4,5,7,8}. Answer: {1,2,3,4,5,7,8} (equal to part (v): distributive law).

Q1. (viii) (A intersection B) union (A intersection C).

Step 1: A intersection B = {4,5}; A intersection C = {}. Step 2: union = {4,5}. Answer: {4,5} (equal to part (vi): distributive law).

Q2. If U = {21,22,23,...,40}, X = {21,23,25,...,39}, Y = {22,24,26,...,40} then find: (i) (X union Y)^c.

Step 1: X union Y = {21,22,...,40} = U. Step 2: complement of U in U = {}. Answer: {}.

Q2. (ii) (X intersection Y)^c.

Step 1: X and Y are disjoint (odd vs even), so X intersection Y = {}. Step 2: complement of {} in U = U. Answer: {21,22,...,40}.

Q2. (iii) X^c intersection Y^c.

Step 1: X^c = Y (the evens) and Y^c = X (the odds). Step 2: Y intersection X = {}. Answer: {} (De Morgan: equals (X union Y)^c).

Q2. (iv) X^c union Y^c.

Step 1: X^c = Y and Y^c = X. Step 2: Y union X = U. Answer: {21,22,...,40} (De Morgan: equals (X intersection Y)^c).

Q2. (v) (X - Y)^c.

Step 1: X - Y = X (X and Y are disjoint). Step 2: complement of X = Y. Answer: {22,24,26,...,40}.

Q2. (vi) (Y - X)^c.

Step 1: Y - X = Y. Step 2: complement of Y = X. Answer: {21,23,25,...,39}.

Q3. If U = {11,12,13,...,30}, C = {11,13,15,...,29}, D = {12,14,16,...,30} then show that: (i) C - D = C intersection D^c.

Step 1: C - D = C, since C (odds) and D (evens) are disjoint. Step 2: D^c = U - D = the odd numbers = C. Step 3: C intersection D^c = C intersection C = C. Step 4: both sides equal C. Hence shown.

Q3. (ii) D - C = D intersection C^c.

Step 1: D - C = D, since D and C are disjoint. Step 2: C^c = U - C = the even numbers = D. Step 3: D intersection C^c = D intersection D = D. Step 4: both sides equal D. Hence shown.

Q4. If U = {0,1,2,3,4}, A = phi, B = {0,1}, then prove that: (i) (A union B)^c = A^c intersection B^c.

Step 1: A union B = {0,1}; (A union B)^c = U - {0,1} = {2,3,4}. Step 2: A^c = U - phi = U

= {0,1,2,3,4}; B^c = U - {0,1} = {2,3,4}. Step 3: A^c intersection B^c = {2,3,4}. Step 4: both sides equal {2,3,4}. Hence proved.

Q4. (ii) (A intersection B)^c = A^c union B^c.

Step 1: A intersection B = phi; (A intersection B)^c = U = {0,1,2,3,4}. Step 2: A^c union B^c = {0,1,2,3,4} union {2,3,4} = {0,1,2,3,4}. Step 3: both sides equal U. Hence proved.

Q5. If D = {0,1,2}, E = {1,2}, U = {0,1,2,3,4}, then verify the following by Venn diagram: (i) D union E = E union D.

Step 1: D union E = {0,1,2}; E union D = {0,1,2}. Step 2: both sides are equal, and their Venn diagrams shade exactly the same region (both circles). Hence verified.

Q5. (ii) D intersection E = E intersection D.

Step 1: D intersection E = {1,2}; E intersection D = {1,2}. Step 2: both sides are equal, and their Venn diagrams shade the same overlapping lens. Hence verified.

Q5. (iii) (D union E)^c = D^c intersection E^c.

Step 1: (D union E)^c = {0,1,2}^c = {3,4}. Step 2: D^c = {3,4}; E^c = {0,3,4}; D^c intersection E^c = {3,4}. Step 3: both sides equal {3,4}. Hence verified.

Q5. (iv) (D intersection E)^c = D^c union E^c.

Step 1: (D intersection E)^c = {1,2}^c = {0,3,4}. Step 2: D^c union E^c = {3,4} union {0,3,4} = {0,3,4}. Step 3: both sides equal {0,3,4}. Hence verified.

Q6. If U = {1,2,3,4,5,6}, A = {1,2,3,4}, B = {3,4,5}, C = {4,5,6}, then verify the following by using Venn diagrams: (i) A union (B union C) = (A union B) union C.

Step 1: B union C = {3,4,5,6}; A union (B union C) = {1,2,3,4,5,6}. Step 2: A union B = {1,2,3,4,5}; (A union B) union C = {1,2,3,4,5,6}. Step 3: both sides equal U, and the Venn diagrams shade the same region. Hence verified.

Q6. (ii) A intersection (B intersection C) = (A intersection B) intersection C.

Step 1: B intersection C = {4,5}; A intersection {4,5} = {4}. Step 2: A intersection B = {3,4}; {3,4} intersection C = {4}. Step 3: both sides equal {4}. Hence verified.

Q6. (iii) A union (B intersection C) = (A union B) intersection (A union C).

Step 1: B intersection C = {4,5}; A union {4,5} = {1,2,3,4,5}. Step 2: A union B = {1,2,3,4,5}; A union C = {1,2,3,4,5,6}; their intersection = {1,2,3,4,5}. Step 3: both sides equal {1,2,3,4,5}. Hence verified.

Q6. (iv) A intersection (B union C) = (A intersection B) union (A intersection C).

Step 1: B union C = {3,4,5,6}; A intersection that = {3,4}. Step 2: A intersection B = {3,4}; A intersection C = {4}; their union = {3,4}. Step 3: both sides equal {3,4}. Hence verified.

Q6. (v) (A union B)^c = A^c intersection B^c.

Step 1: A union B = {1,2,3,4,5}; (A union B)^c = {6}. Step 2: A^c = {5,6}; B^c = {1,2,6}; A^c intersection B^c = {6}. Step 3: both sides equal {6}. Hence verified.

Q6. (vi) (A intersection B)^c = A^c union B^c.

Step 1: A intersection B = {3,4}; (A intersection B)^c = {1,2,5,6}. Step 2: A^c union B^c

= {5,6} union {1,2,6} = {1,2,5,6}. Step 3: both sides equal {1,2,5,6}. Hence verified.

Q7. If U = W, A = phi, B = N, then verify De-Morgan's laws.

Step 1: recall from the book (p.48): N = {1,2,3,...} and W = {0,1,2,...}, so N^c in W is {0}. Step 2: (A union B)^c = (phi union N)^c = N^c = {0}; A^c intersection B^c = W intersection N^c = {0}. Equal. Step 3: (A intersection B)^c = (phi)^c = W; A^c union B^c = W union N^c = W. Equal. Hence De-Morgan's laws are verified.

Q8. In a survey, 60% of people use app A, 50% use app B, and 10% use both. What percentage of people use exactly one app?

Step 1: let A and B be the sets of users. Step 2: |A union B| = |A| + |B| - |A intersection B| = 60% + 50% - 10% = 100%. Step 3: exactly one app = |A union B| - |A intersection B| = 100% - 10% = 90%. Answer: 90%. Note: the book's worked Examples 3.9 and 3.18 print the union value (100% and 90%) as the 'only one' answer; the correct exactly-one value subtracts the overlap once more, as done here.

Q9. In a survey, 50% of people enjoy strategy games, 50% enjoy role-playing games and 30% enjoy both. What percentage of people enjoy at least one type of game.

Step 1: at least one = |A union B| = |A| + |B| - |A intersection B|. Step 2: 50% + 50% - 30% = 70%. Answer: 70%.

Q10. On a world map with 100 countries, 25 are islands and 10 are both island and located in the southern hemisphere. How many countries are not islands?

Step 1: countries that are islands = 25 (the 'both' figure is not needed for this question). Step 2: not islands = 100 - 25 = 75. Answer: 75 countries.

EXERCISE 3.3 (book pp.71-72)

Q1. If A = {-2,-1,0,1}, B = {1,2}, then find two binary relations in A x B. Also write its domain and range. (Example relations: any two subsets of A x B are valid.)

Step 1: A x B = {(-2,1),(-2,2),(-1,1),(-1,2),(0,1),(0,2),(1,1),(1,2)}. Step 2: take R1 = {(-2,1),(-1,2)}; domain of R1 = {-2,-1}, range of R1 = {1,2}. Step 3: take R2 = {(0,1),(1,2)}; domain of R2 = {0,1}, range of R2 = {1,2}. Answer: R1 = {(-2,1),(-1,2)} (domain {-2,-1}, range {1,2}); R2 = {(0,1),(1,2)} (domain {0,1}, range {1,2}).

Q1. (continued) Find two binary relations in B x A. Also write its domain and range. (Example relations: any two subsets of B x A are valid.)

Step 1: B x A = {(1,-2),(1,-1),(1,0),(1,1),(2,-2),(2,-1),(2,0),(2,1)}. Step 2: take S1 = {(1,-2),(2,1)}; domain of S1 = {1,2}, range of S1 = {-2,1}. Step 3: take S2 = {(1,0),(2,-1)}; domain of S2 = {1,2}, range of S2 = {0,-1}. Answer: S1 = {(1,-2),(2,1)} (domain {1,2}, range {-2,1}); S2 = {(1,0),(2,-1)} (domain {1,2}, range {0,-1}).

Q2. If U = {x | x in W, x <= 4} and V = {y | y in N, y <= 3} then find the following

relation from U to V: (i) R1 = {(x,y) | x > y}. Also write domain and range of each relation.

Step 1: U = {0,1,2,3,4} and V = {1,2,3}. Step 2: pairs with x > y: (2,1), (3,1), (3,2), (4,1), (4,2), (4,3). Step 3: domain = first components = {2,3,4}; range = second components = {1,2,3}. Answer: R1 = {(2,1),(3,1),(3,2),(4,1),(4,2),(4,3)}; domain {2,3,4}, range {1,2,3}.

Q2. (ii) R2 = {(x,y) | x = y}.

Step 1: U = {0,1,2,3,4}, V = {1,2,3}. Step 2: pairs with x = y: (1,1), (2,2), (3,3). Step 3: domain = {1,2,3}, range = {1,2,3}. Answer: R2 = {(1,1),(2,2),(3,3)}; domain {1,2,3}, range {1,2,3}.

Q2. (iii) R3 = {(x,y) | x + y = 4}.

Step 1: U = {0,1,2,3,4}, V = {1,2,3}. Step 2: pairs with sum 4: (1,3), (2,2), (3,1). Step 3: domain = {1,2,3}, range = {1,2,3}. Answer: R3 = {(1,3),(2,2),(3,1)}; domain {1,2,3}, range {1,2,3}.

Q2. (iv) R4 = {(x,y) | x - y = 3}.

Step 1: U = {0,1,2,3,4}, V = {1,2,3}. Step 2: pairs with difference 3: only (4,1). Step 3: domain = {4}, range = {1}. Answer: R4 = {(4,1)}; domain {4}, range {1}.

Q3. If set A has 3 elements, then find the number of binary relations in A.

Step 1: A x A has 3 x 3 = 9 ordered pairs. Step 2: a binary relation is any subset of A x A. Step 3: number of subsets of a 9-element set = 2^9 = 512. Answer: 512.

Q4. If R = {(x,y) | x,y in N, x + y = 4}, then find domain and range of R.

Step 1: N = {1,2,3,...}, so x and y are positive. Step 2: pairs with sum 4: (1,3), (2,2), (3,1). Step 3: domain = {1,2,3}, range = {1,2,3}. Answer: domain {1,2,3}, range {1,2,3}.

Q5. If A = {3,1,-1,-3}, B = {2,4} then find R1 = {(x,y) | x + y < 0}, R2 = {(x,y) | x + y > 0}.

Step 1: A x B = {(3,2),(3,4),(1,2),(1,4),(-1,2),(-1,4),(-3,2),(-3,4)}. Step 2: sums: 5,7,3,5,1,3,-1,1. Step 3: only (-3,2) has a negative sum. Answer: R1 = {(-3,2)}; R2 = {(3,2),(3,4),(1,2),(1,4),(-1,2),(-1,4),(-3,4)}.

Q6. If R = {(x,y) | y = 4x + 5} and -2 <= x <= 1 then find its domain and range. (x taken as integers in the interval, as in the book's relation examples.)

Step 1: x = -2 gives y = 4(-2) + 5 = -3. Step 2: x = -1 gives y = 1. Step 3: x = 0 gives y = 5. Step 4: x = 1 gives y = 9. Answer: R = {(-2,-3),(-1,1),(0,5),(1,9)}; domain {-2,-1,0,1}, range {-3,1,5,9}.

Q7. If R = {(x,y) | x^2 + y^2 = 9}, x,y in Z, then find relations in the form of table, order pairs and draw graph.

Step 1: integer pairs with x^2 + y^2 = 9: (3,0), (-3,0), (0,3), (0,-3). Step 2: ordered-pair form: R = {(3,0),(-3,0),(0,3),(0,-3)}. Step 3: table form: x: 3, -3, 0, 0 against y: 0, 0, 3, -3. Step 4: graph: the four points (3,0), (-3,0), (0,3), (0,-3) plotted on the coordinate axes. Answer: R = {(3,0),(-3,0),(0,3),(0,-3)}; domain {-3,0,3}, range {-3,0,3}.

REVIEW EXERCISE 3 (book pp.72-73)

Q1. Choose the correct answer: (i) A set having no element is called:

(a) subset

(b) empty set ✓

(c) power set

(d) singleton set

Q1. (ii) The set {1,2,3,...,10} is:

(a) Empty set

(b) subset

(c) finite set ✓

(d) infinite set

Q1. (iii) A set {p/q | p,q in Z, q not = 0} is called a set of:

(a) Rational numbers ✓

(b) Irrational numbers

(c) Integers

(d) Whole numbers

Q1. (iv) Power set of an empty set is:

(a) phi

(b) {phi} ✓

(c) {0}

(d) {phi,{phi}}

Q1. (v) A set having only one element is called:

(a) Null set

(b) power set

(c) Singleton set ✓

(d) infinite set

Q1. (vi) If A is a subset of B then A intersection B =:

(a) A ✓

(b) B

(c) A intersection B

(d) A union B

Q1. (vii) If set A has 4 elements and set B has 2 elements then A x B has ..... Ordered pairs.

(a) 2

(b) 4

(c) 6

(d) 8 ✓

Q1. (viii) If B is a subset of A then B - A is equal to:

(a) A

(b) B

(c) phi ✓

(d) {phi}

Q1. (ix) The domain of {(0,1),(1,2),(2,3)} is:

(a) {1,2,3}

(b) {0,2,3}

(c) {0,1,2,3}

(d) {0,1,2} ✓

Q2. If U = {1,2,3,...,10}, A = {1,3,5,7,9}, B = {2,4,6,8,10}, then find: (i) (A union B)^c.

Step 1: A union B = {1,2,3,4,5,6,7,8,9,10} = U. Step 2: complement of U in U = {}. Answer: {}.

Q2. (ii) A^c union B^c.

Step 1: A^c = U - A = {2,4,6,8,10} = B. Step 2: B^c = U - B = {1,3,5,7,9} = A. Step 3: A^c union B^c = B union A = {1,2,3,4,5,6,7,8,9,10}. Answer: {1,2,3,4,5,6,7,8,9,10}.

Q3. If A = {1,2,3,4}, B = {3,4,5,6}, C = {5,6,7,8} then show that: (i) A union (B intersection C) = (A union B) intersection (A union C).

Step 1: B intersection C = {5,6}; A union {5,6} = {1,2,3,4,5,6}. Step 2: A union B = {1,2,3,4,5,6}; A union C = {1,2,3,4,5,6,7,8}; their intersection = {1,2,3,4,5,6}. Step 3: both sides equal {1,2,3,4,5,6}. Hence shown.

Q3. (ii) A intersection (B union C) = (A intersection B) union (A intersection C).

Step 1: B union C = {3,4,5,6,7,8}; A intersection that = {3,4}. Step 2: A intersection B = {3,4}; A intersection C = {}; their union = {3,4}. Step 3: both sides equal {3,4}. Hence shown.

Q4. If (u - 3, v + 4) = (3u - 5, 6v - 11), then find the values of u and v.

Step 1: equate first components: u - 3 = 3u - 5. Step 2: bring terms together: -3 + 5 = 3u - u, so 2 = 2u, u = 1. Step 3: equate second components: v + 4 = 6v - 11. Step 4: 4 + 11 = 6v - v, so 15 = 5v, v = 3. Answer: u = 1, v = 3.

Q5. If A = {x | x in N, x <= 5} and B = {y | y in W, y <= 4} then find the following relation from A to B: (i) R1 = {(x,y) | x = y}.

Step 1: A = {1,2,3,4,5} and B = {0,1,2,3,4}. Step 2: pairs with x = y: (1,1), (2,2), (3,3), (4,4). Answer: R1 = {(1,1),(2,2),(3,3),(4,4)}.

Q5. (ii) R2 = {(x,y) | x > y}.

Step 1: A = {1,2,3,4,5}, B = {0,1,2,3,4}. Step 2: x = 1 gives (1,0); x = 2 gives (2,0), (2,1); x = 3 gives (3,0), (3,1), (3,2); x = 4 gives (4,0), (4,1), (4,2), (4,3); x = 5 gives (5,0), (5,1), (5,2), (5,3), (5,4). Answer: R2 = {(1,0),(2,0),(2,1),(3,0),(3,1),(3,2),(4,0),(4,1),(4,2),(4,3),(5,0),(5,1),(5,2),(5,3),(5,4)} (15

pairs).

Q5. (iii) R3 = {(x,y) | x + y = 6}.

Step 1: A = {1,2,3,4,5}, B = {0,1,2,3,4}. Step 2: pairs with sum 6: (2,4), (3,3), (4,2), (5,1).

Answer: R3 = {(2,4),(3,3),(4,2),(5,1)}.

Q5. (iv) R4 = {(x,y) | x - y = 2}.

Step 1: A = {1,2,3,4,5}, B = {0,1,2,3,4}. Step 2: pairs with difference 2: (2,0), (3,1), (4,2), (5,3). Answer: R4 = {(2,0),(3,1),(4,2),(5,3)}.

Unit 4: Factorization and Algebraic Manipulation

EXERCISE 4.1 (book pp.77-78)

Q1. Factorize the following algebraic expressions: (i) 4x - 8(y - 2z).

Step 1: remove the bracket: 4x - 8y + 16z. Step 2: take 4 common to all terms. Answer: 4(x - 2y + 4z).

Q1. (ii) 3xy^2 + 27yx^2 - 81xy.

Step 1: each term contains 3xy. Step 2: 3xy^2 = 3xy(y); 27yx^2 = 3xy(9x); 81xy = 3xy(27). Step 3: take 3xy common. Answer: 3xy(y + 9x - 27).

Q1. (iii) ab(x^2 + 1) + x(a^2 + b^2).

Step 1: expand: abx^2 + ab + a^2x + b^2x. Step 2: group: (abx^2 + a^2x) + (b^2x + ab) = ax(bx + a) + b(bx + a). Step 3: take (bx + a) common. Answer: (bx + a)(ax + b).

Q1. (iv) (x + y)(a + b) - (y + z)(a + b).

Step 1: (a + b) is common to both terms. Step 2: take it out: (a + b)[(x + y) - (y + z)]. Step 3: simplify inside: x + y - y - z = x - z. Answer: (a + b)(x - z).

Q1. (v) 7(x + 2y)^2 - 21(x + 2y).

Step 1: take 7(x + 2y) common. Step 2: 7(x + 2y)[(x + 2y) - 3]. Answer: 7(x + 2y)(x + 2y - 3).

Q1. (vi) 12xy + 20x - 15 - 9y.

Step 1: group: (12xy + 20x) - (9y + 15) = 4x(3y + 5) - 3(3y + 5). Step 2: take (3y + 5) common. Answer: (3y + 5)(4x - 3).

Q2. Factorize the following algebraic expressions: (i) x(3a + b) + z(b + 3a).

Step 1: b + 3a is the same as 3a + b. Step 2: take (3a + b) common. Answer: (3a + b)(x + z).

Q2. (ii) 3a^3 - 2a^2 + 3a - 2.

Step 1: group: (3a^3 - 2a^2) + (3a - 2) = a^2(3a - 2) + 1(3a - 2). Step 2: take (3a - 2) common. Answer: (3a - 2)(a^2 + 1).

Q2. (iii) x^2 - 2xy + xz - 2yz.

Step 1: group: (x^2 - 2xy) + (xz - 2yz) = x(x - 2y) + z(x - 2y). Step 2: take (x - 2y) common. Answer: (x - 2y)(x + z).

Q2. (iv) 9ky - 4hx + 12hy - 3kx.

Step 1: group k-terms and h-terms: (9ky - 3kx) + (12hy - 4hx) = 3k(3y - x) + 4h(3y - x). Step 2: take (3y - x) common. Answer: (3y - x)(3k + 4h).

Q2. (v) 4xp - 12qx - 2py + 6qy.

Step 1: group: (4xp - 12qx) - (2py - 6qy) = 4x(p - 3q) - 2y(p - 3q). Step 2: take (p - 3q) common: (p - 3q)(4x - 2y). Step 3: take 2 out of the second bracket. Answer: 2(p - 3q)(2x - y).

Q2. (vi) a^n(x - 2z) + b^(n-1)(x - 2z) + c^(n-2)(2z - x).

Step 1: 2z - x = -(x - 2z). Step 2: the expression becomes (x - 2z)(a^n + b^(n-1) - c^(n-2)).

Answer: (x - 2z)(a^n + b^(n-1) - c^(n-2)).

Q3. Factorize: (i) 1 + 12ab + 36a^2b^2.

Step 1: 1 = (1)^2, 36a^2b^2 = (6ab)^2, and 2(1)(6ab) = 12ab. Step 2: perfect square a^2 + 2ab + b^2 = (a + b)^2. Answer: (1 + 6ab)^2.

Q3. (ii) x^4y^4 - 28xy + 196/(x^2y^2).

Step 1: x^4y^4 = (x^2y^2)^2 and 196/(x^2y^2) = (14/(xy))^2. Step 2: 2(x^2y^2)(14/(xy)) = 28xy. Step 3: perfect square (a - b)^2 form. Answer: (x^2y^2 - 14/(xy))^2.

Q3. (iii) x^2 + 0.6xy + 0.09y^2.

Step 1: 0.09y^2 = (0.3y)^2 and 2(x)(0.3y) = 0.6xy. Step 2: perfect square form. Answer: (x + 0.3y)^2.

Q3. (iv) 16 + (x + 4y)^2 - 8(4y + x).

Step 1: write as (x + 4y)^2 - 8(x + 4y) + 16. Step 2: this is a^2 - 2ab + b^2 with a = x + 4y, b = 4. Answer: (x + 4y - 4)^2.

Q3. (v) x^4 - 18x^2 + 81.

Step 1: (x^2)^2 - 2(x^2)(9) + 9^2 = (x^2 - 9)^2. Step 2: x^2 - 9 = (x - 3)(x + 3). Answer: (x - 3)^2(x + 3)^2.

Q3. (vi) x^5 - 8x^3y^2 + 16xy^4.

Step 1: take x common: x(x^4 - 8x^2y^2 + 16y^4). Step 2: inside: (x^2)^2 - 2(x^2)(4y^2) + (4y^2)^2 = (x^2 - 4y^2)^2. Step 3: x^2 - 4y^2 = (x - 2y)(x + 2y). Answer: x(x - 2y)^2(x + 2y)^2.

Q4. Factorize: (i) x^2y^2 - 256.

Step 1: 256 = 16^2, so a^2 - b^2 form with a = xy, b = 16. Answer: (xy - 16)(xy + 16).

Q4. (ii) (x - 3)^2 - 3.

Step 1: 3 = (sqrt(3))^2, so a^2 - b^2 form with a = x - 3, b = sqrt(3). Answer: (x - 3 - sqrt(3))(x - 3 + sqrt(3)).

Q4. (iii) (9x^2)/(4y^2) - (25y^2)/(16x^2).

Step 1: (9x^2)/(4y^2) = (3x/(2y))^2 and (25y^2)/(16x^2) = (5y/(4x))^2. Step 2: a^2 - b^2 form. Answer: (3x/(2y) - 5y/(4x))(3x/(2y) + 5y/(4x)).

Q4. (iv) 3x^4y^3 - 27y.

Step 1: take 3y common: 3y(x^4y^2 - 9). Step 2: x^4y^2 - 9 = (x^2y)^2 - 3^2 = (x^2y - 3)(x^2y + 3). Answer: 3y(x^2y - 3)(x^2y + 3).

Q4. (v) z^(2n) - y^(2n).

Step 1: z^(2n) = (z^n)^2 and y^(2n) = (y^n)^2. Step 2: a^2 - b^2 form. Answer: (z^n - y^n)(z^n + y^n).

Q4. (vi) a^2 + b^2 - 2cd - c^2 - d^2 + 2ab.

Step 1: rearrange: (a^2 + 2ab + b^2) - (c^2 + 2cd + d^2) = (a + b)^2 - (c + d)^2. Step 2: a^2 - b^2 form. Answer: (a + b + c + d)(a + b - c - d).

Q4. (vii) x^4 - 256y^4.

Step 1: (x^2)^2 - (16y^2)^2 = (x^2 - 16y^2)(x^2 + 16y^2). Step 2: x^2 - 16y^2 = (x - 4y)(x + 4y). Answer: (x - 4y)(x + 4y)(x^2 + 16y^2).

Q5. Factorize: (i) 36x^2 - 9y^2 - 24x + 4.

Step 1: group: (36x^2 - 24x + 4) - 9y^2 = (6x - 2)^2 - (3y)^2. Step 2: a^2 - b^2 form. Answer: (6x - 2 - 3y)(6x - 2 + 3y).

Q5. (ii) 2x^2 + 2y^2 - 4xy - 288z^2.

Step 1: take 2 common: 2[(x^2 - 2xy + y^2) - 144z^2] = 2[(x - y)^2 - (12z)^2]. Step 2: a^2 - b^2 form. Answer: 2(x - y - 12z)(x - y + 12z).

Q5. (iii) a + b + c + a^2 - b^2 - c^2 - 2bc.

Step 1: group: (a + b + c) + (a^2 - (b^2 + 2bc + c^2)) = (a + b + c) + (a^2 - (b + c)^2). Step 2: a^2 - (b + c)^2 = (a - b - c)(a + b + c). Step 3: take (a + b + c) common. Answer: (a + b + c)(1 + a - b - c).

Q5. (iv) 4s^2 - 1 - 2t - t^2.

Step 1: 4s^2 - (1 + 2t + t^2) = (2s)^2 - (1 + t)^2. Step 2: a^2 - b^2 form. Answer: (2s - 1 - t)(2s + 1 + t).

Q5. (v) 49x^2 - y^2 + 121z^2 - 154xz.

Step 1: group: (49x^2 - 154xz + 121z^2) - y^2 = (7x - 11z)^2 - y^2. Step 2: a^2 - b^2 form. Answer: (7x - 11z - y)(7x - 11z + y).

Unit 4: Factorization and Algebraic Manipulation

(continued: Exercise 4.2)

EXERCISE 4.2 (book pp.84-85)

Q1. Factorize: (i) x^4 + y^4/4.

Step 1: complete the square: x^4 + y^4/4 = (x^2)^2 + 2(x^2)(y^2/2) + (y^2/2)^2 - x^2y^2 = (x^2 + y^2/2)^2 - (xy)^2. Step 2: a^2 - b^2 form. Answer: (x^2 + xy + y^2/2)(x^2 - xy + y^2/2).

Q1. (ii) 4x^4 + 81y^4.

Step 1: (2x^2)^2 + 2(2x^2)(9y^2) + (9y^2)^2 - 36x^2y^2 = (2x^2 + 9y^2)^2 - (6xy)^2. Step 2: a^2 - b^2 form. Answer: (2x^2 + 6xy + 9y^2)(2x^2 - 6xy + 9y^2).

Q1. (iii) x^4 + 36 + 8x^2.

Step 1: write as (x^2 + 6)^2 - 4x^2, since (x^2 + 6)^2 = x^4 + 12x^2 + 36. Step 2: a^2 - b^2 form with a = x^2 + 6, b = 2x. Answer: (x^2 - 2x + 6)(x^2 + 2x + 6).

Q1. (iv) 16a^4 - 97a^2b^2 + 81b^4.

Step 1: as a quadratic in a^2: (16a^2 - 81b^2)(a^2 - b^2) = 16a^4 - 97a^2b^2 + 81b^4. Step 2: both brackets are differences of squares. Answer: (4a - 9b)(4a + 9b)(a - b)(a + b).

Q1. (v) 9a^4 + 16b^4 - 28a^2b^2.

Step 1: (3a^2 - 4b^2)^2 = 9a^4 - 24a^2b^2 + 16b^4. Step 2: the given expression = (3a^2 - 4b^2)^2 - 4a^2b^2 = (3a^2 - 4b^2)^2 - (2ab)^2. Step 3: a^2 - b^2 form. Answer: (3a^2 - 2ab - 4b^2)(3a^2 + 2ab - 4b^2).

Q1. (vi) x^4 - y^4 - 2x^3y + 2xy^3.

Step 1: group: (x^4 - y^4) - 2xy(x^2 - y^2) = (x^2 - y^2)(x^2 + y^2) - 2xy(x^2 - y^2). Step 2: take (x^2 - y^2) common: (x^2 - y^2)(x^2 - 2xy + y^2) = (x^2 - y^2)(x - y)^2. Step 3: x^2 - y^2 = (x - y)(x + y). Answer: (x - y)^3(x + y).

Q2. Factorize: (i) x^2 + 15x + 36.

Step 1: find two numbers with product 36 and sum 15: 3 and 12. Step 2: split the middle term: x^2 + 3x + 12x + 36 = x(x + 3) + 12(x + 3). Answer: (x + 3)(x + 12).

Q2. (ii) x^2 - 2x - 143.

Step 1: factors of 143 with difference 2: 11 and 13; bigger number takes the sign of -2, so -13 and +11. Step 2: x^2 - 13x + 11x - 143 = x(x - 13) + 11(x - 13). Answer: (x - 13)(x + 11).

Q2. (iii) x^2 + 6x - 160.

Step 1: factors of 160 with difference 6: 10 and 16; bigger takes the + sign: +16 and -10. Step 2: x^2 + 16x - 10x - 160 = x(x + 16) - 10(x + 16). Answer: (x + 16)(x - 10).

Q2. (iv) x^2 - 10x + 16.

Step 1: factors of 16 with sum 10: 2 and 8, both negative. Step 2: x^2 - 2x - 8x + 16 = x(x - 2) - 8(x - 2). Answer: (x - 2)(x - 8).

Q3. Factorize: (i) 2x^2 + 12x - 110.

Step 1: take 2 common: 2(x^2 + 6x - 55). Step 2: factors of 55 with difference 6: 5 and 11; bigger takes + sign: x^2 + 11x - 5x - 55 = (x + 11)(x - 5). Answer: 2(x + 11)(x - 5).

Q3. (ii) sqrt(2)x^2 + 7x + 6sqrt(2).

Step 1: a = sqrt(2), c = 6sqrt(2), so ac = 12; factors of 12 with sum 7: 3 and 4. Step 2: sqrt(2)x^2 + 3x + 4x + 6sqrt(2) = x(sqrt(2)x + 3) + 2sqrt(2)(sqrt(2)x + 3). Answer: (sqrt(2)x + 3)(x + 2sqrt(2)).

Q3. (iii) 3x^2 - 17x + 14.

Step 1: ac = 42; factors with sum 17: 14 and 3. Step 2: 3x^2 - 14x - 3x + 14 = x(3x - 14) - 1(3x - 14). Answer: (3x - 14)(x - 1).

Q3. (iv) 14s^2 + 33s - 5.

Step 1: ac = -70; factors with difference 33: 35 and -2. Step 2: 14s^2 + 35s - 2s - 5 = 7s(2s + 5) - 1(2s + 5). Answer: (2s + 5)(7s - 1).

Q3. (v) 30x^2 + 7xy - 15y^2.

Step 1: ac = -450; factors with difference 7: 25 and -18. Step 2: 30x^2 + 25xy - 18xy - 15y^2 = 5x(6x + 5y) - 3y(6x + 5y). Answer: (6x + 5y)(5x - 3y).

Q3. (vi) 2x^2 + 12x - 110. (Printed twice in the book: identical to Q3(i).)

Step 1: take 2 common: 2(x^2 + 6x - 55). Step 2: (x + 11)(x - 5) as in Q3(i). Answer: 2(x + 11)(x - 5).

Q4. Factorize: (i) (x^2 - 2x - 35)(x^2 - 2x - 3) - 144.

Step 1: let y = x^2 - 2x: (y - 35)(y - 3) - 144 = y^2 - 38y + 105 - 144 = y^2 - 38y - 39. Step 2: y^2 - 38y - 39 = (y - 39)(y + 1). Step 3: back-substitute: (x^2 - 2x - 39)(x^2 - 2x + 1) = (x^2 - 2x - 39)(x - 1)^2. Answer: (x^2 - 2x - 39)(x - 1)^2.

Q4. (ii) (x^2 + 7x - 8)(x^2 + 7x + 10) - 880.

Step 1: let y = x^2 + 7x: (y - 8)(y + 10) - 880 = y^2 + 2y - 80 - 880 = y^2 + 2y - 960. Step 2: y^2 + 2y - 960 = (y + 32)(y - 30). Step 3: back-substitute: (x^2 + 7x + 32)(x^2 + 7x - 30) = (x^2 + 7x + 32)(x + 10)(x - 3). Answer: (x^2 + 7x + 32)(x + 10)(x - 3).

Q4. (iii) (x + 1)(x + 2)(x + 3)(x + 4) - 24.

Step 1: pair first with last and middle two: (x + 1)(x + 4) = x^2 + 5x + 4; (x + 2)(x + 3) = x^2 + 5x + 6. Step 2: let y = x^2 + 5x: (y + 4)(y + 6) - 24 = y^2 + 10y + 24 - 24 = y(y + 10). Step 3: back-substitute: (x^2 + 5x)(x^2 + 5x + 10) = x(x + 5)(x^2 + 5x + 10). Answer: x(x + 5)(x^2 + 5x + 10).

Q4. (iv) (x^2 - x - 12)(x^2 - x - 12) - x^2.

Step 1: this is (x^2 - x - 12)^2 - x^2, a difference of squares. Step 2: (x^2 - x - 12 - x)(x^2 - x - 12 + x) = (x^2 - 2x - 12)(x^2 - 12). Answer: (x^2 - 2x - 12)(x^2 - 12).

Q5. Factorize: (i) x^3 - 125 - 15x^2 + 75x.

Step 1: rearrange: x^3 - 15x^2 + 75x - 125. Step 2: this matches (a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 with a = x, b = 5. Answer: (x - 5)^3.

Q5. (ii) 8x^3 + 72x^2y + 216xy^2 + 216y^3.

Step 1: matches (a + b)^3 with a = 2x, b = 6y: (2x)^3 = 8x^3; 3(2x)^2(6y) = 72x^2y; 3(2x)(6y)^2 = 216xy^2; (6y)^3 = 216y^3. Answer: (2x + 6y)^3 = 8(x + 3y)^3.

Q5. (iii) x^3 + x + 1/(3x) + 1/(27x^3).

Step 1: matches (a + b)^3 with a = x, b = 1/(3x): 3x^2(1/(3x)) = x; 3x(1/(9x^2)) = 1/(3x); (1/(3x))^3 = 1/(27x^3). Answer: (x + 1/(3x))^3.

Q6. Factorize: (i) x^3 + 81.

Step 1: 81 is not a perfect cube, so this does not factor over the rationals. Step 2: over the reals, 81 = (3 x 3rd-root(3))^3, since (3 x 3rd-root(3))^3 = 27 x 3 = 81. Step 3: a^3 + b^3 form with b = 3 x 3rd-root(3). Answer: (x + 3 x 3rd-root(3))(x^2 - 3 x 3rd-root(3) x + 9 x 3rd-root(9)). Note: the book prints 81 here although 81 is not a cube; the factorisation above is exact but uses cube roots.

Q6. (ii) 8x^3 + 216y^3.

Step 1: (2x)^3 + (6y)^3, so a^3 + b^3 form. Step 2: (2x + 6y)((2x)^2 - (2x)(6y) + (6y)^2). Answer: (2x + 6y)(4x^2 - 12xy + 36y^2) = 8(x + 3y)(x^2 - 3xy + 9y^2).

Q6. (iii) x^3 - 1/64.

Step 1: (x)^3 - (1/4)^3, so a^3 - b^3 form. Answer: (x - 1/4)(x^2 + x/4 + 1/16).

Q6. (iv) 1 - 216y^3.

Step 1: (1)^3 - (6y)^3, so a^3 - b^3 form. Answer: (1 - 6y)(1 + 6y + 36y^2).

Q6. (v) x^6y^6 - z^6.

Step 1: (x^2y^2)^3 - (z^2)^3, so a^3 - b^3 form: (x^2y^2 - z^2)(x^4y^4 + x^2y^2z^2 + z^4). Step 2: x^2y^2 - z^2 = (xy - z)(xy + z). Answer: (xy - z)(xy + z)(x^4y^4 + x^2y^2z^2 + z^4).

Q6. (vi) (x - 3y)^3 + 125z^3.

Step 1: (x - 3y)^3 + (5z)^3, so a^3 + b^3 form. Step 2: (x - 3y + 5z)((x - 3y)^2 - 5z(x - 3y) + (5z)^2). Step 3: expand the second bracket: x^2 - 6xy + 9y^2 - 5xz + 15yz + 25z^2. Answer: (x - 3y + 5z)(x^2 - 6xy + 9y^2 - 5xz + 15yz + 25z^2).

Unit 4: Factorization and Algebraic Manipulation

(continued: Exercises 4.3-4.4)

EXERCISE 4.3 (book p.86) — Find the H.C.F. of the following polynomials by factorization

Q1. 5x^3y^4, 20x^2y^3.

Step 1: factor each: 5x^3y^4 = 5 x x^2 x y^3 x y; 20x^2y^3 = 5 x 4 x x^2 x y^3. Step 2: common factors: 5, x^2, y^3. Answer: H.C.F. = 5x^2y^3.

Q2. 26a^3b^4c^2, 65a^2b^2c^4.

Step 1: gcd of 26 and 65 is 13. Step 2: lowest powers common: a^2, b^2, c^2. Answer: H.C.F. = 13a^2b^2c^2.

Q3. 18x^2y^2c^4, 24x^2y^3c^4.

Step 1: gcd of 18 and 24 is 6. Step 2: lowest powers common: x^2, y^2, c^4. Answer: H.C.F. = 6x^2y^2c^4.

Q4. x^2 - y^2, x^2 - 3xy + 2y^2.

Step 1: x^2 - y^2 = (x - y)(x + y). Step 2: x^2 - 3xy + 2y^2 = (x - y)(x - 2y). Step 3: the only common factor is (x - y). Answer: H.C.F. = (x - y).

Q5. x^6 + 64, x^4 + 7x^2 + 12.

Step 1: x^6 + 64 = (x^2)^3 + 4^3 = (x^2 + 4)(x^4 - 4x^2 + 16). Step 2: x^4 + 7x^2 + 12 = (x^2 + 3)(x^2 + 4). Step 3: common factor is (x^2 + 4). Answer: H.C.F. = x^2 + 4.

Q6. 2x^3 + 2, 2x^2 + 6x + 4.

Step 1: 2x^3 + 2 = 2(x^3 + 1) = 2(x + 1)(x^2 - x + 1). Step 2: 2x^2 + 6x + 4 = 2(x^2 + 3x + 2) = 2(x + 1)(x + 2). Step 3: common factors: 2 and (x + 1). Answer: H.C.F. = 2(x + 1).

Q7. 6a^3 + 24a^2 + 6a - 36, 4a^3 - 8a^2 - 20a + 24.

Step 1: first = 6(a^3 + 4a^2 + a - 6) = 6(a - 1)(a + 2)(a + 3), since a = 1 is a root and a^3 + 4a^2 + a - 6 = (a - 1)(a^2 + 5a + 6). Step 2: second = 4(a^3 - 2a^2 - 5a + 6) = 4(a - 1)(a - 3)(a + 2), since a = 1 is a root and a^3 - 2a^2 - 5a + 6 = (a - 1)(a^2 - a - 6). Step 3: common factors: 2, (a - 1), (a + 2). Answer: H.C.F. = 2(a - 1)(a + 2).

Q8. y^2 - 36, y^2 - 10y + 24.

Step 1: y^2 - 36 = (y - 6)(y + 6). Step 2: y^2 - 10y + 24 = (y - 4)(y - 6). Step 3: common factor is (y - 6). Answer: H.C.F. = (y - 6).

Q9. y^3 - 8, y^2 - 7y + 10.

Step 1: y^3 - 8 = (y - 2)(y^2 + 2y + 4). Step 2: y^2 - 7y + 10 = (y - 2)(y - 5). Step 3: common factor is (y - 2). Answer: H.C.F. = (y - 2).

Q10. m^2 - n^2, m^4 - n^4, m^6 - n^6.

Step 1: m^2 - n^2 = (m - n)(m + n). Step 2: m^4 - n^4 = (m^2 - n^2)(m^2 + n^2) = (m - n)(m + n)(m^2 + n^2). Step 3: m^6 - n^6 = (m^2 - n^2)(m^4 + m^2n^2 + n^4) = (m - n)(m + n)(m^4 + m^2n^2 + n^4). Step 4: common to all three: (m - n)(m + n). Answer: H.C.F. = m^2 - n^2.

EXERCISE 4.4 (book p.88) — Find H.C.F. by division method

Q1. x^3 + 4x^2 - 3x - 2, x^3 - 7x^2 - 5x + 11.

Step 1: factor (division gives the same result): x^3 + 4x^2 - 3x - 2 = (x - 1)(x^2 + 5x + 2), since x = 1 is a root. Step 2: x^3 - 7x^2 - 5x + 11 = (x - 1)(x^2 - 6x - 11), since x = 1 is a root. Step 3: the quadratics share no factor, so the last non-zero divisor is (x - 1). Answer: H.C.F. = (x - 1).

Q2. x^3 + 6x^2 + 7x + 2, 5x^3 + 56x^2 + 50x - 1.

Step 1: x^3 + 6x^2 + 7x + 2 = (x + 1)(x^2 + 5x + 2), since x = -1 is a root. Step 2: 5x^3 + 56x^2 + 50x - 1 = (x + 1)(5x^2 + 51x - 1), since x = -1 is a root. Step 3: the quadratics share no factor. Answer: H.C.F. = (x + 1).

Q3. x^4 - x^3 - 6x^2, x^4 - 9x^2.

Step 1: x^4 - x^3 - 6x^2 = x^2(x^2 - x - 6) = x^2(x - 3)(x + 2). Step 2: x^4 - 9x^2 = x^2(x^2 - 9) = x^2(x - 3)(x + 3). Step 3: common factors: x^2 and (x - 3). Answer: H.C.F. = x^2(x - 3).

Q4. 2y^3 - 9y^2 + 9y - 7, y^3 - 5y^2 + 5y - 4.

Step 1: 2y^3 - 9y^2 + 9y - 7 = (2y - 7)(y^2 - y + 1), since y = 7/2 is a root. Step 2: y^3 - 5y^2 + 5y - 4 = (y - 4)(y^2 - y + 1), since y = 4 is a root. Step 3: common factor is (y^2 - y + 1). Answer: H.C.F. = (y^2 - y + 1).

Q5. 2y^3 - 54, 2y^4 + 18y^2 + 162.

Step 1: 2y^3 - 54 = 2(y^3 - 27) = 2(y - 3)(y^2 + 3y + 9). Step 2: 2y^4 + 18y^2 + 162 = 2(y^4 + 9y^2 + 81) = 2(y^2 + 3y + 9)(y^2 - 3y + 9), by completing the square. Step 3: common factors: 2 and (y^2 + 3y + 9). Answer: H.C.F. = 2(y^2 + 3y + 9).

Q6. 25y^4 - 5y^3 - y - 1, 20y^4 - y^2 - 1.

Step 1: 25y^4 - 5y^3 - y - 1 = (5y^2 - y - 1)(5y^2 + 1). Step 2: 20y^4 - y^2 - 1 = (5y^2 + 1)(4y^2 - 1). Step 3: common factor is (5y^2 + 1). Answer: H.C.F. = (5y^2 + 1).

Q7. x^5 - y^5, x^6 - y^6.

Step 1: x^5 - y^5 = (x - y)(x^4 + x^3y + x^2y^2 + xy^3 + y^4). Step 2: x^6 - y^6 = (x - y)(x + y)(x^2 + xy + y^2)(x^2 - xy + y^2). Step 3: the quartic factor of the first shares no factor with the remaining factors of the second, so the only common factor is (x - y). Answer: H.C.F. = (x - y).

Q8. x^5 - 13x + 12, x^4 - 5x^2 + 4.

Step 1: x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4) = (x - 1)(x + 1)(x - 2)(x + 2). Step 2: test these roots in x^5 - 13x + 12: x = 1 gives 1 - 13 + 12 = 0; x = -1 gives -1 + 13 + 12 = 24; x = 2

gives 32 - 26 + 12 = 18; x = -2 gives -32 + 26 + 12 = 6. Step 3: only (x - 1) is common.

Answer: H.C.F. = (x - 1).

Unit 4: Factorization and Algebraic Manipulation

(continued: Exercise 4.5)

EXERCISE 4.5 (book p.90) — Find L.C.M of the following by factorization

Q1. x^3y^2z^2, x^2y^3z^2, x^2y^2z^3.

Step 1: take the highest power of each variable present: x^3, y^3, z^3. Answer: L.C.M. = x^3y^3z^3.

Q2. x^2 - 16, x^2 - 11x + 28.

Step 1: x^2 - 16 = (x - 4)(x + 4). Step 2: x^2 - 11x + 28 = (x - 4)(x - 7). Step 3: common factor (x - 4) taken once; uncommon factors (x + 4) and (x - 7). Answer: L.C.M. = (x - 4)(x + 4)(x - 7).

Q3. x^2 - 5x + 6, x^2 - 10x + 16.

Step 1: x^2 - 5x + 6 = (x - 2)(x - 3). Step 2: x^2 - 10x + 16 = (x - 2)(x - 8). Answer: L.C.M. = (x - 2)(x - 3)(x - 8).

Q4. x^2 + 7x + 12, x^2 + 10x + 24.

Step 1: x^2 + 7x + 12 = (x + 3)(x + 4). Step 2: x^2 + 10x + 24 = (x + 4)(x + 6). Answer: L.C.M. = (x + 3)(x + 4)(x + 6).

Q5. x^3 - 1, x^3 - 2x^2 + x.

Step 1: x^3 - 1 = (x - 1)(x^2 + x + 1). Step 2: x^3 - 2x^2 + x = x(x - 1)^2. Step 3: highest powers: x, (x - 1)^2, (x^2 + x + 1). Answer: L.C.M. = x(x - 1)^2(x^2 + x + 1).

Q6. a^3 - 8, a^2 - 4.

Step 1: a^3 - 8 = (a - 2)(a^2 + 2a + 4). Step 2: a^2 - 4 = (a - 2)(a + 2). Answer: L.C.M. = (a - 2)(a + 2)(a^2 + 2a + 4).

Q7. x^2 - x - 2, x^2 - 2x - 3, x^2 - 7x - 8.

Step 1: x^2 - x - 2 = (x - 2)(x + 1). Step 2: x^2 - 2x - 3 = (x - 3)(x + 1). Step 3: x^2 - 7x - 8 = (x - 8)(x + 1). Answer: L.C.M. = (x + 1)(x - 2)(x - 3)(x - 8).

Q8. x^2 - x - 12, x^2 - x - 56, x^2 - x - 72.

Step 1: x^2 - x - 12 = (x - 4)(x + 3). Step 2: x^2 - x - 56 = (x - 8)(x + 7). Step 3: x^2 - x - 72 = (x - 9)(x + 8). Step 4: no common factors, so the L.C.M. is the product. Answer: L.C.M. = (x - 4)(x + 3)(x - 8)(x + 7)(x - 9)(x + 8).

Q9. x^3 - y^3, x^4 - y^4, x^6 - y^6.

Step 1: x^3 - y^3 = (x - y)(x^2 + xy + y^2). Step 2: x^4 - y^4 = (x - y)(x + y)(x^2 + y^2). Step 3: x^6 - y^6 = (x - y)(x + y)(x^2 + xy + y^2)(x^2 - xy + y^2). Step 4: collect highest powers of each distinct factor. Answer: L.C.M. = (x - y)(x + y)(x^2 + y^2)(x^2 + xy + y^2)(x^2 - xy + y^2).

Q10. x^3 - 1, x^4 + x^2 + 1, (x^2 - x + 1)^2.

Step 1: x^3 - 1 = (x - 1)(x^2 + x + 1). Step 2: x^4 + x^2 + 1 = (x^2 + x + 1)(x^2 - x + 1). Step 3: the third expression is (x^2 - x + 1)^2. Step 4: highest powers: (x - 1), (x^2 + x + 1), (x^2 - x + 1)^2. Answer: L.C.M. = (x - 1)(x^2 + x + 1)(x^2 - x + 1)^2.

Unit 4: Factorization and Algebraic Manipulation

(continued: Exercise 4.6)

EXERCISE 4.6 (book p.93) — Find L.C.M of the following polynomials with the help of H.C.F.

Q1. x^3 + x^2 + x + 1, x^3 + 6x^2 + 7x + 2.

Step 1: x^3 + x^2 + x + 1 = (x + 1)(x^2 + 1) by grouping. Step 2: x^3 + 6x^2 + 7x + 2 = (x + 1)(x^2 + 5x + 2), since x = -1 is a root. Step 3: H.C.F. = (x + 1). Step 4: L.C.M. = A x B / H = (x + 1)(x^2 + 1)(x^2 + 5x + 2). Answer: L.C.M. = (x + 1)(x^2 + 1)(x^2 + 5x + 2).

Q2. x^3 - 3x^2 - 4x + 12, x^3 - 3x - 2.

Step 1: x^3 - 3x^2 - 4x + 12 = (x - 3)(x^2 - 4) = (x - 3)(x - 2)(x + 2). Step 2: x^3 - 3x - 2 = (x + 1)(x^2 - x - 2) = (x + 1)^2(x - 2), since x = -1 is a root. Step 3: H.C.F. = (x - 2). Step 4: L.C.M. = A x B / H = (x - 3)(x - 2)(x + 2)(x + 1)^2. Answer: L.C.M. = (x - 3)(x - 2)(x + 2)(x + 1)^2.

Q3. 2x^3 - 5x^2 + 9x - 6, 4x^3 - 7x^2 + 10x - 7.

Step 1: 2x^3 - 5x^2 + 9x - 6 = (x - 1)(2x^2 - 3x + 6), since x = 1 is a root. Step 2: 4x^3 - 7x^2 + 10x - 7 = (x - 1)(4x^2 - 3x + 7), since x = 1 is a root. Step 3: the two quadratics share no factor, so H.C.F. = (x - 1). Step 4: L.C.M. = A x B / H = (x - 1)(2x^2 - 3x + 6)(4x^2 - 3x + 7). Answer: L.C.M. = (x - 1)(2x^2 - 3x + 6)(4x^2 - 3x + 7).

Q4. 3x^2 + 9x^2 - 84x, x^6 - 5x^5 + 4x^4. (Note: the book prints two x^2 terms; taken as printed, the first expression combines to 12x^2 - 84x.)

Step 1: first expression = 12x^2 - 84x = 12x(x - 7). Step 2: second = x^4(x^2 - 5x + 4) = x^4(x - 1)(x - 4). Step 3: H.C.F. = x. Step 4: L.C.M. = A x B / H = 12x(x - 7) x x^4(x - 1)(x - 4) / x = 12x^4(x - 7)(x - 1)(x - 4). Answer: L.C.M. = 12x^4(x - 7)(x - 1)(x - 4).

Q5. x^4 - x^3 - x + 1, 2x^4 + 7x^2 + 3x - 12.

Step 1: x^4 - x^3 - x + 1 = (x - 1)(x^3 - 1) = (x - 1)^2(x^2 + x + 1). Step 2: 2x^4 + 7x^2 + 3x - 12 = (x - 1)(2x^3 + 2x^2 + 9x + 12), since x = 1 is a root. Step 3: the cubic shares no factor with (x - 1) or (x^2 + x + 1), so H.C.F. = (x - 1). Step 4: L.C.M. = A x B / H = (x - 1)^2(x^2 + x + 1)(2x^3 + 2x^2 + 9x + 12). Answer: L.C.M. = (x - 1)^2(x^2 + x + 1)(2x^3 + 2x^2 + 9x + 12).

Q6. H.C.F. and L.C.M. of two polynomials are (x - 3) and x^3 + 11x^2 + 12x - 12 respectively. Where one polynomial is x^2 - x - 6, find the other polynomial.

Step 1: A = x^2 - x - 6 = (x - 3)(x + 2). Step 2: B = H x L / A = (x - 3)(x^3 + 11x^2 + 12x - 12) / ((x - 3)(x + 2)) = (x^3 + 11x^2 + 12x - 12) / (x + 2). Step 3: divide: (x + 2)(x^2 + 9x - 6) = x^3 + 11x^2 + 12x - 12. Answer: the other polynomial is x^2 + 9x - 6.

Q7. Find B if A = x^2 - 5x + 6, H = x - 3, L = x^3 - 9x^2 + 26x - 24.

Step 1: A = (x - 2)(x - 3). Step 2: B = H x L / A = (x - 3)(x^3 - 9x^2 + 26x - 24) / ((x - 2)(x - 3)) = (x^3 - 9x^2 + 26x - 24) / (x - 2). Step 3: divide: (x - 2)(x^2 - 7x + 12) = x^3 - 9x^2 + 26x - 24. Answer: B = x^2 - 7x + 12 = (x - 3)(x - 4).

Q8. The product of two polynomials is x^4 - 12x^3 + 53x^2 - 102x + 72 and their H.C.F. is (x - 3), find L.C.M.

Step 1: L.C.M. = product / H.C.F. = (x^4 - 12x^3 + 53x^2 - 102x + 72) / (x - 3). Step 2: synthetic division by 3: coefficients 1, -12, 53, -102, 72 give quotient 1, -9, 26, -24 with remainder 0. Answer: L.C.M. = x^3 - 9x^2 + 26x - 24.

Q9. If B = x^2 - 9, H = x - 3 and L = (x - 3)(x + 3)(x + 1), find the other polynomial A.

Step 1: A = H x L / B = (x - 3)(x - 3)(x + 3)(x + 1) / ((x - 3)(x + 3)). Step 2: cancel (x - 3)(x + 3). Answer: A = (x - 3)(x + 1) = x^2 - 2x - 3.

Q10. If A = (x^2 - 1)(x + 2), H = (x - 2)(x + 2), L = (x - 1)(x + 1)(x + 2)(x + 3)(x + 4), then find B.

Step 1: A = (x - 1)(x + 1)(x + 2). Step 2: B = H x L / A = (x - 2)(x + 2)(x - 1)(x + 1)(x + 2)(x + 3)(x + 4) / ((x - 1)(x + 1)(x + 2)). Step 3: cancel (x - 1)(x + 1)(x + 2). Answer: B = (x - 2)(x + 2)(x + 3)(x + 4).

Unit 4: Factorization and Algebraic Manipulation

(continued: Exercise 4.7)

EXERCISE 4.7 (book pp.97-98)

Q1. Find the square root by factorization: (i) 4x^2 + 12xy + 9y^2.

Step 1: (2x)^2 + 2(2x)(3y) + (3y)^2 = (2x + 3y)^2. Answer: square root = +-(2x + 3y).

Q1. (ii) a^2 - 8ab + 16b^2.

Step 1: (a)^2 - 2(a)(4b) + (4b)^2 = (a - 4b)^2. Answer: square root = +-(a - 4b).

Q1. (iii) a^4 - 6a^2x^2 + 9x^4.

Step 1: (a^2)^2 - 2(a^2)(3x^2) + (3x^2)^2 = (a^2 - 3x^2)^2. Answer: square root = +-(a^2 - 3x^2).

Q1. (iv) (x^2 + 4x + 3)(x^2 + 5x + 6)(x^2 + 3x + 2).

Step 1: factor each: (x + 1)(x + 3); (x + 2)(x + 3); (x + 1)(x + 2). Step 2: the product is (x + 1)^2(x + 2)^2(x + 3)^2 = [(x + 1)(x + 2)(x + 3)]^2. Answer: square root = +-(x + 1)(x + 2)(x + 3).

Q1. (v) y(y + 2)(y + 4)(y + 6) + 16.

Step 1: pair first with last: y(y + 6) = y^2 + 6y; pair middle two: (y + 2)(y + 4) = y^2 + 6y + 8. Step 2: let t = y^2 + 6y: t(t + 8) + 16 = t^2 + 8t + 16 = (t + 4)^2. Answer: square root = +-(y^2 + 6y + 4).

Q1. (vi) x^2 + 1/x^2 - 4(x + 1/x) + 6, (x not = 0).

Step 1: let a = x + 1/x; then x^2 + 1/x^2 = a^2 - 2. Step 2: the expression becomes a^2 - 2 - 4a + 6 = a^2 - 4a + 4 = (a - 2)^2. Answer: square root = +-(x + 1/x - 2).

Q1. (vii) (x + 1/x)^2 - 4(x + 1/x) + 4, (x not = 0).

Step 1: let a = x + 1/x; the expression is a^2 - 4a + 4 = (a - 2)^2. Answer: square root = +-(x + 1/x - 2). (Same result as Q1(vi), by the same substitution.)

Q2. Find square root by division method: (i) x^4 - 2x^3 + 3x^2 - 2x + 1.

Step 1: (x^2 - x + 1)^2 = x^4 - 2x^3 + 3x^2 - 2x + 1, so the division method gives quotient x^2 - x + 1 with remainder 0. Answer: square root = +-(x^2 - x + 1).

Q2. (ii) 9x^4 - 12x^3 + 16x^2 - 8x + 4.

Step 1: (3x^2 - 2x + 2)^2 = 9x^4 - 12x^3 + 16x^2 - 8x + 4, so the division method gives quotient 3x^2 - 2x + 2 with remainder 0. Answer: square root = +-(3x^2 - 2x + 2).

Q2. (iii) 4x^4 + 20x^3 + 41x^2 + 40x + 16.

Step 1: (2x^2 + 5x + 4)^2 = 4x^4 + 20x^3 + 41x^2 + 40x + 16, so the division method gives quotient 2x^2 + 5x + 4 with remainder 0. Answer: square root = +-(2x^2 + 5x + 4).

Q2. (iv) x^6 + 2x^4 - 4x^3 + x^2 - 4x + 4.

Step 1: seek (x^3 + ax^2 + bx + c)^2: the x^5 term forces a = 0; the x^4 term gives b =

1; the x^3 term gives c = -2; then the x^2, x and constant terms check out as 1, -4 and 4.

Step 2: the division method gives quotient x^3 + x - 2 with remainder 0. Answer: square root = +-(x^3 + x - 2).

Q2. (v) x^2 + 1/x^2 + 4(x + 1/x) + 6, (x not = 0).

Step 1: let a = x + 1/x; then x^2 + 1/x^2 = a^2 - 2. Step 2: the expression becomes a^2 - 2 + 4a + 6 = a^2 + 4a + 4 = (a + 2)^2. Answer: square root = +-(x + 1/x + 2).

Q2. (vi) x^6 + 1/x^6 - 4(x^3 + 1/x^3) + 6, (x not = 0).

Step 1: let b = x^3 + 1/x^3; then x^6 + 1/x^6 = b^2 - 2. Step 2: the expression becomes b^2 - 2 - 4b + 6 = b^2 - 4b + 4 = (b - 2)^2. Answer: square root = +-(x^3 + 1/x^3 - 2).

Q3. Find the value of 'h' for which the given expressions are perfect squares: (a) 9x^4 + 30x^3 + 49x^2 + hx + 16.

Step 1: the square root must be (3x^2 + 5x + 4), since (3x^2)^2 = 9x^4 and 4^2 = 16.

Step 2: (3x^2 + 5x + 4)^2 = 9x^4 + 30x^3 + 49x^2 + 40x + 16. Step 3: compare the x term: h = 40. Answer: h = 40.

Q3. (b) x^4 - 6x^3 + 13x^2 - 12x + h.

Step 1: the square root must be (x^2 - 3x + c). Step 2: (x^2 - 3x + c)^2 = x^4 - 6x^3 + (9 + 2c)x^2 - 6cx + c^2. Step 3: 9 + 2c = 13 gives c = 2, and -6c = -12 checks. Step 4: h = c^2 = 4. Answer: h = 4.

Q4. To make the expression 49x^4 - 42x^3 + 79x^2 - 27x + 16 a perfect square: (a) What should be added?

Step 1: by the division method the root begins 7x^2 - 3x + 5, and (7x^2 - 3x + 5)^2 = 49x^4 - 42x^3 + 79x^2 - 30x + 25. Step 2: subtract the given expression: (49x^4 - 42x^3 + 79x^2 - 30x + 25) - (49x^4 - 42x^3 + 79x^2 - 27x + 16) = -3x + 9. Answer: add 9 - 3x.

Q4. (b) What should be subtracted?

Step 1: the same difference, with sign changed: the given expression exceeds the perfect square by 3x - 9 in the x and constant terms. Answer: subtract 3x - 9.

Q4. (c) What will be the value of x?

Step 1: the added/subtracted adjustment vanishes when 3x - 9 = 0, i.e. x = 3. Step 2: check: at x = 3 the given expression equals 49(81) - 42(27) + 79(9) - 27(3) + 16 = 3481 = 59^2, and 7x^2 - 3x + 5 = 59. Answer: x = 3.

Unit 4: Factorization and Algebraic Manipulation

(continued: Exercise 4.8 and Review Exercise 4)

EXERCISE 4.8 (book p.104) — Applications of factorization

Q1. The height, y metres, of a model rocket launched directly upwards from the ground can be modelled by y = 96t - 8t^2, where t is the time in seconds after it leaves the ground. (i) Find the height of the rocket 12 seconds after it leaves the ground. (ii) At what time will the rocket strike the ground again?

Step 1: (i) put t = 12: y = 96(12) - 8(12)^2 = 1152 - 1152 = 0. Step 2: (ii) set y = 0: 96t - 8t^2 = 8t(12 - t) = 0, so t = 0 or t = 12. Answer: (i) 0 m (it is back on the ground); (ii) after 12 seconds.

Q2. A rectangular field, 70 m long and 50 m wide, is surrounded by a concrete path of uniform width; given that the area of the path is 1024 m^2, find the width of the path.

Step 1: let the width be w m. Step 2: area of field plus path = (70 + 2w)(50 + 2w) = 3500 + 240w + 4w^2. Step 3: path area = 240w + 4w^2 = 1024, so w^2 + 60w - 256 = 0. Step 4: factor: (w + 64)(w - 4) = 0, so w = 4 (w = -64 rejected). Answer: the path is 4 m wide.

Q3. The length of a side and the corresponding height of a triangle are (x + 3) cm and (2x - 5) cm respectively. Given that the area of the triangle is 20 cm^2, find the value of x.

Step 1: area = (1/2)(x + 3)(2x - 5) = 20, so (x + 3)(2x - 5) = 40. Step 2: expand: 2x^2 + x - 15 = 40, so 2x^2 + x - 55 = 0. Step 3: factor: (2x + 11)(x - 5) = 0, so x = 5 (x = -11/2 rejected, since 2x - 5 must be positive). Answer: x = 5.

Q4. A piece of wire 44 cm long is cut into two parts, each part is bent to form a square. Given that the total area of the two squares is 65 cm^2, find the perimeter of each square.

Step 1: let the sides be a cm and b cm. Step 2: 4a + 4b = 44, so a + b = 11; and a^2 + b^2 = 65. Step 3: (a + b)^2 = 121 = a^2 + 2ab + b^2, so 2ab = 56, ab = 28. Step 4: a and b are roots of t^2 - 11t + 28 = (t - 4)(t - 7) = 0, so the sides are 4 cm and 7 cm. Answer: perimeters 16 cm and 28 cm.

Q5. The figure shows an ancient coin which was once used in China. The coin is the shape of a circle of radius 3 cm with a square of side x cm removed from its centre. The area of each face of the coin is 7pi cm^2. (i) Form an equation in x and show that it reduces to 2pi - x^2 = 0. (ii) Solve the equation 2pi - x^2 = 0. (iii) Find the perimeter of the square.

Step 1: (i) area = circle area - square area = pi(3)^2 - x^2 = 9pi - x^2. Step 2: set equal to

7pi: 9pi - x^2 = 7pi, so 2pi - x^2 = 0. Step 3: (ii) x^2 = 2pi, so x = sqrt(2pi) cm (positive root). Step 4: (iii) perimeter = 4x. Answer: (i) 2pi - x^2 = 0; (ii) x = sqrt(2pi) cm; (iii) perimeter = 4sqrt(2pi) cm.

Q6. The vertical height y (in meters) of an object launched straight upward from the ground is modeled by y = 17t - 5t^2, where t is the time in seconds after launch. (i) Determine the height of the object 2.5 seconds after launch. (ii) Find the time at which the object will return to the ground.

Step 1: (i) put t = 2.5: y = 17(2.5) - 5(2.5)^2 = 42.5 - 31.25 = 11.25. Step 2: (ii) set y = 0: 17t - 5t^2 = t(17 - 5t) = 0, so t = 0 or t = 17/5 = 3.4. Answer: (i) 11.25 m; (ii) after 3.4 seconds.

Q7. Length of a rectangular bridge is 5 m more than its breadth. If breadth is x m then find area of rectangular bridge in terms of x. If area of rectangular bridge is 84 m^2, find length and width of the rectangular bridge.

Step 1: length = (x + 5) m, so area = x(x + 5) = x^2 + 5x m^2. Step 2: x^2 + 5x = 84, so x^2 + 5x - 84 = 0. Step 3: factor: (x + 12)(x - 7) = 0, so x = 7 (x = -12 rejected). Answer: area = x^2 + 5x; breadth 7 m, length 12 m.

Q8. An open rectangular box is to be constructed by cutting square corners from a 60 x 60 inch piece of cardboard and folding up the flaps. Find volume in terms of x. Also find the value of x when V(x) = 0.

Step 1: cutting squares of side x leaves a base of (60 - 2x) by (60 - 2x) and height x. Step 2: V(x) = x(60 - 2x)^2 cubic inches. Step 3: V(x) = 0 gives x = 0 or 60 - 2x = 0, i.e. x = 30. Answer: V(x) = x(60 - 2x)^2; V(x) = 0 when x = 0 or x = 30 inches (x = 30 is the cut that flattens the box).

Q9. A small factory produces hand-crafted wooden chairs. The profit P(x) in thousands of Rupees, depending on the number of chairs x (in tens) produced and sold, is modeled by P(x) = x^3 - 6x^2 + 11x - 6. (i) Find the production levels (in tens of chairs) where the company makes no profit (break-even points). (ii) If the company produces 50 chairs, find the profit.

Step 1: (i) set P(x) = 0: x = 1 is a root since 1 - 6 + 11 - 6 = 0, so P(x) = (x - 1)(x^2 - 5x + 6) = (x - 1)(x - 2)(x - 3). Step 2: break-even at x = 1, 2, 3, i.e. 10, 20 and 30 chairs. Step 3: (ii) 50 chairs means x = 5: P(5) = 125 - 150 + 55 - 6 = 24 (thousand Rupees). Answer: (i) 10, 20 and 30 chairs; (ii) Rs. 24,000 profit.

REVIEW EXERCISE 4 (book p.105)

Q1. (i) Factors of x^2 - 16 are:

(a) (x + 4)(x - 4) ✓

(b) (x + 4)^2

(c) (x + 4)(x + 5)

(d) (x - 4)^2

Q1. (ii) Factors of x^2 + 8x + 15 are:

(a) (x + 5)(x - 3)

(b) (x + 5)(x + 3) ✓

(c) (x + 6)(x + 8)

(d) (x - 3)(x - 5)

Q1. (iii) What should be added to 3x^2 + 8y^2 to complete the square?

(a) 2sqrt(24)xy ✓

(b) 24xy

(c) 3sqrt(24)xy

(d) sqrt(24)xy

Q1. (iv) Factors of 64/x^3 - x^3 are:

(a) (4/x - x)(16/x^2 + 4 + x^2) ✓

(b) (4/x - x)(16/x^2 - x)(16/x^2 - 4 + x^2)

(c) (x - 4/x)(x^2 + 4 + 16/x^2)

(d) (4/x + x)(16/x^2 + 4 - x^2)

Q1. (v) H.C.F. of x^3 - y^3 and x^4 - y^4 is:

(a) x - y ✓

(b) x + y

(c) x^2 - y^2

(d) x^2 + y^2

Q1. (vi) sqrt(4x^2 + 4xy + y^2) is:

(a) 2x - y

(b) 2x + y ✓

(c) 2x + 6y

(d) x + 34y

Q2. Factorize the following: (i) r^8 - 256y^8.

Step 1: (r^4)^2 - (16y^4)^2 = (r^4 - 16y^4)(r^4 + 16y^4). Step 2: r^4 - 16y^4 = (r^2 - 4y^2)(r^2 + 4y^2) = (r - 2y)(r + 2y)(r^2 + 4y^2). Answer: (r - 2y)(r + 2y)(r^2 + 4y^2)(r^4 + 16y^4).

Q2. (ii) a^6 - 2a^3 - 15.

Step 1: treat as a quadratic in a^3: factors of -15 with sum -2 are -5 and +3. Answer: (a^3 - 5)(a^3 + 3).

Q3. Factorize: (i) (x + 1/x)^2 - 7(x + 1/x) + 12.

Step 1: let a = x + 1/x; then a^2 - 7a + 12 = (a - 3)(a - 4). Answer: (x + 1/x - 3)(x + 1/x - 4).

Q3. (ii) 64x^3 - 27y^3.

Step 1: (4x)^3 - (3y)^3, so a^3 - b^3 form. Answer: (4x - 3y)(16x^2 + 12xy + 9y^2).

Q4. Find H.C.F. of the following expressions: x^6 - y^6 and x^4 - y^4.

Step 1: x^6 - y^6 = (x - y)(x + y)(x^2 + xy + y^2)(x^2 - xy + y^2). Step 2: x^4 - y^4 = (x - y)(x + y)(x^2 + y^2). Step 3: common factors: (x - y)(x + y). Answer: H.C.F. = x^2 - y^2.

Q5. Find square root of (u^4 + 1/u^4) - 4(u^2 + 1/u^2) + 6, u not = 0.

Step 1: let c = u^2 + 1/u^2; then u^4 + 1/u^4 = c^2 - 2. Step 2: the expression becomes c^2 - 2 - 4c + 6 = c^2 - 4c + 4 = (c - 2)^2. Answer: square root = +-(u^2 + 1/u^2 - 2).

Unit 5: Linear Equations and Inequalities

EXERCISE 5.1 (book p.111) — Solve the following equations, and check your solutions

Q1. (i) (3x - 1)/5 = (7 - x)/3.

Step 1: cross-multiply: 3(3x - 1) = 5(7 - x). Step 2: 9x - 3 = 35 - 5x. Step 3: 14x = 38. Step 4: x = 19/7. Check: both sides equal 10/7. Answer: solution set = {19/7}.

Q1. (ii) 5x - 3 = x + 9.

Step 1: 5x - x = 9 + 3. Step 2: 4x = 12. Answer: x = 3; solution set = {3}.

Q1. (iii) (2/3)x - 1 = (2/3)(x + 1/2).

Step 1: expand the right side: (2/3)x + 1/3. Step 2: the equation becomes (2/3)x - 1 = (2/3)x + 1/3. Step 3: cancelling (2/3)x gives -1 = 1/3, which is false. Answer: no solution; solution set = {}.

Q1. (iv) 3(2x - 1) - 5(2 - x) = 4(x - 1) + 5.

Step 1: expand: 6x - 3 - 10 + 5x = 4x - 4 + 5. Step 2: 11x - 13 = 4x + 1. Step 3: 7x = 14. Answer: x = 2; solution set = {2}.

Q1. (v) (3/8)x + 1 + 4/5 = (1/4)x + 1 + 3/10.

Step 1: combine constants: (3/8)x + 9/5 = (1/4)x + 13/10. Step 2: multiply by 40: 15x + 72 = 10x + 52. Step 3: 5x = -20. Answer: x = -4; check: both sides equal 3/10.

Q1. (vi) 2x/(2x + 5) = 2/3 - 5/(4x + 10), x not = -5/2.

Step 1: note 4x + 10 = 2(2x + 5). Step 2: multiply both sides by 3(2x + 5): 6x = 2(2x + 5) - 15/2. Step 3: 6x = 4x + 5/2, so 2x = 5/2. Answer: x = 5/4 (allowed, since 2x + 5 = 15/2 is not zero).

Q1. (vii) 2x/(x - 1) + 1/3 = 5/6 + 2/(x - 1), x not = 1.

Step 1: bring the fractions with denominator (x - 1) together: 2x/(x - 1) - 2/(x - 1) = 5/6 - 1/3. Step 2: 2(x - 1)/(x - 1) = 1/2, i.e. 2 = 1/2, which is false. Answer: no solution; solution set = {}.

Q1. (viii) (x + 5)/6 - [(14 - x)/2 - 1/4] = (2x - 7)/12.

Step 1: simplify the bracket: (14 - x)/2 - 1/4 = (27 - 2x)/4. Step 2: multiply the whole equation by 12: 2(x + 5) - 3(27 - 2x) = 2x - 7. Step 3: 2x + 10 - 81 + 6x = 2x - 7, so 8x - 71 = 2x - 7. Step 4: 6x = 64. Answer: x = 32/3; check: both sides equal 43/36.

EXERCISE 5.2 (book p.115)

Q1. Solve the following equations. Also check the solution: (i) sqrt(x - 8) = 3.

Step 1: square both sides: x - 8 = 9. Step 2: x = 17. Check: sqrt(9) = 3. Answer: solution set = {17}.

Q1. (ii) sqrt(x + 5) = sqrt(2x + 7).

Step 1: square both sides: x + 5 = 2x + 7. Step 2: x = -2. Check: both sides equal sqrt(3). Answer: solution set = {-2}.

Q1. (iii) 3rd-root(2x - 4) - 2 = 0.

Step 1: 3rd-root(2x - 4) = 2. Step 2: cube both sides: 2x - 4 = 8. Step 3: x = 6. Check: 3rd-root(8) - 2 = 0. Answer: solution set = {6}.

Q1. (iv) 3rd-root(2x + 3) = 3rd-root(3x - 2).

Step 1: cube both sides: 2x + 3 = 3x - 2. Step 2: x = 5. Check: both sides equal 3rd-root(13). Answer: solution set = {5}.

Q1. (v) (sqrt(2x + 3) - 14)/6 = -2.

Step 1: multiply by 6: sqrt(2x + 3) - 14 = -12. Step 2: sqrt(2x + 3) = 2. Step 3: square: 2x + 3 = 4, so x = 1/2. Check: (2 - 14)/6 = -2. Answer: solution set = {1/2}.

Q1. (vi) sqrt(2/(2x + 3)) = sqrt(3/(2x + 1)).

Step 1: square both sides: 2/(2x + 3) = 3/(2x + 1). Step 2: cross-multiply: 2(2x + 1) = 3(2x + 3), so 4x + 2 = 6x + 9, giving x = -7/2. Step 3: CHECK: at x = -7/2, 2x + 3 = -4 and 2x + 1 = -6, so both radicands 2/(2x + 3) = -1/2 and 3/(2x + 1) = -1/2 are negative and the square roots are undefined over the reals. Step 4: the formal solution is extraneous. Answer: no real solution; solution set = {}.

Q2. Solve the following absolute equations: (i) |x + 5| = 20.

Step 1: x + 5 = 20 or x + 5 = -20. Answer: solution set = {15, -25}.

Q2. (ii) |15 + (3x - 2)| = 5x + 10.

Step 1: simplify: |3x + 13| = 5x + 10. Step 2: case 1: 3x + 13 = 5x + 10 gives x = 3/2; check: both sides equal 35/2. Step 3: case 2: 3x + 13 = -(5x + 10) gives x = -23/8; check: the right side 5x + 10 = -35/8 is negative while the left side is non-negative, so this root is rejected. Answer: solution set = {3/2}.

Q2. (iii) |2x - 1| = |4x + 3|.

Step 1: 2x - 1 = 4x + 3 gives x = -2; check: both sides equal 5. Step 2: 2x - 1 = -(4x + 3) gives 6x = -2, x = -1/3; check: both sides equal 5/3. Answer: solution set = {-2, -1/3}.

Q2. (iv) |(4x - 2)/5| = |(6x + 3)/2|.

Step 1: case 1: (4x - 2)/5 = (6x + 3)/2 gives 2(4x - 2) = 5(6x + 3), so 8x - 4 = 30x + 15, x = -19/22. Step 2: case 2: (4x - 2)/5 = -(6x + 3)/2 gives 2(4x - 2) = -5(6x + 3), so 8x - 4 = -30x - 15, x = -11/38. Answer: solution set = {-19/22, -11/38}.

Q2. (v) 2 - 4|2x + 3| = -18.

Step 1: 4|2x + 3| = 20, so |2x + 3| = 5. Step 2: 2x + 3 = 5 gives x = 1; 2x + 3 = -5 gives x = -4. Answer: solution set = {1, -4}.

Unit 5: Linear Equations and Inequalities (continued: Exercise 5.3 and Review Exercise 5)

EXERCISE 5.3 (book p.118)

Q1. Solve the following inequalities for which x in R: (i) 2x + 3 > 4x - 9.

Step 1: 3 + 9 > 4x - 2x. Step 2: 12 > 2x. Answer: solution set = {x in R | x < 6}.

Q1. (ii) (x + 3)/2 >= 7.

Step 1: multiply by 2: x + 3 >= 14. Answer: solution set = {x in R | x >= 11}.

Q1. (iii) -2(x - 4) < -8.

Step 1: -2x + 8 < -8. Step 2: -2x < -16. Step 3: divide by -2 and reverse the sign. Answer: solution set = {x in R | x > 8}.

Q1. (iv) 6x - 3 > 5x + 2x + 1.

Step 1: 6x - 3 > 7x + 1. Step 2: -3 - 1 > 7x - 6x. Answer: solution set = {x in R | x < -4}.

Q1. (v) 0.5x + 3 > 7.

Step 1: 0.5x > 4. Answer: solution set = {x in R | x > 8}.

Q1. (vi) (-1/3)x > 5 - 2x.

Step 1: multiply by 3: -x > 15 - 6x. Step 2: 6x - x > 15. Answer: solution set = {x in R | x > 3}.

Q2. Solve the following inequalities for which x in R and also show on number line: (i) -4 <= 2x <= 3x - 2.

Step 1: from -4 <= 2x: x >= -2. Step 2: from 2x <= 3x - 2: 2 <= x. Step 3: combine: x >= 2. Answer: solution set = {x in R | x >= 2}; on the number line, a closed dot at 2 with the ray to the right.

Q2. (ii) -3 > x - 9 > 2x.

Step 1: from x - 9 < -3: x < 6. Step 2: from x - 9 > 2x: -9 > x. Step 3: combine: x < -9. Answer: solution set = {x in R | x < -9}; on the number line, an open dot at -9 with the ray to the left.

Q2. (iii) (1 - x) < -2 <= 3 - x.

Step 1: from 1 - x < -2: x > 3. Step 2: from -2 <= 3 - x: x <= 5. Answer: solution set = {x in R | 3 < x <= 5}; on the number line, an open dot at 3, a closed dot at 5, shaded between.

Q2. (iv) 6 >= 4x >= 3x - 1.

Step 1: from 4x <= 6: x <= 3/2. Step 2: from 4x >= 3x - 1: x >= -1. Answer: solution set = {x in R | -1 <= x <= 3/2}; on the number line, closed dots at -1 and 3/2, shaded between.

Q2. (v) 2x <= x + 6 < 3x + 5.

Step 1: from 2x <= x + 6: x <= 6. Step 2: from x + 6 < 3x + 5: 1 < 2x, so x > 1/2. Answer: solution set = {x in R | 1/2 < x <= 6}; on the number line, an open dot at 1/2, a closed dot

at 6, shaded between.

Q2. (vi) 3x - 2 >= 10 >= x + 4.

Step 1: from 3x - 2 >= 10: x >= 4. Step 2: from 10 >= x + 4: x <= 6. Answer: solution set = {x in R | 4 <= x <= 6}; on the number line, closed dots at 4 and 6, shaded between.

Q3. A student's monthly budget for mobile and internet is Rs. 2,500. He wants to keep at least Rs. 700 unused for emergencies. If he spends x rupees on data packages, write and solve the inequality.

Step 1: amount spent plus amount kept unused must not exceed the budget: x + 700 <= 2500. Step 2: x <= 1800. Answer: he can spend at most Rs. 1,800 on data packages.

Q4. A student in Quetta earns Rs. 12,000 monthly from part-time work. He must spend at least Rs. 7,000 on food and transport. He wants to save between Rs. 2,000 and Rs. 4,000 for tuition. If x is his expenses, write and solve the inequality. (Reading: x = total monthly expenses, so savings = 12000 - x.)

Step 1: savings = 12000 - x. Step 2: 2000 <= 12000 - x <= 4000. Step 3: subtract 12000: -10000 <= -x <= -8000. Step 4: multiply by -1 and reverse both signs: 8000 <= x <= 10000. Answer: his total monthly expenses must be between Rs. 8,000 and Rs. 10,000 (which is consistent with spending at least Rs. 7,000 on food and transport).

REVIEW EXERCISE 5 (book pp.118-119)

Q1. (i) Which of the following inequalities is the solution of 7a - 5 < 2a + 18?

(a) a < 23/5 ✓

(b) a > 13

(c) a <= 23

(d) a >= 13

Q1. (ii) The solution of inequality 5 - (3y/4) <= 6 is:

(a) y <= -4/3

(b) y >= -4/3 ✓

(c) y <= -1/3

(d) y <= -4

Q1. (iii) If 1/x < 1/y, then: (for positive x and y)

(a) x > y ✓

(b) y > x

(c) y = 1

(d) x = 1

Q1. (iv) If 4 + (5y/2) = 19, what is the value of y?

(a) 6 ✓

(b) 30

(c) 15

(d) 75

Q1. (v) If 6 - a > 7 then:

(a) a > 1

(b) a > -1

(c) a < -1 ✓

(d) a < 1

Q3. Find solution set of the following inequalities for x, z in R: (i) x/5 - 5/3 < 3.

Step 1: x/5 < 3 + 5/3 = 14/3. Step 2: x < 70/3. Answer: solution set = {x in R | x < 70/3}.

Q3. (ii) (1/2)x - 5 = 1/3.

Step 1: x/2 = 5 + 1/3 = 16/3. Step 2: x = 32/3. Answer: solution set = {32/3}.

Q3. (iii) 2x + 5 = 15.

Step 1: 2x = 10. Answer: x = 5; solution set = {5}.

Q3. (iv) (6z + 10)/(3z + 5) = -2.

Step 1: multiply by (3z + 5), with 3z + 5 not = 0: 6z + 10 = -2(3z + 5) = -6z - 10. Step 2: 12z = -20, so z = -5/3. Step 3: CHECK: at z = -5/3 the denominator 3z + 5 = 0, so the original expression is undefined. Step 4: the formal root is extraneous. Answer: no solution; solution set = {}.

Q3. (v) (2z + 8)/(z - 6) < 11.

Step 1: bring all to one side: (2z + 8)/(z - 6) - 11 < 0, i.e. (-9z + 74)/(z - 6) < 0, i.e. (9z - 74)/(z - 6) > 0. Step 2: the quotient is positive when both factors have the same sign: z < 6 or z > 74/9 (z = 6 excluded, z = 74/9 excluded). Answer: solution set = {z in R | z < 6 or z > 74/9}.

Q4. Solve the following inequalities (x, y in R): (i) (2y + 5)/2 <= (7y + 17)/2.

Step 1: multiply by 2: 2y + 5 <= 7y + 17. Step 2: -12 <= 5y. Answer: solution set = {y in R | y >= -12/5}.

Q4. (ii) (2x - 5)/3 <= 3.

Step 1: multiply by 3: 2x - 5 <= 9. Step 2: 2x <= 14. Answer: solution set = {x in R | x <= 7}.

Unit 6: Trigonometry and Bearing

EXERCISE 6.1 (book p.126)

Q1. Convert the following angles in radians: (i) 120 degrees.

Step 1: multiply by pi/180: 120 x pi/180. Step 2: simplify: 120/180 = 2/3. Answer: 2pi/3 radians.

Q1. (ii) 105 degrees.

Step 1: 105 x pi/180. Step 2: 105/180 = 7/12. Answer: 7pi/12 radians.

Q1. (iii) 270 degrees.

Step 1: 270 x pi/180. Step 2: 270/180 = 3/2. Answer: 3pi/2 radians.

Q1. (iv) -135 degrees.

Step 1: -135 x pi/180. Step 2: -135/180 = -3/4. Answer: -3pi/4 radians.

Q1. (v) 60 degrees.

Step 1: 60 x pi/180. Step 2: 60/180 = 1/3. Answer: pi/3 radians.

Q1. (vi) 40 degrees.

Step 1: 40 x pi/180. Step 2: 40/180 = 2/9. Answer: 2pi/9 radians.

Q1. (vii) 225 degrees.

Step 1: 225 x pi/180. Step 2: 225/180 = 5/4. Answer: 5pi/4 radians.

Q1. (viii) 315 degrees.

Step 1: 315 x pi/180. Step 2: 315/180 = 7/4. Answer: 7pi/4 radians.

Q1. (ix) 32 degrees.

Step 1: 32 x pi/180. Step 2: 32/180 = 8/45. Answer: 8pi/45 radians.

Q1. (x) -50 degrees.

Step 1: -50 x pi/180. Step 2: -50/180 = -5/18. Answer: -5pi/18 radians.

Q2. Convert the following angles in degree: (i) 3pi/5.

Step 1: multiply by 180/pi: (3pi/5) x (180/pi). Step 2: 3 x 180/5 = 108. Answer: 108 degrees.

Q2. (ii) 7pi/2.

Step 1: (7pi/2) x (180/pi) = 7 x 90. Answer: 630 degrees.

Q2. (iii) pi/6.

Step 1: (pi/6) x (180/pi) = 180/6. Answer: 30 degrees.

Q2. (iv) -3pi/8.

Step 1: (-3pi/8) x (180/pi) = -3 x 180/8. Answer: -67.5 degrees.

Q2. (v) 5pi.

Step 1: 5pi x (180/pi) = 5 x 180. Answer: 900 degrees.

Q2. (vi) 2pi/3.

Step 1: (2pi/3) x (180/pi) = 2 x 60. Answer: 120 degrees.

Q2. (vii) 3.5pi.

Step 1: 3.5pi x (180/pi) = 3.5 x 180. Answer: 630 degrees.

Q2. (viii) 9pi/2.

Step 1: (9pi/2) x (180/pi) = 9 x 90. Answer: 810 degrees.

Q2. (ix) -15.4pi.

Step 1: -15.4pi x (180/pi) = -15.4 x 180. Answer: -2772 degrees.

Q2. (x) 2.5pi.

Step 1: 2.5pi x (180/pi) = 2.5 x 180. Answer: 450 degrees.

Q3. Convert the following angles into minutes: (i) 31.5 degrees.

Step 1: 1 degree = 60 minutes. Step 2: 31.5 x 60 = 1890. Answer: 1890 minutes.

Q3. (ii) -15.25 degrees.

Step 1: -15.25 x 60. Answer: -915 minutes.

Q3. (iii) 135 degrees.

Step 1: 135 x 60. Answer: 8100 minutes.

Q3. (iv) -21.5 degrees.

Step 1: -21.5 x 60. Answer: -1290 minutes.

Q3. (v) 13.65 degrees.

Step 1: 13.65 x 60. Answer: 819 minutes.

Q4. Convert the following angles into degree upto 2 decimal places: (i) 15 degrees 35 minutes.

Step 1: 35 minutes = 35/60 degrees = 0.5833 degrees. Step 2: 15 + 0.5833 = 15.5833. Answer: 15.58 degrees.

Q4. (ii) 250 degrees 5 minutes.

Step 1: 5 minutes = 5/60 degrees = 0.0833 degrees. Step 2: 250 + 0.0833 = 250.0833. Answer: 250.08 degrees.

Q4. (iii) 30 degrees 30 minutes.

Step 1: 30 minutes = 30/60 degrees = 0.5 degrees. Step 2: 30 + 0.5 = 30.5. Answer: 30.50 degrees.

Q4. (iv) 15 degrees 40 minutes.

Step 1: 40 minutes = 40/60 degrees = 0.6667 degrees. Step 2: 15 + 0.6667 = 15.6667. Answer: 15.67 degrees.

Q4. (v) 210 degrees 15 minutes.

Step 1: 15 minutes = 15/60 degrees = 0.25 degrees. Step 2: 210 + 0.25 = 210.25. Answer: 210.25 degrees.

Unit 6: Trigonometry and Bearing (continued: Exercise 6.2)

EXERCISE 6.2 (book pp.128-129) — Values read from the printed figures; trig values to 1-2 decimal places

Q1. Find the value of the unknown in each of the following right-angled triangles: (i) hypotenuse 15 cm, angle 67 degrees, unknown side a opposite the angle.

Step 1: sin 67 degrees = a/15. Step 2: a = 15 x sin 67 degrees = 15 x 0.9205. Answer: a = 13.81 cm (approx).

Q1. (ii) side 9.7 cm opposite a 15-degree angle, unknown hypotenuse b.

Step 1: sin 15 degrees = 9.7/b. Step 2: b = 9.7/sin 15 degrees = 9.7/0.2588. Answer: b = 37.48 cm (approx).

Q1. (iii) hypotenuse 13.5 m, angle 36 degrees, unknown side a adjacent to the angle.

Step 1: cos 36 degrees = a/13.5. Step 2: a = 13.5 x cos 36 degrees = 13.5 x 0.8090. Answer: a = 10.92 m (approx).

Q1. (iv) side 17 m adjacent to a 61-degree angle, unknown hypotenuse b.

Step 1: cos 61 degrees = 17/b. Step 2: b = 17/cos 61 degrees = 17/0.4848. Answer: b = 35.07 m (approx).

Q1. (v) side 7 cm adjacent to a 44.2-degree angle, unknown side c opposite the angle and hypotenuse f.

Step 1: tan 44.2 degrees = c/7, so c = 7 x tan 44.2 degrees = 7 x 0.9725. Step 2: cos 44.2 degrees = 7/f, so f = 7/cos 44.2 degrees = 7/0.7169. Answer: c = 6.81 cm (approx), f = 9.76 cm (approx).

Q1. (vi) side 8.9 cm opposite a 21.5-degree angle, unknown hypotenuse h and other side g.

Step 1: sin 21.5 degrees = 8.9/h, so h = 8.9/sin 21.5 degrees = 8.9/0.3665. Step 2: tan 21.5 degrees = 8.9/g, so g = 8.9/tan 21.5 degrees = 8.9/0.3939. Answer: h = 24.28 cm (approx), g = 22.59 cm (approx).

Q2. In triangle ABC, AB = 8.9 m and angle ABC = 56 degrees. H lies on BC such that AH is perpendicular to BC. The figure marks angle A as a right angle. Find the length of: (i) AH.

Step 1: in right triangle ABH, sin 56 degrees = AH/AB. Step 2: AH = 8.9 x sin 56 degrees = 8.9 x 0.8290. Answer: AH = 7.38 m (approx).

Q2. (ii) HC.

Step 1: in right triangle ABH, BH = AB x cos 56 degrees = 8.9 x 0.5592 = 4.98 m. Step 2: in

triangle ABC (right-angled at A), BC = AB/cos 56 degrees = 8.9/0.5592 = 15.92 m. Step 3: HC = BC - BH = 15.92 - 4.98. Answer: HC = 10.94 m (approx).

Q3. In the figure, QS = 2.5 cm, angle QSR = 45 degrees and angle QST = 60 degrees. T lies on PS with ST perpendicular to TQ, angle SQR is marked as a right angle, and P, Q, R are collinear. Find the length of: (i) TQ.

Step 1: in right triangle STQ, the hypotenuse is SQ = 2.5 cm and angle TSQ = 60 degrees. Step 2: TQ = SQ x sin 60 degrees = 2.5 x 0.8660. Answer: TQ = 2.17 cm (approx).

Q3. (ii) PT.

Step 1: in right triangle STQ, angle SQT = 30 degrees. Step 2: since P, Q, R are collinear and angle SQR = 90 degrees, angle PQT = 180 degrees - 30 degrees - 90 degrees = 60 degrees. Step 3: in right triangle PTQ, tan 60 degrees = PT/TQ, so PT = 2.17 x 1.7321. Answer: PT = 3.75 cm (approx).

Q3. (iii) PR.

Step 1: in right triangle PTQ, PQ = TQ/cos 60 degrees = 2.17/0.5 = 4.33 cm. Step 2: in right triangle SQR, QR = SQ x tan 45 degrees = 2.5 x 1 = 2.5 cm. Step 3: PR = PQ + QR = 4.33 + 2.5. Answer: PR = 6.83 cm (approx).

Q4. Find the value of the unknown in each of the following right-angled triangles: (i) side 12 cm opposite angle a, hypotenuse 26 cm.

Step 1: sin a = 12/26 = 0.4615. Step 2: a = sin^-1(0.4615). Answer: a = 27.5 degrees (approx).

Q4. (ii) side 10 cm adjacent to angle b, hypotenuse 17 cm.

Step 1: cos b = 10/17 = 0.5882. Step 2: b = cos^-1(0.5882). Answer: b = 54.0 degrees (approx).

Q4. (iii) side 11 cm opposite angle c, other side 27 cm.

Step 1: tan c = 11/27 = 0.4074. Step 2: c = tan^-1(0.4074). Answer: c = 22.2 degrees (approx).

Q4. (iv) side 15 cm opposite angle c, hypotenuse 22.7 cm.

Step 1: sin c = 15/22.7 = 0.6608. Step 2: c = sin^-1(0.6608). Answer: c = 41.4 degrees (approx).

Q4. (v) side 12.5 cm opposite angle f, other side 14 cm.

Step 1: tan f = 12.5/14 = 0.8929. Step 2: f = tan^-1(0.8929). Answer: f = 41.8 degrees (approx).

Unit 6: Trigonometry and Bearing (continued: Exercise 6.3)

EXERCISE 6.3 (book pp.133-135)

Q1. Find the angle of elevation in the following figures: (i) vertical side k, base sqrt(3)k.

Step 1: tan theta = k/(sqrt(3)k) = 1/sqrt(3). Step 2: theta = tan^-1(1/sqrt(3)). Answer: theta = 30 degrees.

Q1. (ii) hypotenuse 2k, vertical side sqrt(3)k.

Step 1: the base = sqrt((2k)^2 - (sqrt(3)k)^2) = k. Step 2: tan theta = sqrt(3)k/k = sqrt(3). Step 3: theta = tan^-1(sqrt(3)). Answer: theta = 60 degrees.

Q1. (iii) hypotenuse 2 m, vertical side 1 m.

Step 1: sin theta = 1/2. Step 2: theta = sin^-1(1/2). Answer: theta = 30 degrees.

Q1. (iv) vertical side sqrt(3)k, base 3k.

Step 1: tan theta = sqrt(3)k/(3k) = 1/sqrt(3). Step 2: theta = tan^-1(1/sqrt(3)). Answer: theta = 30 degrees.

Q2. Find the angle of depression in the following figures (the angle of depression equals the angle at the ground by alternate angles): (i) vertical side 5 cm, base 6 cm.

Step 1: tan theta = 5/6 = 0.8333. Step 2: theta = tan^-1(0.8333). Answer: theta = 39.8 degrees (approx).

Q2. (ii) hypotenuse 8 cm, base 4 cm.

Step 1: cos theta = 4/8 = 0.5. Step 2: theta = cos^-1(0.5). Answer: theta = 60 degrees.

Q2. (iii) hypotenuse 5 cm, vertical side 4 cm.

Step 1: sin theta = 4/5 = 0.8. Step 2: theta = sin^-1(0.8). Answer: theta = 53.1 degrees (approx).

Q2. (iv) hypotenuse 8 cm, vertical side 4 cm.

Step 1: sin theta = 4/8 = 0.5. Step 2: theta = sin^-1(0.5). Answer: theta = 30 degrees.

Q3. A balloon is connected to a meteorological ground station by a cable of length 215 m inclined at 60 degrees to the horizontal. Determine the height of the balloon from the earth.

Step 1: height = 215 x sin 60 degrees. Step 2: 215 x sqrt(3)/2 = 215 x 0.8660. Answer: height = 186.20 m (approx).

Q4. The angle of elevation of the top of a tower at a distance of 150 meters from its foot on a horizontal plane is found to be 30 degrees. Find the height of the tower.

Step 1: height = 150 x tan 30 degrees. Step 2: 150 x (1/sqrt(3)) = 150 x 0.5774. Answer: height = 86.60 m (approx).

Q5. An observer 1.5 m tall is 20 m away from a tower 30 m high. Determine the angle of elevation from his eye to the top of the tower.

Step 1: height above the eye = 30 - 1.5 = 28.5 m. Step 2: tan theta = 28.5/20 = 1.425. Step 3: theta = tan^-1(1.425). Answer: theta = 54.9 degrees (approx).

Q6. The angle of depression of a 37 m high building from the top of a tower 117 m high is 38 degrees. Calculate the distance between the building and tower.

Step 1: vertical difference = 117 - 37 = 80 m. Step 2: tan 38 degrees = 80/distance. Step 3: distance = 80/tan 38 degrees = 80/0.7813. Answer: distance = 102.40 m (approx).

Q7. From the top of a cliff, 201 m high, the angle of depression of the top and bottom of a tower are observed to be 30 degrees and 60 degrees. Find the height of the tower.

Step 1: let the distance be d and the tower height be h. Step 2: tan 60 degrees = 201/d, so d = 201/sqrt(3). Step 3: tan 30 degrees = (201 - h)/d, so 201 - h = d x tan 30 degrees = (201/sqrt(3)) x (1/sqrt(3)) = 201/3 = 67. Step 4: h = 201 - 67. Answer: height of the tower = 134 m.

Q8. The angle of elevation of a tower from a point is 30 degrees. On walking 100 m towards the tower, the angle of elevation becomes 60 degrees. Find the height of the tower.

Step 1: let the height be h. Step 2: distance at 30 degrees = h/tan 30 degrees = h x sqrt(3); distance at 60 degrees = h/tan 60 degrees = h/sqrt(3). Step 3: h x sqrt(3) - h/sqrt(3) = 100, so h x (2/sqrt(3)) = 100. Step 4: h = 50 x sqrt(3). Answer: height = 86.60 m (approx).

Q9. A string attached to a flying kite is 200 m long, and its angle of elevation is 60 degrees. Find the height of the kite above the hand, taking the string to be fully stretched.

Step 1: height = 200 x sin 60 degrees. Step 2: 200 x sqrt(3)/2 = 100 x sqrt(3). Answer: height = 173.21 m (approx).

Unit 6: Trigonometry and Bearing (continued: Exercise 6.4)

EXERCISE 6.4 (book pp.139-140) — Prove the following identities

Q1. Prove: (1 - tan^2 theta)/(1 + tan^2 theta) = cos^2 theta - sin^2 theta.

Step 1: multiply numerator and denominator by cos^2 theta. Step 2: (1 - tan^2 theta) x cos^2 theta = cos^2 theta - sin^2 theta. Step 3: (1 + tan^2 theta) x cos^2 theta = cos^2 theta + sin^2 theta = 1. Step 4: the ratio is cos^2 theta - sin^2 theta. Hence proved.

Q2. Prove: sin theta/(sec theta - 1) + sin theta/(sec theta + 1) = 2 cot theta.

Step 1: combine over (sec^2 theta - 1) = tan^2 theta: numerator = sin theta x (sec theta + 1 + sec theta - 1) = 2 sin theta sec theta. Step 2: 2 sin theta x (1/cos theta) divided by (sin^2 theta/cos^2 theta) = 2 cos theta/sin theta. Answer: 2 cot theta. Hence proved.

Q3. Prove: sin theta/(cosec theta - cot theta) = 1 + cos theta.

Step 1: cosec theta - cot theta = (1 - cos theta)/sin theta. Step 2: sin theta divided by (1 - cos theta)/sin theta = sin^2 theta/(1 - cos theta). Step 3: sin^2 theta = 1 - cos^2 theta = (1 - cos theta)(1 + cos theta). Step 4: cancelling (1 - cos theta) gives 1 + cos theta. Hence proved.

Q4. Prove: cos theta/(1 + sin theta) = (1 - sin theta)/cos theta.

Step 1: cross-multiply: cos^2 theta = (1 - sin theta)(1 + sin theta). Step 2: (1 - sin theta)(1 + sin theta) = 1 - sin^2 theta = cos^2 theta. Hence proved.

Q5. Prove: (sin theta - 2 sin^3 theta)/(2 cos^3 theta - cos theta) = tan theta.

Step 1: numerator = sin theta (1 - 2 sin^2 theta). Step 2: denominator = cos theta (2 cos^2 theta - 1) = cos theta (2(1 - sin^2 theta) - 1) = cos theta (1 - 2 sin^2 theta). Step 3: cancelling (1 - 2 sin^2 theta) gives sin theta/cos theta = tan theta. Hence proved.

Q6. Prove: (tan theta - cot theta)/(sin theta cos theta) = tan^2 theta - cot^2 theta.

Step 1: tan theta - cot theta = (sin^2 theta - cos^2 theta)/(sin theta cos theta). Step 2: dividing by sin theta cos theta gives (sin^2 theta - cos^2 theta)/(sin^2 theta cos^2 theta) = 1/cos^2 theta - 1/sin^2 theta = sec^2 theta - cosec^2 theta. Step 3: sec^2 theta - cosec^2 theta = (1 + tan^2 theta) - (1 + cot^2 theta) = tan^2 theta - cot^2 theta. Hence proved.

Q7. Prove: cos theta/(1 - sin theta) + cos theta/(1 + sin theta) = 2 sec theta.

Step 1: combine over (1 - sin^2 theta) = cos^2 theta: numerator = cos theta x (1 + sin theta + 1 - sin theta) = 2 cos theta. Step 2: 2 cos theta/cos^2 theta = 2/cos theta = 2 sec theta. Hence proved.

Q8. Prove: cos theta/(cosec theta + 1) + cos theta/(cosec theta - 1) = 2 tan theta.

Step 1: combine over (cosec^2 theta - 1) = cot^2 theta: numerator = cos theta x (cosec theta - 1 + cosec theta + 1) = 2 cos theta cosec theta. Step 2: 2 cos theta x (1/sin theta) divided by (cos^2 theta/sin^2 theta) = 2 sin theta/cos theta = 2 tan theta. Hence proved.

Q9. Prove: sin theta/(1 - cos theta) + tan theta/(1 + cos theta) = sec theta cosec theta + cot theta.

Step 1: sin theta/(1 - cos theta) = sin theta (1 + cos theta)/sin^2 theta = (1 + cos theta)/sin theta = cosec theta + cot theta. Step 2: cosec theta + tan theta/(1 + cos theta) = [cos theta (1 + cos theta) + sin^2 theta]/(sin theta cos theta (1 + cos theta)) = (1 + cos theta)/(sin theta cos theta (1 + cos theta)) = 1/(sin theta cos theta) = sec theta cosec theta. Step 3: adding the cot theta from Step 1 gives sec theta cosec theta + cot theta. Hence proved.

Q10. Prove: sqrt((1 - sin theta)/(1 + sin theta)) = sec theta - tan theta.

Step 1: multiply inside the root by (1 - sin theta)/(1 - sin theta): (1 - sin theta)^2/(1 - sin^2 theta) = (1 - sin theta)^2/cos^2 theta. Step 2: square root = (1 - sin theta)/cos theta = sec theta - tan theta (for cos theta positive). Hence proved.

Q11. Prove: (1 + cos theta)/(1 - cos theta) = (cosec theta + cot theta)^2.

Step 1: cosec theta + cot theta = (1 + cos theta)/sin theta. Step 2: squaring gives (1 + cos theta)^2/sin^2 theta = (1 + cos theta)^2/((1 - cos theta)(1 + cos theta)). Step 3: cancelling (1 + cos theta) gives (1 + cos theta)/(1 - cos theta). Hence proved.

Q12. Prove: (1 + tan^2 theta)(1 - sin theta)(1 + sin theta) = 1.

Step 1: 1 + tan^2 theta = sec^2 theta = 1/cos^2 theta. Step 2: (1 - sin theta)(1 + sin theta) = 1 - sin^2 theta = cos^2 theta. Step 3: (1/cos^2 theta) x cos^2 theta = 1. Hence proved.

Q13. Prove: 1/(sec theta - tan theta) = sec theta + tan theta.

Step 1: multiply numerator and denominator by (sec theta + tan theta). Step 2: denominator = sec^2 theta - tan^2 theta = 1. Step 3: the result is sec theta + tan theta. Hence proved.

Q14. Prove: sqrt((1 + sin theta)/(1 - sin theta)) + sqrt((1 - sin theta)/(1 + sin theta)) = 2 sec theta.

Step 1: first root = (1 + sin theta)/cos theta and second root = (1 - sin theta)/cos theta (as in Q10). Step 2: sum = (1 + sin theta + 1 - sin theta)/cos theta = 2/cos theta = 2 sec theta. Hence proved.

Q15. If sin theta = 4/5, then find the remaining trigonometric ratios using trigonometric identities. (Acute theta.)

Step 1: cos theta = sqrt(1 - sin^2 theta) = sqrt(1 - 16/25) = 3/5. Step 2: tan theta = sin theta/cos theta = 4/3. Step 3: cosec theta = 5/4, sec theta = 5/3, cot theta = 3/4. Answer: cos = 3/5, tan = 4/3, cosec = 5/4, sec = 5/3, cot = 3/4.

Q16. If tan theta = 1/sqrt(3), then find the remaining trigonometric ratios. (Acute

theta.)

Step 1: sec^2 theta = 1 + tan^2 theta = 1 + 1/3 = 4/3, so sec theta = 2/sqrt(3) and cos theta = sqrt(3)/2. Step 2: sin theta = tan theta x cos theta = (1/sqrt(3)) x (sqrt(3)/2) = 1/2. Step 3: cosec theta = 2, cot theta = sqrt(3). Answer: sin = 1/2, cos = sqrt(3)/2, cosec = 2, sec = 2/sqrt(3), cot = sqrt(3).

Q17. If cos theta = 1/2, then find: (i) 1 - sec^2 theta, (ii) sqrt(1 + cosec^2 theta), (iii) sqrt(1 + tan^2 theta).

Step 1: sec theta = 1/cos theta = 2, so (i) 1 - sec^2 theta = 1 - 4 = -3. Step 2: sin theta = sqrt(1 - 1/4) = sqrt(3)/2, so cosec theta = 2/sqrt(3) and cosec^2 theta = 4/3; (ii) sqrt(1 + 4/3) = sqrt(7/3). Step 3: (iii) sqrt(1 + tan^2 theta) = sec theta = 2. Answer: (i) -3, (ii) sqrt(7/3), (iii) 2.

Q18. A surveyor stands on level ground and observes the top of a tower at an angle theta. If the distance from the surveyor to the tower is 12 m and the height of the tower is 5 m, use the identity 1 + cot^2 theta = cosec^2 theta to find cosec theta.

Step 1: tan theta = 5/12, so cot theta = 12/5. Step 2: cosec^2 theta = 1 + cot^2 theta = 1 + 144/25 = 169/25. Step 3: cosec theta = 13/5. Answer: cosec theta = 13/5.

Q19. A ladder leans against a wall making an angle theta with the ground. If the ratio of the height of the wall to the base of the ladder is 3:4, find the value of sec theta using the identity 1 + tan^2 theta = sec^2 theta.

Step 1: tan theta = 3/4. Step 2: sec^2 theta = 1 + tan^2 theta = 1 + 9/16 = 25/16. Step 3: sec theta = 5/4. Answer: sec theta = 5/4.

Unit 6: Trigonometry and Bearing (continued: Exercise 6.5)

EXERCISE 6.5 (book pp.145-146) — Bearings are measured clockwise from north, written as three figures

Q1. The figure shows the positions of O, A, B, C and D. A is 33 degrees east of north, B is 28 degrees south of east, C is 44 degrees south of west, D is 39 degrees west of north. Find the bearing of: (i) A from O.

Step 1: A is 33 degrees clockwise from north. Answer: bearing of A from O = 033 degrees.

Q1. (ii) B from O.

Step 1: east is 090 degrees; B is 28 degrees further towards south. Step 2: 90 + 28 = 118. Answer: bearing of B from O = 118 degrees.

Q1. (iii) C from O.

Step 1: west is 270 degrees; C is 44 degrees towards south, i.e. 44 degrees back. Step 2: 270 - 44 = 226. Answer: bearing of C from O = 226 degrees.

Q1. (iv) D from O.

Step 1: north is 360 degrees; D is 39 degrees towards west, i.e. 39 degrees back. Step 2: 360 - 39 = 321. Answer: bearing of D from O = 321 degrees.

Q2. The figure shows the position of P, A, B and C. A is 35 degrees north of east, B is 15 degrees east of south, C is 47 degrees north of west. Find the bearing of: (i) A from P.

Step 1: east is 090 degrees; A is 35 degrees towards north, i.e. 35 degrees back. Step 2: 90 - 35 = 55. Answer: bearing of A from P = 055 degrees.

Q2. (ii) B from P.

Step 1: south is 180 degrees; B is 15 degrees towards east, i.e. 15 degrees back. Step 2: 180 - 15 = 165. Answer: bearing of B from P = 165 degrees.

Q2. (iii) C from P.

Step 1: west is 270 degrees; C is 47 degrees towards north, i.e. 47 degrees further clockwise. Step 2: 270 + 47 = 317. Answer: bearing of C from P = 317 degrees.

Q2. (iv) P from A.

Step 1: reverse bearing = forward bearing + 180 degrees. Step 2: 55 + 180 = 235. Answer: bearing of P from A = 235 degrees.

Q3. The figure shows the positions of P, Q and R. At P, PR is 36 degrees from north and PQ is 37 degrees from PR; at R the interior angle PRQ is 100 degrees. Find the bearing of: (i) Q from P.

Step 1: bearing of R from P = 036 degrees. Step 2: Q is 37 degrees further clockwise from

PR. Step 3: 36 + 37 = 73. Answer: bearing of Q from P = 073 degrees.

Q3. (ii) R from P.

Step 1: PR is 36 degrees clockwise from north. Answer: bearing of R from P = 036 degrees.

Q3. (iii) P from R.

Step 1: reverse bearing = 36 + 180. Answer: bearing of P from R = 216 degrees.

Q3. (iv) Q from R.

Step 1: from R, P lies at bearing 216 degrees; Q is 100 degrees anticlockwise from RP (towards the south-east). Step 2: 216 - 100 = 116. Answer: bearing of Q from R = 116 degrees. (Check: bearing of R from Q = 116 + 180 = 296 degrees, and the interior angle at Q is 43 degrees, since 37 + 100 + 43 = 180.)

Q4. A, B, C and D are the four corners of a rectangular plot marked out on level ground. Given that the bearing of B from A is 040 degrees and that the bearing of C from A is 090 degrees, find the bearing of: (i) B from C.

Step 1: in rectangle ABCD, angle ABC = 90 degrees and angle BAC = 90 - 40 = 50 degrees, so angle BCA = 40 degrees. Step 2: CA points west (270 degrees) from C; CB is 40 degrees towards north from CA. Step 3: 270 + 40 = 310. Answer: bearing of B from C = 310 degrees.

Q4. (ii) A from C.

Step 1: C lies due east of A, so A lies due west of C. Answer: bearing of A from C = 270 degrees.

Q4. (iii) D from C.

Step 1: CD is parallel to AB but in the opposite direction. Step 2: 40 + 180 = 220. Answer: bearing of D from C = 220 degrees.

Q5. P, Q and R are three points on level ground. Given that the bearing of R from P is 135 degrees, angle PQR = 55 degrees and angle PRQ = 48 degrees, find the bearing of: (i) P from R.

Step 1: reverse bearing = 135 + 180 = 315. Answer: bearing of P from R = 315 degrees.

Q5. (ii) Q from R.

Step 1: at R, Q lies 48 degrees clockwise from RP (the configuration with Q north-east of the PR line). Step 2: 315 + 48 = 363 = 003. Answer: bearing of Q from R = 003 degrees. NOTE: the printed data fix only the triangle's shape, not its mirror image; the mirror configuration gives Q from R = 267 degrees instead.

Q5. (iii) P from Q.

Step 1: angle QPR = 180 - 55 - 48 = 77 degrees, so the bearing of Q from P = 135 - 77 = 058 degrees. Step 2: reverse bearing = 58 + 180 = 238. Answer: bearing of P from Q = 238 degrees. (Mirror configuration: 032 degrees.)

Q6. A, B and C are three points on level ground. Given that the bearing of B from A is 122 degrees, angle CAB = 32 degrees and angle ABC = 86 degrees, find the

possible bearing(s) of C from B.

Step 1: bearing of A from B = 122 + 180 = 302 degrees. Step 2: at B, C lies 86 degrees from BA on either side. Step 3: 302 - 86 = 216 and 302 + 86 = 388 = 028. Answer: the possible bearings of C from B are 216 degrees or 028 degrees.

Q7. The figure shows four points on level ground. A is due north of D, B is due east of D and angle DBC = 37 degrees. Given AD = 34 m, AB = 57 m and BC = 28 m, find: (i) the bearing of B from A.

Step 1: in right triangle ADB, cos(angle DAB) = 34/57 = 0.5965, so angle DAB = 53.4 degrees. Step 2: B lies south-east of A, so its bearing = 180 - 53.4. Answer: bearing of B from A = 126.6 degrees (approx).

Q7. (ii) The shortest distance from C to BD.

Step 1: the shortest distance is the perpendicular from C to BD. Step 2: distance = BC x sin 37 degrees = 28 x 0.6018. Answer: 16.85 m (approx).

Q7. (iii) The bearing of D from C.

Step 1: BD = sqrt(57^2 - 34^2) = 45.75 m. Step 2: with D = (0, 0) and B = (45.75, 0), C = (23.39, -16.85), so D - C = (-23.39, 16.85). Step 3: bearing = 360 - tan^-1(23.39/16.85) = 360 - 54.2. Answer: bearing of D from C = 305.8 degrees (approx).

Unit 6: Trigonometry and Bearing (continued: Review Exercise 6)

REVIEW EXERCISE 6 (book pp.146-148)

Q1. (i) 2pi/3 radian is equal to:

(a) 140 degrees

(b) 120 degrees ✓

(c) 100 degrees

(d) 180 degrees

Q1. (ii) 1 degree is equal to:

(a) 0.175 radian

(b) 0.0175 radian ✓

(c) 175 radian

(d) 0.00175 radian

Q1. (iii) sin 30 degrees is equal to:

(a) 1/2 ✓

(b) sqrt(3)/2

(c) 1/sqrt(3)

(d) 2

Q1. (iv) If tan theta = 3/4, then sin theta =

(a) 2/5

(b) 3/5 ✓

(c) 4/5

(d) 1

Q1. (v) If sin 30 degrees = 1/2 then cos 30 degrees =

(a) sqrt(3)/2 ✓

(b) 1/sqrt(2)

(c) sqrt(3)

(d) 1/2

Q1. (vi) Bearings are written in figures:

(a) two

(b) three ✓

(c) four

(d) five

Q1. (vii) Bearings start from:

(a) east

(b) west

(c) north ✓

(d) south

Q2. A person standing 30 m away from a tower has an eye level 1.5 m above the ground. The angle of elevation from his eyes to the top of the tower is 45 degrees. Determine the height of the tower.

Step 1: height above eye level = 30 x tan 45 degrees = 30 x 1 = 30 m. Step 2: total height = 30 + 1.5. Answer: height of the tower = 31.5 m.

Q3. Prove that: sec^4 theta - tan^4 theta = 1 + 2 tan^2 theta.

Step 1: sec^4 theta - tan^4 theta = (sec^2 theta - tan^2 theta)(sec^2 theta + tan^2 theta). Step 2: sec^2 theta - tan^2 theta = 1. Step 3: sec^2 theta + tan^2 theta = (1 + tan^2 theta) + tan^2 theta = 1 + 2 tan^2 theta. Hence proved.

Q4. In figure, AD = 45 m, angle DAC = 30 degrees and angle DBC = 60 degrees and angle ACD = 90 degrees. Find: (i) DC, (ii) AB, (iii) angle ADB.

Step 1: in right triangle ADC, DC = AD x sin 30 degrees = 45 x 0.5 = 22.5 m, and AC = AD x cos 30 degrees = 45 x sqrt(3)/2 = 38.97 m. Step 2: in right triangle DBC, BC = DC/tan 60 degrees = 22.5/sqrt(3) = 12.99 m. Step 3: AB = AC - BC = 38.97 - 12.99 = 25.98 m. Step 4: angle ADC = 60 degrees and angle BDC = 30 degrees, so angle ADB = 60 - 30 = 30 degrees. Answer: (i) DC = 22.5 m, (ii) AB = 25.98 m (approx), (iii) angle ADB = 30 degrees.

Q5. Three points A, B and C lie on a horizontal ground. T is the top of the vertical tower standing on A. The bearings of B and C from A are 135 degrees and 225 degrees respectively and the bearing of C from B is 250 degrees. If BC = 50 m and angle of elevation of T from C is 35 degrees, calculate the angle of elevation of T from C. NOTE: the printed question repeats the given 35 degrees as its ask, so the notes give the quantities the data actually determine.

Step 1: angle BAC = 225 - 135 = 90 degrees. Step 2: bearing of B from C = 250 - 180 = 070 degrees and bearing of A from C = 225 - 180 = 045 degrees, so angle BCA = 70 - 45 = 25 degrees. Step 3: angle ABC = 180 - 90 - 25 = 65 degrees. Step 4: AC = BC x sin 65 degrees = 50 x 0.9063 = 45.32 m. Step 5: tower height = AC x tan 35 degrees = 45.32 x 0.7002 = 31.73 m. Step 6: AB = BC x sin 25 degrees = 50 x 0.4226 = 21.13 m, so the angle of elevation of T from B = tan^-1(31.73/21.13) = 56.3 degrees (approx). Answer: tower height = 31.73 m (approx); angle of elevation of T from B = 56.3 degrees (approx); the angle of elevation of T from C is the given 35 degrees.

Unit 7: Coordinate Geometry

EXERCISE 7.1 (book p.160)

Q1. Find distance between the following points: (i) (2, 5) and (5, 10).

Step 1: d = sqrt((5 - 2)^2 + (10 - 5)^2). Step 2: d = sqrt(9 + 25). Answer: d = sqrt(34) units.

Q1. (ii) (-4, 5) and (-7, -8).

Step 1: d = sqrt((-7 + 4)^2 + (-8 - 5)^2). Step 2: d = sqrt(9 + 169). Answer: d = sqrt(178) units.

Q1. (iii) (3a, a) and (19a, 10a).

Step 1: d = sqrt((19a - 3a)^2 + (10a - a)^2) = sqrt(256a^2 + 81a^2). Step 2: d = sqrt(337a^2). Answer: d = a x sqrt(337) units (for a positive; strictly sqrt(337) x |a|).

Q1. (iv) (1/2, 3) and (5, -1/2).

Step 1: d = sqrt((5 - 1/2)^2 + (-1/2 - 3)^2) = sqrt((9/2)^2 + (-7/2)^2). Step 2: d = sqrt(81/4 + 49/4) = sqrt(130/4). Answer: d = sqrt(130)/2 units.

Q1. (v) (4.5, -2.3) and (5.2, 1.2).

Step 1: d = sqrt((5.2 - 4.5)^2 + (1.2 + 2.3)^2) = sqrt(0.49 + 12.25). Step 2: d = sqrt(12.74). Answer: d = 3.57 units (approx).

Q2. Show that whether A(1, -1), B(2, 2) and C(4, 3) are collinear or not.

Step 1: AB = sqrt(1 + 9) = sqrt(10); BC = sqrt(4 + 1) = sqrt(5); AC = sqrt(9 + 16) = 5. Step 2: sqrt(10) + sqrt(5) = 5.40 (approx), which is not equal to 5, and no other sum of two sides equals the third. Answer: the points are NOT collinear.

Q3. If two vertices of an equilateral triangle are A(-3, 0) and B(3, 0), find the third vertex.

Step 1: AB = 6 units, so the third vertex lies on the perpendicular bisector x = 0. Step 2: height = (6 x sqrt(3))/2 = 3 x sqrt(3). Answer: the third vertex is (0, 3 x sqrt(3)) or (0, -3 x sqrt(3)).

Q4. Show that the points P(-2, 8), Q(3, 20) and (11, 8) are the vertices of an isosceles triangle.

Step 1: PQ = sqrt(25 + 144) = 13. Step 2: QR = sqrt(64 + 144) = sqrt(208) = 4 x sqrt(13). Step 3: PR = sqrt(169 + 0) = 13. Step 4: PQ = PR = 13, so two sides are equal. Answer: the points are the vertices of an isosceles triangle.

Q5. Show that the points A(0, 2), B(sqrt(3), -1) and C(0, -2) are the vertices of a right angled triangle.

Step 1: AB^2 = 3 + 9 = 12; BC^2 = 3 + 1 = 4; AC^2 = 0 + 16 = 16. Step 2: AB^2 + BC^2 = 12 + 4 = 16 = AC^2. Answer: by Pythagoras, the triangle is right angled at B.

Q6. Show that the points A(5, 2), B(-2, 3), C(-3, -4) and D(4, -5) are the vertices of

the square.

Step 1: AB^2 = 49 + 1 = 50; BC^2 = 1 + 49 = 50; CD^2 = 49 + 1 = 50; DA^2 = 1 + 49 = 50. Step 2: diagonals AC^2 = 64 + 36 = 100 and BD^2 = 36 + 64 = 100. Step 3: all four sides are equal, the diagonals are equal, and 50 + 50 = 100. Answer: the points are the vertices of a square.

Q7. Show that P(0, 7), Q(6, 5), R(5, 2) and S(-1, 4) are the vertices of a rectangle.

Step 1: PQ^2 = 36 + 4 = 40; QR^2 = 1 + 9 = 10; RS^2 = 36 + 4 = 40; SP^2 = 1 + 9 = 10. Step 2: diagonals PR^2 = 25 + 25 = 50 and QS^2 = 49 + 1 = 50. Step 3: opposite sides are equal and the diagonals are equal. Answer: the points are the vertices of a rectangle.

Q8. Find the coordinates of the midpoint of the line segment joining the following pair of points: (i) (6, 7) and (4, 3).

Step 1: x = (6 + 4)/2 = 5. Step 2: y = (7 + 3)/2 = 5. Answer: midpoint = (5, 5).

Q8. (ii) (2t, 8a) and (8t, 9a).

Step 1: x = (2t + 8t)/2 = 5t. Step 2: y = (8a + 9a)/2 = 17a/2. Answer: midpoint = (5t, 17a/2).

Q8. (iii) (a + 1, 2a + 3) and (a - 1, 2a - 1).

Step 1: x = (a + 1 + a - 1)/2 = a. Step 2: y = (2a + 3 + 2a - 1)/2 = 2a + 1. Answer: midpoint = (a, 2a + 1).

Q8. (iv) (p + 2q, 2p + 13q) and (5p - 2q, -2p - 7q).

Step 1: x = (p + 2q + 5p - 2q)/2 = 3p. Step 2: y = (2p + 13q - 2p - 7q)/2 = 3q. Answer: midpoint = (3p, 3q).

Q9. A(-2, 1) and B(6, 5) are opposite ends of the diameter of a circle. Find coordinates of its midpoint.

Step 1: the centre is the midpoint of the diameter. Step 2: x = (-2 + 6)/2 = 2; y = (1 + 5)/2 = 3. Answer: midpoint (centre) = (2, 3).

Q10. The coordinate of the midpoint of the line joining A(p^2, p) and B(-q^2, q) are (5, 1). Find the possible values of p and q.

Step 1: (p^2 - q^2)/2 = 5, so p^2 - q^2 = 10. Step 2: (p + q)/2 = 1, so p + q = 2. Step 3: p^2 - q^2 = (p - q)(p + q) = 2(p - q) = 10, so p - q = 5. Step 4: solving p + q = 2 and p - q = 5 gives p = 7/2 and q = -3/2. Answer: p = 7/2, q = -3/2.

Q11. Find the value of P so that (-2, 2.5) is the midpoint between (P, 2) and (-1, 3).

Step 1: (P - 1)/2 = -2, so P - 1 = -4. Step 2: P = -3. Step 3: check: (2 + 3)/2 = 2.5. Answer: P = -3.

Unit 7: Coordinate Geometry (continued: Exercise 7.2)

EXERCISE 7.2 (book pp.168-169)

Q1. Find the gradient of the line passing through each of the following pairs of points: (i) A(0, 0), B(-5, 2).

Step 1: m = (2 - 0)/(-5 - 0) = 2/(-5). Answer: m = -2/5.

Q1. (ii) C(5, 3), D(1, 8).

Step 1: m = (8 - 3)/(1 - 5) = 5/(-4). Answer: m = -5/4.

Q1. (iii) E(-4, 6), F(8, 10).

Step 1: m = (10 - 6)/(8 + 4) = 4/12. Answer: m = 1/3.

Q1. (iv) G(3, 5), H(9, -8).

Step 1: m = (-8 - 5)/(9 - 3) = -13/6. Answer: m = -13/6.

Q1. (v) P(-6, -2), Q(2, 6).

Step 1: m = (6 + 2)/(2 + 6) = 8/8. Answer: m = 1.

Q1. (vi) R(-7, 10), S(6, 9).

Step 1: m = (9 - 10)/(6 + 7) = -1/13. Answer: m = -1/13.

Q2. If the gradient of a line joining the points (k, -5) and (2, -4k) is -4, find the value of k.

Step 1: (-4k + 5)/(2 - k) = -4. Step 2: -4k + 5 = -8 + 4k. Step 3: 13 = 8k. Answer: k = 13/8.

Q3. Check whether the following pairs of points joining the line are parallel, perpendicular or neither: (i) (5, 7), (7, 8) and (7, -5), (19, 1).

Step 1: m1 = (8 - 7)/(7 - 5) = 1/2. Step 2: m2 = (1 + 5)/(19 - 7) = 6/12 = 1/2. Step 3: m1 = m2. Answer: the lines are parallel.

Q3. (ii) (1, -5), (3, 7) and (5/2, 7/4), (-3/2, 9/4).

Step 1: m1 = (7 + 5)/(3 - 1) = 6. Step 2: m2 = (9/4 - 7/4)/(-3/2 - 5/2) = (1/2)/(-4) = -1/8. Step 3: m1 is not equal to m2, and m1 x m2 = -3/4, not -1. Answer: the lines are neither parallel nor perpendicular.

Q3. (iii) (3, 7), (-6, 9) and (3, 18), (5, 27).

Step 1: m1 = (9 - 7)/(-6 - 3) = -2/9. Step 2: m2 = (27 - 18)/(5 - 3) = 9/2. Step 3: m1 x m2 = -1. Answer: the lines are perpendicular.

Q4. The points A(2, -3), B(3, -2) and C(8, z) are collinear, find the value of z.

Step 1: slope of AB = (-2 + 3)/(3 - 2) = 1. Step 2: slope of BC = (z + 2)/(8 - 3) = (z + 2)/5. Step 3: for collinearity (z + 2)/5 = 1, so z + 2 = 5. Answer: z = 3.

Q5. The co-ordinates of four points O(0, 0), P(2, 1), Q(3, 9) and R(3h, 2h + 7). Find the value(s) of h if: (i) O, P and R are collinear.

Step 1: slope of OP = 1/2. Step 2: slope of OR = (2h + 7)/(3h). Step 3: (2h + 7)/(3h) = 1/2 gives 4h + 14 = 3h. Answer: h = -14.

Q5. (ii) OP is parallel to QR.

Step 1: slope of QR = (2h + 7 - 9)/(3h - 3) = (2h - 2)/(3h - 3) = 2/3 (for h not equal to 1).

Step 2: slope of OP = 1/2, which is never equal to 2/3. Answer: no value of h makes OP parallel to QR (for h = 1, R coincides with Q, so QR is undefined).

Q5. (iii) OP is perpendicular to QR.

Step 1: slope of QR = 2/3 (as above) and slope of OP = 1/2. Step 2: (1/2) x (2/3) = 1/3, never -1. Answer: no value of h makes OP perpendicular to QR.

Q6. Two airports are located on a straight line where A(4, 5) and B(10, 13). What is the distance between the airports?

Step 1: d = sqrt((10 - 4)^2 + (13 - 5)^2) = sqrt(36 + 64). Answer: d = 10 units.

Q7. A city park is located at a point (4, 7) on a map. A library is located at (12, 15). What is the distance between the city park and the library? Additionally, what is the midpoint of the park and the library?

Step 1: distance = sqrt((12 - 4)^2 + (15 - 7)^2) = sqrt(64 + 64) = sqrt(128) = 8 x sqrt(2).

Step 2: midpoint = ((4 + 12)/2, (7 + 15)/2) = (8, 11). Answer: distance = 8 x sqrt(2) units (11.31 approx); midpoint = (8, 11).

Q8. The school is located at (5, 8) and a student's home is at (11, 12). How far is the student's home from the school and what is the midpoint of his route?

Step 1: distance = sqrt((11 - 5)^2 + (12 - 8)^2) = sqrt(36 + 16) = sqrt(52) = 2 x sqrt(13).

Step 2: midpoint = ((5 + 11)/2, (8 + 12)/2) = (8, 10). Answer: distance = 2 x sqrt(13) units (7.21 approx); midpoint = (8, 10).

Q9. A new road is being constructed between town A(3, 4) and B(9, 14). The engineer needs to determine the gradient of the road. What is the gradient of the road between town A and B?

Step 1: m = (14 - 4)/(9 - 3) = 10/6. Answer: m = 5/3.

Unit 7: Coordinate Geometry (continued: Review Exercise 7)

REVIEW EXERCISE 7 (book p.169)

Q1. (i) Distance between A(4, 5) and B(1, 9) is:

(a) 2.5

(b) 5 ✓

(c) 7.5

(d) 10

Q1. (ii) Midpoint of A(-2, 5), B(2, 3) is:

(a) M(0, 4) ✓

(b) M(4, 0)

(c) (0, -4)

(d) (0, -4)

Q1. (iii) If the slopes of two lines are equal then the lines are:

(a) parallel ✓

(b) intersecting

(c) 2

(d) 3

Q1. (iv) If m1 x m2 = -1 then the lines are:

(a) parallel

(b) intersecting

(c) perpendicular ✓

(d) none of these

Q1. (v) If A(0, -5) and B(6, 1) are two points then its slope is:

(a) 0

(b) 1 ✓

(c) 2

(d) 3

Q2. Find the gradient of the following points joining the line: (i) A(-6, 0), B(-4, 5).

Step 1: m = (5 - 0)/(-4 + 6) = 5/2. Answer: m = 5/2.

Q2. (ii) C(3.8, 4.5), D(4, 6.5).

Step 1: m = (6.5 - 4.5)/(4 - 3.8) = 2/0.2. Answer: m = 10.

Q3. Show that the points A(1, 2), B(3, 6) and (5, 10) are collinear.

Step 1: slope of AB = (6 - 2)/(3 - 1) = 2. Step 2: slope of the second pair = (10 - 6)/(5 - 3) = 2. Step 3: equal slopes with B common. Answer: the points are collinear.

Q4. Show that the points joining the lines A(3, 5), B(7, 8) and C(5, 8), D(8, 12) are perpendicular. NOTE: as printed this claim does NOT hold — the working below proves the opposite, and the notes do not fake the proof.

Step 1: slope of AB = (8 - 5)/(7 - 3) = 3/4. Step 2: slope of CD = (12 - 8)/(8 - 5) = 4/3. Step 3: (3/4) x (4/3) = 1, not -1. Answer: the lines are NOT perpendicular as printed (their slopes are reciprocals with product +1); the book's statement is an error.

Q5. Two locations in a park are located at points P(2, 3) and Q(10, 7). What is the midpoint and distance between them?

Step 1: midpoint = ((2 + 10)/2, (3 + 7)/2) = (6, 5). Step 2: distance = sqrt((10 - 2)^2 + (7 - 3)^2) = sqrt(64 + 16) = sqrt(80) = 4 x sqrt(5). Answer: midpoint = (6, 5); distance = 4 x sqrt(5) units (8.94 approx).

Unit 8: Geometry of Straight Lines

EXERCISE 8.1 (book p.179)

Q1. Find the equation of straight line using slope-intercept form if it has: (i) gradient 7 and y-intercept -8.

Step 1: y = mx + c with m = 7 and c = -8. Answer: y = 7x - 8.

Q1. (ii) slope -5 and passing through (-1, 6).

Step 1: y = -5x + c and the point (-1, 6) lies on it. Step 2: 6 = -5 x (-1) + c, so c = 1. Answer: y = -5x + 1.

Q2. Find the equation of straight line using point-slope form if it has: (i) slope -3 and passing through (7, 3).

Step 1: y - 3 = -3(x - 7). Step 2: y - 3 = -3x + 21. Answer: 3x + y - 24 = 0.

Q2. (ii) slope -1/2 and passing through (-4, -7).

Step 1: y + 7 = (-1/2)(x + 4). Step 2: 2y + 14 = -x - 4. Answer: x + 2y + 18 = 0.

Q3. Find the equation of the line passing through (7, 11) and perpendicular to a line through (-1, 5) and (7, 6).

Step 1: slope of the given line = (6 - 5)/(7 + 1) = 1/8. Step 2: slope of the required line = -8 (negative reciprocal). Step 3: y - 11 = -8(x - 7). Answer: 8x + y - 67 = 0.

Q4. Find the equation of line passing through (3, -2) making an angle 135 degrees with x-axis.

Step 1: slope m = tan 135 degrees = -1. Step 2: y + 2 = -1(x - 3). Answer: x + y - 1 = 0.

Q5. Find equation of line passing two points: (i) (-1, 5) and (2, 8).

Step 1: slope = (8 - 5)/(2 + 1) = 1. Step 2: y - 5 = 1(x + 1). Answer: x - y + 6 = 0.

Q5. (ii) (3, 1) and (5, -7).

Step 1: slope = (-7 - 1)/(5 - 3) = -4. Step 2: y - 1 = -4(x - 3). Answer: 4x + y - 13 = 0.

Q5. (iii) (18, 10) and (21, 8).

Step 1: slope = (8 - 10)/(21 - 18) = -2/3. Step 2: y - 10 = (-2/3)(x - 18). Answer: 2x + 3y - 66 = 0.

Q6. Find the equation of the line passing through (8, 3) and also passing the midpoint of the line segment joining (3, 5) and (8, 10).

Step 1: midpoint = ((3 + 8)/2, (5 + 10)/2) = (11/2, 15/2). Step 2: slope = (15/2 - 3)/(11/2 - 8) = (9/2)/(-5/2) = -9/5. Step 3: y - 3 = (-9/5)(x - 8). Answer: 9x + 5y - 87 = 0.

Q7. Find the equation of the line passing through (-8, -1) and making equal intercepts on the coordinate axes.

Step 1: with equal intercepts a, x/a + y/a = 1, i.e. x + y = a. Step 2: -8 + (-1) = a, so a = -9. Answer: x + y + 9 = 0.

Q8. A line makes the positive intercepts on coordinate axes whose sum is 9 and

passes through (-3, 5), then find the equation of the straight line.

Step 1: let the intercepts be a and 9 - a (both positive), so x/a + y/(9 - a) = 1. Step 2: -3/a + 5/(9 - a) = 1 gives a^2 - a - 27 = 0. Step 3: a = (1 + sqrt(109))/2 (the other root is negative and rejected since the intercepts are positive). Step 4: the other intercept is (17 - sqrt(109))/2. Answer: x/((1 + sqrt(109))/2) + y/((17 - sqrt(109))/2) = 1, i.e. (17 - sqrt(109))x + (1 + sqrt(109))y = 8 x sqrt(109) - 46 (approx: 3.28x + 5.72y = 18.76).

Unit 8: Geometry of Straight Lines (continued: Exercise 8.2)

EXERCISE 8.2 (book pp.185-186)

Q1. Reduce the equation 3x - 8y = 24 to the: (i) slope-intercept form, (ii) intercepts form, (iii) normal form.

Step 1: (i) 8y = 3x - 24, so y = (3/8)x - 3. Step 2: (ii) dividing by 24: x/8 + y/(-3) = 1. Step 3: (iii) dividing by sqrt(9 + 64) = sqrt(73): (3/sqrt(73))x - (8/sqrt(73))y = 24/sqrt(73), i.e. x cos alpha + y sin alpha = p with cos alpha = 3/sqrt(73), sin alpha = -8/sqrt(73), p = 24/sqrt(73). Answer: (i) y = (3/8)x - 3; (ii) x/8 - y/3 = 1; (iii) (3x - 8y)/sqrt(73) = 24/sqrt(73).

Q2. Reduce the equation 3x + 5y = 15 to the: (i) slope-intercept form, (ii) intercepts form, (iii) symmetric form, (iv) normal form.

Step 1: (i) 5y = -3x + 15, so y = (-3/5)x + 3. Step 2: (ii) dividing by 15: x/5 + y/3 = 1. Step 3: (iii) the x-intercept is (5, 0) and slope = -3/5 = tan alpha, so cos alpha = 5/sqrt(34) and sin alpha = -3/sqrt(34); symmetric form: (x - 5)/(5/sqrt(34)) = y/(-3/sqrt(34)). Step 4: (iv) dividing by sqrt(34): (3/sqrt(34))x + (5/sqrt(34))y = 15/sqrt(34). Answer: (i) y = (-3/5)x + 3; (ii) x/5 + y/3 = 1; (iii) (x - 5)/(5/sqrt(34)) = y/(-3/sqrt(34)); (iv) (3x + 5y)/sqrt(34) = 15/sqrt(34).

Q3. If P is the perpendicular distance from the origin to a line whose intercepts on the axes are a and b, show that: 1/P^2 = 1/a^2 + 1/b^2.

Step 1: the line is x/a + y/b = 1, i.e. (1/a)x + (1/b)y - 1 = 0. Step 2: distance from the origin P = | -1 |/sqrt((1/a)^2 + (1/b)^2) = 1/sqrt(1/a^2 + 1/b^2). Step 3: squaring and taking reciprocals: 1/P^2 = 1/a^2 + 1/b^2. Hence proved.

Q4. Three villages A, B and C are connected by straight roads with vertices (3, 5), (9, 7) and (-3, 5) respectively. Find: (i) the distance between the village A and B; (ii) the middle point between the village B and C.

Step 1: (i) AB = sqrt((9 - 3)^2 + (7 - 5)^2) = sqrt(36 + 4) = sqrt(40) = 2 x sqrt(10) units. Step 2: (ii) midpoint of B and C = ((9 - 3)/2, (7 + 5)/2) = (3, 6). Answer: (i) 2 x sqrt(10) units; (ii) (3, 6).

Q5. The value of a machine is expected to decrease: its value 1 year after the date of purchase will be Rs.90,000 and its value after 5 years is expected to be Rs.60,000. (i) Determine the slope intercept equation (V = mt + K) which relates the value V of the machine to its age t, stated in years. (ii) Interpret the meaning of the slope and V intercept. (iii) Determine the t-intercept and interpret the meaning.

Step 1: (i) slope m = (60000 - 90000)/(5 - 1) = -7500. Step 2: V = -7500t + K and 90000 =

-7500 + K, so K = 97500. Answer: (i) V = -7500t + 97500. Step 3: (ii) the slope -7500 means the machine loses Rs.7,500 of value per year; the V-intercept 97500 is the value at purchase (t = 0). Step 4: (iii) V = 0 gives t = 97500/7500 = 13. Answer: t-intercept = 13 years, i.e. the machine's value reaches zero after 13 years.

Q6. If C represents temperature in degrees Celsius and F represents temperature in degrees Fahrenheit, assume that the relationship between the two scales is linear and is graphed with F on vertical axis. Two temperature pairs on this line are (5, 41) and (25, 77). Using these points, find the equation of the straight line (in the form C = mF + K). Also, explain the meaning of slope, the C-intercept and F-intercept.

Step 1: the pairs are (C, F) = (5, 41) and (25, 77). Step 2: slope m = (25 - 5)/(77 - 41) = 20/36 = 5/9. Step 3: C = (5/9)F + K and 5 = (5/9)(41) + K, so K = 5 - 205/9 = -160/9. Answer: C = (5/9)F - 160/9. Step 4: the slope 5/9 means one Fahrenheit degree equals 5/9 Celsius degrees; the C-intercept -160/9 (about -17.78) is the Celsius temperature when F = 0; the F-intercept is 32 (putting C = 0 gives F = 32, the freezing point of water).

Q7. A water tank is being filled at a constant rate. After 2 hours, the water level reaches 150 litres, and the filling rate is 30 litres per hour. Write the equation of water level y (in litres) after x hours and find the water after 5 hours.

Step 1: y = 30x + b and 150 = 30 x 2 + b, so b = 90. Step 2: y = 30x + 90. Step 3: after 5 hours y = 30 x 5 + 90 = 240. Answer: y = 30x + 90; water after 5 hours = 240 litres.

Q8. A straight road passes through two towns, A(2, 3) and B(8, 9) on a coordinate map. Find the equation of the road and also check whether a third town C(5, 6) lies on the same road.

Step 1: slope = (9 - 3)/(8 - 2) = 1. Step 2: y - 3 = 1(x - 2), i.e. x - y + 1 = 0. Step 3: for C(5, 6): 5 - 6 + 1 = 0. Answer: equation x - y + 1 = 0; C(5, 6) DOES lie on the same road.

Unit 8: Geometry of Straight Lines (continued: Exercise 8.3)

EXERCISE 8.3 (book pp.191-192)

Q1. Find the acute angle between the lines 7x - 3y = 0 and 3x - 11y + 5 = 0.

Step 1: slopes m1 = 7/3 and m2 = 3/11. Step 2: tan theta = |(m2 - m1)/(1 + m1m2)| = |(3/11 - 7/3)/(1 + 21/33)| = 34/27. Step 3: theta = tan^-1(34/27). Answer: theta = 51.5 degrees (approx).

Q2. Find the acute angle between the lines 8x - 3y - 1 = 0 and 11x - 5y + 15 = 0.

Step 1: slopes m1 = 8/3 and m2 = 11/5. Step 2: tan theta = |(11/5 - 8/3)/(1 + 88/15)| = 7/103. Step 3: theta = tan^-1(7/103). Answer: theta = 3.9 degrees (approx).

Q3. Show that the angle measured from the line l1: a1x + b1y + c1 = 0 to the line l2: a2x + b2y + c2 = 0 is tan^-1((a1b2 - a2b1)/(a1a2 + b1b2)). Also deduce the conditions for parallelism and perpendicularity.

Step 1: slopes m1 = -a1/b1 and m2 = -a2/b2. Step 2: (m2 - m1)/(1 + m1m2) = (-a2/b2 + a1/b1)/(1 + a1a2/(b1b2)) = (a1b2 - a2b1)/(a1a2 + b1b2). Step 3: angle = tan^-1((a1b2 - a2b1)/(a1a2 + b1b2)). Step 4: parallel when the angle is 0, i.e. a1b2 - a2b1 = 0; perpendicular when the denominator is 0, i.e. a1a2 + b1b2 = 0. Hence proved.

Q4. Find interior angles of the triangle whose vertices are A(-4, 5), B(0, 7) and C(5, -5).

Step 1: slopes AB = 1/2, BC = -12/5, CA = -10/9. Step 2: tan A = |(CA slope - AB slope)/(1 + product)| = 29/8, so A = 74.6 degrees (approx). Step 3: tan B = 29/2, so B = 86.1 degrees (approx). Step 4: C = 180 - 74.6 - 86.1 = 19.4 degrees (approx). Answer: angles are 74.6, 86.1 and 19.4 degrees (approx).

Q5. Find the angles of the triangle with slopes of its sides are -7/3, 5/2 and 4/3.

Step 1: the pair (5/2, 4/3) gives tan = (7/6)/(13/3) = 7/26, angle = 15.1 degrees (approx). Step 2: the pair (5/2, -7/3) gives tan = 1, angle = 45 degrees. Step 3: the pair (-7/3, 4/3) gives acute angle tan^-1(33/19) = 60.1 degrees (approx), so the interior angle is 180 - 60.1 = 119.9 degrees (approx). Answer: angles are 45, 15.1 and 119.9 degrees (approx).

Q6. What is the equation of the line that passes through (2, -2) and the point of intersection of 2x + 3y - 5 = 0 and 7x - 5y - 2 = 0?

Step 1: solving the two lines: 7(5 - 3y)/2 - 5y - 2 = 0 gives y = 1 and x = 1. Step 2: line through (2, -2) and (1, 1) has slope (1 + 2)/(1 - 2) = -3. Step 3: y + 2 = -3(x - 2). Answer: 3x + y - 4 = 0.

Q7. Find the equation of a line through (1, 3) and the point of intersection of lines 2x - 3y - 5 = 0 and 4x + 5y - 7 = 0.

Step 1: solving the two lines: 11y + 3 = 0 gives y = -3/11 and x = 23/11. Step 2: line through (1, 3) and (23/11, -3/11) has slope (-3/11 - 3)/(23/11 - 1) = -3. Step 3: y - 3 = -3(x - 1). Answer: 3x + y - 6 = 0.

Q8. Find the equation of line which passes through the point of intersection of the lines 7x - 3y - 19 = 0 and 3x + 2y + 5 = 0 and that line is parallel to the line y = 2x + 7.

Step 1: solving the two lines: 23x - 23 = 0 gives x = 1 and y = -4. Step 2: parallel to y = 2x + 7, so slope = 2. Step 3: y + 4 = 2(x - 1). Answer: 2x - y - 6 = 0.

Unit 8: Geometry of Straight Lines (continued: Review Exercise 8)

REVIEW EXERCISE 8 (book p.192)

Q1. (i) A point p lies on x-axis with vertices:

(a) (2, 3)

(b) (0, 4)

(c) (5, 0) ✓

(d) (-2, 3)

Q1. (ii) If m = 3, c = -5 then y =

(a) 2x - 5

(b) 3x - 5 ✓

(c) -3x + 5

(d) -3x - 5

Q1. (iii) If m1 = 2 and m2 = -1/2 are slopes of two straight lines then angle between them is:

(a) 30 degrees

(b) 60 degrees

(c) 90 degrees ✓

(d) 120 degrees

Q1. (iv) Angle between two intersecting lines can be calculated by the formula:

(a) theta = tan^-1((m2 - m1)/(1 + m2m1)) ✓

(b) theta = tan^-1((m2 + m1)/(1 + m2m1))

(c) theta = tan^-1((m2 - m1)/(1 - m2m1))

(d) none of these

Q1. (v) A triangle having two equal sides is called:

(a) Right angled triangle

(b) Isosceles triangle ✓

(c) Scalene triangle

(d) Equilateral triangle

Q2. Find the equation of line passing through (3, 5) and (2, 4).

Step 1: slope = (4 - 5)/(2 - 3) = 1. Step 2: y - 5 = 1(x - 3). Answer: x - y + 2 = 0.

Q3. Find the angles of a triangle if slopes of sides are 3, -1/3 and 2 respectively.

Step 1: the pair (3, -1/3) has product -1, so the angle is 90 degrees. Step 2: the pair (3, 2) gives tan = 1/7, angle = 8.1 degrees (approx). Step 3: the pair (2, -1/3) gives tan = 7, angle = 81.9 degrees (approx). Answer: angles are 90, 8.1 and 81.9 degrees (approx).

Q4. Find the angle between the lines l1: 3x - 5y + 4 = 0 and l2: 5x + 3y - 7 = 0.

Step 1: slopes m1 = 3/5 and m2 = -5/3. Step 2: m1 x m2 = -1. Answer: the lines are perpendicular, so the angle between them is 90 degrees.

Unit 9: Mathematical Logic and Polygons

EXERCISE 9.1 (book p.197)

Q1. Identify the following as mathematical statement, axiom, conjecture and theorem: (i) -5 < 7.

Answer: mathematical statement (it is a sentence that is either true or false; here it is true).

Q1. (ii) Sum of two prime numbers greater than 4 is even.

Answer: theorem. Step 1: every prime greater than 2 is odd. Step 2: the sum of two odd numbers is even. Step 3: hence the sum of any two primes greater than 4 is even — a proved statement.

Q1. (iii) A line can extend to infinity.

Answer: axiom (it is taken to be true as a starting point for reasoning, like Euclid's postulate).

Q1. (iv) 9 = 8.

Answer: mathematical statement (it is a sentence that is either true or false; here it is false).

Q1. (v) Sum of angles in a triangle is 180 degrees.

Answer: theorem (it is proved from known facts).

Q1. (vi) If a = b then a + c = b + c.

Answer: axiom (the addition property of equality, taken as a starting point).

Q1. (vii) -9 > 11.

Answer: mathematical statement (it is a sentence that is either true or false; here it is false).

Q1. (viii) If a = c then ad = cd.

Answer: axiom (the multiplication property of equality, taken as a starting point).

Q1. (ix) 3/5 < 5/3.

Answer: mathematical statement (it is a sentence that is either true or false; here it is true).

Q1. (x) In an isosceles triangle, corresponding angles of same sides are equal.

Answer: theorem (the base-angles result, proved from known facts).

Q2. Prove the following: (i) (m - 1)/5 + (m + 3)/3 = 4(2m + 3)/15.

Step 1: L.H.S = (3(m - 1) + 5(m + 3))/15. Step 2: = (3m - 3 + 5m + 15)/15 = (8m + 12)/15. Step 3: = 4(2m + 3)/15 = R.H.S. Hence proved.

Q2. (i) (continued) (3x - 5)/7 - (3x - 8)/5 = (31 - 6x)/35.

Step 1: L.H.S = (5(3x - 5) - 7(3x - 8))/35. Step 2: = (15x - 25 - 21x + 56)/35 = (31 - 6x)/35 = R.H.S. Hence proved.

Q2. (ii) ((9x^2 - 1)/(3x^2 + 7x + 2)) x ((3x^2 - 4x + 1)/(3x - 1)^2) = (x - 1)/(x + 2).

Step 1: factorise: 9x^2 - 1 = (3x - 1)(3x + 1); 3x^2 + 7x + 2 = (3x + 1)(x + 2); 3x^2 - 4x

+ 1 = (3x - 1)(x - 1). Step 2: the product = (3x - 1)(3x + 1)/((3x + 1)(x + 2)) x (3x - 1)(x - 1)/(3x - 1)^2. Step 3: cancelling (3x + 1) and (3x - 1)^2 gives (x - 1)/(x + 2) = R.H.S. Hence proved.

Q2. (ii) (continued) ((x^2 - 1)/(x^2 - 3x - 4)) divided by ((x^2 - 3x + 2)/(x - 4)^2) = (x - 4)/(x - 2).

Step 1: factorise: x^2 - 1 = (x - 1)(x + 1); x^2 - 3x - 4 = (x - 4)(x + 1); x^2 - 3x + 2 = (x - 1)(x - 2). Step 2: division becomes multiplication by the reciprocal: (x - 1)(x + 1)/((x - 4)(x + 1)) x (x - 4)^2/((x - 1)(x - 2)). Step 3: cancelling (x - 1), (x + 1) and one (x - 4) gives (x - 4)/(x - 2) = R.H.S. Hence proved.

Unit 9: Mathematical Logic and Polygons (continued: Exercise 9.2)

EXERCISE 9.2 (book pp.202-204)

Q1. Find the unknown area of each of the following pairs of similar figures: (i) two triangles with corresponding sides 8 cm and 2 cm, and area of the larger 64 cm^2.

Step 1: A2/A1 = (2/8)^2 = 1/16. Step 2: A2 = 64/16. Answer: A2 = 4 cm^2.

Q1. (ii) two similar shapes of widths 0.4 m and 0.2 m, area of the smaller 0.6 m^2.

Step 1: A1/A2 = (0.4/0.2)^2 = 4. Step 2: A1 = 4 x 0.6. Answer: A1 = 2.4 m^2.

Q1. (iii) two rectangles: area of the first 125 cm^2 with long side 15 cm; long side of the second 6 cm.

Step 1: taking the 6 cm side as corresponding to the 15 cm long side, A2/A1 = (6/15)^2 = 4/25. Step 2: A2 = 125 x 4/25. Answer: A2 = 20 cm^2.

Q1. (iv) two similar shapes of widths 4 m and 6 m, area of the smaller 12 m^2.

Step 1: A2/A1 = (6/4)^2 = 9/4. Step 2: A2 = 12 x 9/4. Answer: A2 = 27 m^2.

Q2. Find the ratio of the areas of two circles whose radii are 4 cm and 7 cm.

Step 1: areas are proportional to the squares of the radii. Step 2: ratio = 4^2 : 7^2. Answer: 16 : 49.

Q3. A triangular plot of land PQR is such that PT = 6 m and PR = 10 m. ST and QR are two water pipes that are parallel to each other. The area of triangle PST is 24 m^2. Find the area of the land occupied by (i) triangle PQR, (ii) SQRT.

Step 1: triangles PST and PQR are similar with linear ratio 10/6 = 5/3. Step 2: (i) area of triangle PQR = 24 x (5/3)^2 = 24 x 25/9 = 200/3. Step 3: (ii) area of SQRT = 200/3 - 24 = 128/3. Answer: (i) 200/3 m^2 (66.67 approx); (ii) 128/3 m^2 (42.67 approx).

Q4. Find the unknown value in each of the following pairs of similar figures: (i) two triangles of areas 24 cm^2 and 6 cm^2, with corresponding sides a cm and 3 cm.

Step 1: (a/3)^2 = 24/6 = 4. Step 2: a/3 = 2. Answer: a = 6 cm.

Q4. (ii) two triangles of areas 10 cm^2 and 90 cm^2, with corresponding sides 5 cm and b cm.

Step 1: (b/5)^2 = 90/10 = 9. Step 2: b/5 = 3. Answer: b = 15 cm.

Q5. The perimeters of two similar regular hexagons are 10 m and 8 m. Given that the area of the larger hexagon is 200 m^2, find the area of the smaller hexagon.

Step 1: area ratio = (8/10)^2 = 16/25. Step 2: smaller area = 200 x 16/25. Answer: 128

m^2.

Q6. In the figure, triangle XYZ is an enlargement of triangle XMN. Given that XM = 6 cm and that the areas of triangles XMN and MYN are 14 cm^2 and 22 cm^2 respectively, find the length of MY.

Step 1: triangles XMN and MYN share the altitude from N to the line XY, so their areas are in the ratio of their bases XM and MY. Step 2: 6/MY = 14/22. Step 3: MY = 6 x 22/14 = 66/7. Answer: MY = 66/7 cm (9.43 approx).

Unit 9: Mathematical Logic and Polygons (continued: Exercise 9.3)

EXERCISE 9.3 (book pp.206-207)

Q1. Find the unknown volume of each of the following pairs of similar solids: (i) two cubes of sides 12 cm and 6 cm, volume of the smaller 72 cm^3.

Step 1: V1/V2 = (12/6)^3 = 8. Step 2: V1 = 72 x 8. Answer: V1 = 576 cm^3.

Q1. (ii) two cones of slant heights 7.5 cm and 2.5 cm, volume of the smaller 12 cm^3.

Step 1: V1/V2 = (7.5/2.5)^3 = 27. Step 2: V1 = 12 x 27. Answer: V1 = 324 cm^3.

Q2. Find the ratio of the volume of: (i) two similar solid cylinders of circumferences 10 cm and 8 cm; (ii) two similar solid cones of heights 9 cm and 12 cm.

Step 1: (i) volume ratio = (10/8)^3 = 1000/512 = 125/64. Step 2: (ii) volume ratio = (9/12)^3 = 729/1728 = 27/64. Answer: (i) 125 : 64; (ii) 27 : 64.

Q3. In a juice shop, a small bottle has a height of 6 cm while a large bottle has a height of 9 cm. If the capacity of the large bottle is 540 cm^3, find the capacity of the small bottle.

Step 1: capacity ratio = (6/9)^3 = 8/27. Step 2: small capacity = 540 x 8/27 = 160. Answer: 160 cm^3.

Q4. Find the unknown values in each of the following pairs of similar solids: (i) two cylinders of volumes 48 cm^3 and 6 cm^3, with lengths a cm and 2 cm.

Step 1: (a/2)^3 = 48/6 = 8. Step 2: a/2 = 2. Answer: a = 4 cm.

Q4. (ii) two cones of volumes 16 cm^3 and 54 cm^3, with slant heights 6 cm and b cm.

Step 1: (b/6)^3 = 54/16 = 27/8. Step 2: b/6 = 3/2. Answer: b = 9 cm.

Q5. The areas of the bases of two similar cones are in the ratio 9 : 16. (i) Find the ratio of the heights of the cones. (ii) Given that the volume of the larger cone is 448 cm^3, find the volume of the smaller cone.

Step 1: (i) base-area ratio is the square of the linear ratio, so the linear (height) ratio = 3 : 4. Step 2: (ii) volume ratio = 27 : 64, so smaller volume = 448 x 27/64 = 189. Answer: (i) 3 : 4; (ii) 189 cm^3.

Q6. Two solid spheres made of the same material have masses of 640 kg and 270 kg respectively. Determine the ratio of their diameters.

Step 1: same material, so volume ratio = mass ratio = 640 : 270 = 64 : 27. Step 2: linear ratio = cube root of 64/27. Answer: 4 : 3.

Q7. A bakery sells similar boxes of cookies in two sizes, small and large. The small box has a mass of 280 g and height of 15 cm. If the large box has a mass of 750 g, find its height.

Step 1: (h/15)^3 = 750/280 = 75/28. Step 2: h = 15 x (75/28)^(1/3). Answer: h = 20.83 cm (approx).

Q8. Two similar solid candy canes have heights 4 cm and 7 cm. (i) Find the ratio of the total surface areas of the candy canes. (ii) Given that the smaller candy cane has a mass of 10 g, find the mass of the larger candy cane.

Step 1: (i) surface-area ratio = (4/7)^2 = 16 : 49. Step 2: (ii) mass ratio = (4/7)^3 = 64 : 343, so larger mass = 10 x 343/64 = 53.59. Answer: (i) 16 : 49; (ii) 53.6 g (approx).

Q9. The volume of one sphere is 4 times that of another sphere. Given that the radius of the smaller sphere is 3 cm, find the radius of the larger sphere.

Step 1: linear ratio = cube root of 4. Step 2: larger radius = 3 x 4^(1/3). Answer: 4.76 cm (approx).

Q10. The mass of a glass figurine of height 6 cm is 500 g. Find the mass of a similar glass figurine if it has a height of 4 cm.

Step 1: mass ratio = (4/6)^3 = 8/27. Step 2: mass = 500 x 8/27. Answer: 148.15 g (approx).

Unit 9: Mathematical Logic and Polygons (continued: Exercise 9.4)

EXERCISE 9.4 (book p.218)

Q1. Find the value of the unknown in each of the following figures: (i) a right-angled triangle with right angle at B, angle at A 39 degrees, angle at C a degrees.

Step 1: a + 90 + 39 = 180. Answer: a = 51.

Q1. (ii) a triangle with angles 64 degrees, 68 degrees and 2b degrees.

Step 1: 2b + 64 + 68 = 180. Step 2: 2b = 48. Answer: b = 24.

Q1. (iii) a triangle with angles 40 degrees, 3c degrees and 4c degrees.

Step 1: 3c + 4c + 40 = 180. Step 2: 7c = 140. Answer: c = 20.

Q2. Given that DAB is a straight line, find the values of the unknown in each of the following figures: (i) a triangle with angle at C 55 degrees, angle at B 47 degrees, exterior angle at A a degrees.

Step 1: exterior angle = sum of the two opposite interior angles. Step 2: a = 55 + 47. Answer: a = 102.

Q2. (ii) a figure with CA perpendicular to DB, angle at D 50 degrees, angle at C split into b degrees and 35 degrees, angle at B c degrees.

Step 1: in the right-angled triangle DCA, b + 50 + 90 = 180, so b = 40. Step 2: in the right-angled triangle CBA, c + 35 + 90 = 180, so c = 55. Answer: b = 40, c = 55.

Q2. (iii) a triangle with exterior angle at A 110 degrees, interior angle at A d degrees, angle at C 2c degrees, angle at B 3c degrees.

Step 1: d = 180 - 110 = 70. Step 2: 70 + 2c + 3c = 180, so 5c = 110. Answer: c = 22, d = 70.

Q3. If the sizes of the angles of a triangle are 3x degrees, 4x degrees and 5x degrees, find the smallest angle of the triangle.

Step 1: 3x + 4x + 5x = 180, so 12x = 180 and x = 15. Step 2: smallest angle = 3x = 45. Answer: 45 degrees.

Q4. In the figure, ABC is a straight line. Given that angle ADB = angle BDC, find: (i) angle BDC, (ii) angle CBD. (The figure shows a right angle at A and angle at C 20 degrees.)

Step 1: in the right-angled triangle ADC, angle ADC = 180 - 90 - 20 = 70 degrees. Step 2: DB bisects angle ADC, so angle BDC = 70/2 = 35 degrees. Step 3: in triangle DBC, angle CBD = 180 - 20 - 35 = 125 degrees. Answer: (i) angle BDC = 35 degrees; (ii) angle CBD = 125 degrees.

Q5. Given that ABC and BED are straight lines, find the value(s) of the unknown(s) in each of the following figures: (i) angle at B between BA and BE is 115 degrees, EF is perpendicular to BC, FG is horizontal with angle at G 32 degrees, unknowns a degrees (angle BEF) and b degrees (angle GED).

Step 1: angle EBC = 180 - 115 = 65 degrees. Step 2: in the right-angled triangle BEF, a = 180 - 90 - 65 = 25 degrees. Step 3: in the right-angled triangle EFG, angle FEG = 180 - 90 - 32 = 58 degrees. Step 4: on the straight line BED, a + 58 + b = 180, so b = 180 - 25 - 58 = 97 degrees. Answer: a = 25, b = 97.

Q5. (ii) ABC and BED are straight lines, E lies on BD, angle at A is 22 degrees, angle at C (BCD) is 89 degrees, angle at D (CDB) is 27 degrees, unknown c degrees (angle AED).

Step 1: in triangle BCD, angle CBD = 180 - 89 - 27 = 64 degrees. Step 2: on the straight line ABC, angle ABE = 180 - 64 = 116 degrees. Step 3: in triangle ABE, angle AEB = 180 - 22 - 116 = 42 degrees. Step 4: on the straight line BED, c = 180 - 42 = 138 degrees. Answer: c = 138.

Unit 9: Mathematical Logic and Polygons (continued: Exercise 9.4, Q6-15)

EXERCISE 9.4 continued (book pp.219-220)

Q6. If the sizes of the angles of a triangle are (x - 35) degrees, (x - 25) degrees and (x/2 - 10) degrees, find the value of x.

Step 1: (x - 35) + (x - 25) + (x/2 - 10) = 180. Step 2: (5/2)x - 70 = 180, so (5/2)x = 250. Answer: x = 100.

Q7. In triangle ABC, angle BAC = 50 degrees and angle BCA = 60 degrees. (i) Find angle ABC. (ii) If AB is produced to D, find angle CBD.

Step 1: (i) angle ABC = 180 - 50 - 60 = 70 degrees. Step 2: (ii) on the straight line ABD, angle CBD = 180 - 70 = 110 degrees. Answer: (i) 70 degrees; (ii) 110 degrees.

Q8. In the figure, ACE and BCD are straight lines. Angle at A is 50 degrees, and the top triangle has angles 61 degrees and 41 degrees. Find: (i) angle ACB, (ii) angle ABC.

Step 1: in the top triangle, angle DCE = 180 - 61 - 41 = 78 degrees. Step 2: (i) angle ACB = angle DCE (vertically opposite) = 78 degrees. Step 3: (ii) in triangle ABC, angle ABC = 180 - 50 - 78 = 52 degrees. Answer: (i) 78 degrees; (ii) 52 degrees.

Q9. Find the values of the unknowns in each of the following parallelograms: (i) angle at A 106 degrees, angle DBC 48 degrees, angle at C a degrees, angle CDB b degrees.

Step 1: opposite angles of a parallelogram are equal, so a = 106. Step 2: in triangle BCD, b = 180 - 106 - 48 = 26. Answer: a = 106, b = 26.

Q9. (ii) a parallelogram with angle at D 50 degrees, angle at C 2d degrees, angle at A 4c degrees.

Step 1: opposite angles are equal, so 4c = 2d. Step 2: adjacent angles are supplementary, so 4c + 50 = 180, giving 4c = 130. Answer: c = 32.5, d = 65.

Q10. The figure shows a rhombus ABCD where the diagonals AC and BD intersect at E. Angle CDB is (3x + 7) degrees and angle BAC is (2x + 53) degrees. Find the value of x.

Step 1: the diagonals of a rhombus are perpendicular, so in the right-angled triangle formed by the diagonals, the two acute angles sum to 90 degrees. Step 2: angle BAC = angle DCA (alternate interior angles, AB parallel to CD). Step 3: (2x + 53) + (3x + 7) = 90, so 5x + 60 = 90 and 5x = 30. Answer: x = 6.

Q11. The figure shows a trapezium ABCD where AB is parallel to CD (the book prints 'AB parallel to BE', a misprint — the parallel arrow marks are on AB and

DC). Angle at D is 5x degrees, at C is y degrees, at A is x degrees, at B is 2.2x degrees. Find the value of x and of y.

Step 1: co-interior angles between the parallel sides: x + 5x = 180, so x = 30. Step 2: 2.2x + y = 180, so y = 180 - 66 = 114. Answer: x = 30, y = 114.

Q12. Find the value of the unknown in each of the following figures: (i) a quadrilateral with angles 84 degrees, 2b degrees, b degrees and 78 degrees.

Step 1: 84 + 2b + b + 78 = 360. Step 2: 3b = 198. Answer: b = 66.

Q12. (ii) a hexagon with angles 108 degrees, 4d degrees, 4d degrees, 5d degrees, 102 degrees and 4d degrees.

Step 1: sum of interior angles of a hexagon = (6 - 2) x 180 = 720 degrees. Step 2: 108 + 4d + 4d + 5d + 102 + 4d = 720, so 17d = 510. Answer: d = 30.

Q13. Find the number of sides of a regular polygon if each exterior angle of the polygon is: (i) 90 degrees; (ii) 45 degrees.

Step 1: n = 360/(exterior angle). Step 2: (i) n = 360/90 = 4. Step 3: (ii) n = 360/45 = 8. Answer: (i) 4 sides; (ii) 8 sides.

Q14. Basement of a godown is in the shape of a regular octagon having side length 10 m and apothem 8 m. Find the perimeter and area of the godown.

Step 1: perimeter = 8 x 10 = 80 m. Step 2: area = (1/2) x perimeter x apothem = (1/2) x 80 x 8 = 320. Answer: perimeter = 80 m; area = 320 m^2.

Q15. A garden is in the shape of an equilateral triangle with side length 12 m. Find its perimeter and also find the cost of fencing at Rs.300 per metre.

Step 1: perimeter = 3 x 12 = 36 m. Step 2: cost = 36 x 300 = 10800. Answer: perimeter = 36 m; cost of fencing = Rs.10,800.

Unit 9: Mathematical Logic and Polygons (continued:

Exercise 9.5)

EXERCISE 9.5 (book pp.225-226)

Q1. Construct, on separate diagrams: (i) the locus of a point that moves so that it is constantly 4 cm from a fixed point A; (ii) the locus of points less than 4 cm from the fixed point A.

Step 1: (i) the locus is the circle of radius 4 cm with centre A. Step 2: (ii) the locus is the interior of that circle (the disc), not including the circumference. Answer: (i) circle radius 4 cm centred at A; (ii) the disc inside that circle.

Q2. A wooden panel is in the shape of triangle ABC, where AB = 4.5 m, BC = 6.5 m and CA = 4 m. NOTE: as printed, Q2 states only the panel's shape and side lengths and gives no further instruction; the notes give the SSS construction.

Step 1: draw AB = 4.5 m. Step 2: with centre A and radius 4 m, and centre B and radius 6.5 m, draw arcs meeting at C. Step 3: join AC and BC. Answer: triangle ABC constructed by SSS (its area by Heron's formula is about 8.87 m^2).

Q3. Two points P and Q are 50 m apart on the south bank of a river flowing due east between parallel banks. R is a point on the north bank such that the bearing of R from P is 042 degrees and the bearing of Q from R is 140 degrees. (i) Using a scale of 2 cm to 10 m, construct triangle PQR. Measure PR and find, correct to the nearest metre, the width of the river.

Step 1: angle at P between PQ (east) and PR (bearing 042 degrees) is 90 - 42 = 48 degrees. Step 2: angle at R between RP (bearing 222 degrees) and RQ (bearing 140 degrees) is 82 degrees. Step 3: angle at Q is 180 - 48 - 82 = 50 degrees. Step 4: by the sine rule, PR = 50 x sin 50 degrees/sin 82 degrees = 38.68 m. Step 5: width = PR x sin 48 degrees = 28.74 m. Answer: PR = 39 m (approx); width of the river = 29 m (nearest metre).

Q3. (ii) A boat B1 is 45 m from P and is equidistant from Q and R. On your diagram, draw the locus which represents points which are: (a) 45 m from P; (b) equidistant from Q and R.

Step 1: (a) the locus is the circle of radius 45 m centred at P. Step 2: (b) the locus is the perpendicular bisector of QR. Step 3: B1 is the intersection of the two loci on the river (at about 41.5 m east and 17.4 m north of P). Answer: B1 is the intersection of the circle (radius 45 m about P) with the perpendicular bisector of QR.

Q3. (iii) A second boat B2 is 15 m from QR and is equidistant from PR and QR. On your diagram, draw the locus which represents points which are: (a) 15 m from QR; (b) equidistant from PR and QR.

Step 1: (a) the locus is the pair of lines parallel to QR at a distance of 15 m. Step 2: (b) the locus is the angle bisector of angle PRQ. Step 3: B2 is the intersection of the two loci in the river (at about 25.5 m east and 5.9 m north of P). Answer: B2 is the intersection of a line parallel to QR at 15 m with the bisector of angle PRQ.

Q3. (iv) By measurement, find the distance between the two boats correct to the nearest metre.

Step 1: from the constructed positions, B1 is about (41.5, 17.4) and B2 about (25.5, 5.9) in metres from P. Step 2: distance = sqrt((41.5 - 25.5)^2 + (17.4 - 5.9)^2) = 19.72 m. Answer: about 20 m (nearest metre).

Q4. A playground is in the shape of a quadrilateral ABCD. It is given that AB = 90 m, AD = 70 m, angle BAD = 115 degrees, angle ABC = 100 degrees and angle ADC = 85 degrees. (i) Using a scale of 1 cm to represent 10 m, make an accurate scale drawing of the playground.

Step 1: draw AB = 9 cm. Step 2: at A draw AD = 7 cm at 115 degrees to AB. Step 3: at B draw BC at 100 degrees to BA, meeting the ray from D that makes 85 degrees at D, giving BC about 11.6 cm (116 m). Step 4: join CD. Answer: quadrilateral ABCD drawn to the scale 1 cm : 10 m.

Q4. (ii) On your diagram, draw (a) the locus which represents points 80 m from B; (b) the locus which represents points equidistant from C and D. Label the point of intersection of these loci with the letter M.

Step 1: (a) the locus is the circle of radius 80 m (8 cm on the drawing) centred at B. Step 2: (b) the locus is the perpendicular bisector of CD. Step 3: their intersection on the playground is M (about 47.9 m east and 68.0 m north of A). Answer: M is the intersection of the circle about B with the perpendicular bisector of CD.

Q4. (iii) A point N on the playground is equidistant from BC and CD and is such that angle ANB = 90 degrees. Represent this point N on your diagram by again using the intersection of two loci.

Step 1: points equidistant from BC and CD lie on the angle bisector of angle BCD. Step 2: points with angle ANB = 90 degrees lie on the circle with diameter AB. Step 3: their intersection on the playground is N (about 51.6 m east and 44.5 m north of A). Answer: N is the intersection of the bisector of angle BCD with the circle on AB as diameter.

Q4. (iv) Measure and write down the length MN.

Step 1: from the constructed positions, M is about (47.9, 68.0) and N about (51.6, 44.5) in metres from A. Step 2: MN = sqrt((51.6 - 47.9)^2 + (44.5 - 68.0)^2) = 23.79 m. Answer: MN = about 24 m (about 2.4 cm on the scale drawing).

Q4. (v) A square region on the playground, if represented on the scale drawing, would have an area of 5 cm^2. Calculate the length, in metres, of a side of the actual square.

Step 1: scale 1 cm = 10 m, so 1 cm^2 = 100 m^2. Step 2: actual area = 5 x 100 = 500 m^2. Step 3: side = sqrt(500). Answer: about 22.36 m.

Q5. A street authority wants to install streetlights at a uniform distance of 3 meters from the edges of a straight road. Find the number of locations where streetlights should be placed, if the street is 30 m long. NOTE: as printed, no spacing between consecutive lights is given, so no definite number follows; the notes give the locus.

Step 1: the locus of points 3 m from a straight road is the pair of lines parallel to the road at 3 m on either side. Step 2: along the 30 m street these are two 30 m lines, one on each side. Answer: the streetlights lie on the two lines 3 m from the edges, running the 30 m length; the question as printed gives no spacing between lights, so the number of locations is not fixed by the data.

Q6. A park has a walking track running in a straight line, and a fence needs to be installed 5 m away from the track on both sides. Where should the fence be placed?

Step 1: the locus of points 5 m from a straight line is the pair of lines parallel to it at 5 m. Answer: the fence is placed on the two lines parallel to the track, 5 m away on both sides.

Unit 9: Mathematical Logic and Polygons (continued: Review Exercise 9)

REVIEW EXERCISE 9 (book pp.226-227)

Q1. (i) A statement which is either true or false is:

(a) axiom

(b) mathematical statement ✓

(c) theorem

(d) conjecture

Q1. (ii) A statement that has not yet been rigorously proved is known as:

(a) theorem

(b) axiom

(c) conjecture ✓

(d) none of these

Q1. (iii) Sum of angles in a triangle is 180 degrees is known as:

(a) theorem ✓

(b) conjecture

(c) axiom

(d) none of these

Q1. (iv) Sum of interior angles of a regular hexagon is:

(a) 540 degrees

(b) 720 degrees ✓

(c) 900 degrees

(d) 1080 degrees

Q1. (v) Each interior angle of a regular octagon is:

(a) 45 degrees

(b) 90 degrees

(c) 135 degrees ✓

(d) 180 degrees

Q1. (vi) If the diagonals of a quadrilateral bisect each other at right angles, then the quadrilateral is:

(a) rectangle

(b) square

(c) parallelogram

(d) rhombus ✓

Q1. (vii) The sum of exterior angles of a regular polygon is:

(a) 90 degrees

(b) 180 degrees

(c) 270 degrees

(d) 360 degrees ✓

Q1. (viii) The locus of points that are equidistant from a given straight line is:

(a) a line parallel to the given line ✓

(b) a pair of perpendicular bisectors

(c) a circle

(d) a triangle

Q1. (ix) The locus of a point that is always at a fixed distance from a given point is:

(a) a straight line

(b) a parabola

(c) a circle ✓

(d) a hyperbola

Q1. (x) The locus of points equidistant from two intersecting lines is:

(a) a straight line

(b) an angle bisector ✓

(c) perpendicular bisector

(d) parallel line to one of the given lines

Q2. Find the ratio of the areas of two similar triangles if the lengths of their corresponding sides are: (i) 3 cm and 5 cm; (ii) 4.5 m and 9 m; (iii) 2 mm and 3 mm.

Step 1: area ratio is the square of the side ratio. Step 2: (i) (3/5)^2 = 9 : 25. Step 3: (ii) (4.5/9)^2 = 1 : 4. Step 4: (iii) (2/3)^2 = 4 : 9. Answer: (i) 9 : 25; (ii) 1 : 4; (iii) 4 : 9.

Q3. Find the ratio of the areas of two similar triangles if their perimeters are 294 cm and 336 cm.

Step 1: linear ratio = 294/336 = 7/8. Step 2: area ratio = (7/8)^2 = 49/64. Answer: 49 : 64.

Q4. (i) If the length of each side of a regular pentagon is doubled, what will happen to its area? (ii) Given that the length of each side of a pentagon, whose area is 25 cm^2, is doubled, find the area of the enlarged pentagon. (iii) If the length of each side of a regular n-sided polygon is tripled, what will happen to its area?

Step 1: (i) area is multiplied by 2^2 = 4. Step 2: (ii) enlarged area = 25 x 4 = 100 cm^2. Step 3: (iii) area is multiplied by 3^2 = 9. Answer: (i) it becomes 4 times; (ii) 100 cm^2; (iii) it becomes 9 times.

Q5. The volumes of two similar cylinders are in the ratio 8 : 27. Find the ratio of the base radii of the cylinders.

Step 1: linear ratio = cube root of 8/27 = 2/3. Answer: 2 : 3.

Q6. The volumes of two similar water jugs are in the ratio 27 : 64. Find: (i) the ratio of the heights of the jugs; (ii) the ratio of the total surface areas of the jugs.

Step 1: (i) linear ratio = cube root of 27/64 = 3/4. Step 2: (ii) surface-area ratio = (3/4)^2 = 9/16. Answer: (i) 3 : 4; (ii) 9 : 16.

Q7. A stop sign is designed in the shape of a regular octagon. What is the measure of each interior angle, and if each side of the octagon is 50 cm and its apothem is 40 cm, what is its area?

Step 1: each interior angle = (8 - 2) x 180/8 = 135 degrees. Step 2: perimeter = 8 x 50 = 400 cm. Step 3: area = (1/2) x 400 x 40 = 8000. Answer: interior angle = 135 degrees; area = 8000 cm^2.

Q8. A gardener wants to design a pentagon shaped flower garden where each side is 6 m. What is the measure of each interior angle, and find the area of the garden if its apothem is 4 m.

Step 1: each interior angle = (5 - 2) x 180/5 = 108 degrees. Step 2: perimeter = 5 x 6 = 30 m. Step 3: area = (1/2) x 30 x 4 = 60. Answer: interior angle = 108 degrees; area = 60 m^2.

Q9. Umair has a hexagonal plot having length of each side 50 m and apothem 42 m. He wants to fix pentagonal tiles with length of each side 12 cm and apothem 10 cm. (i) How many tiles are required for the plot? (ii) If the cost of fixing a tile is Rs.50, what is the total cost to fix the tiles of the plot?

Step 1: area of the plot = (1/2) x (6 x 50) x 42 = 6300 m^2. Step 2: area of one tile = (1/2) x (5 x 12) x 10 = 300 cm^2 = 0.03 m^2. Step 3: (i) number of tiles = 6300/0.03 = 210000. Step 4: (ii) total cost = 210000 x 50 = 10500000. Answer: (i) 210,000 tiles; (ii) Rs.10,500,000.

Q10. A city Mayor wants to build a middle school that is equidistant from two existing schools, P and Q. How should the location be chosen?

Step 1: the locus of points equidistant from P and Q is the perpendicular bisector of PQ. Answer: the school should be built anywhere on the perpendicular bisector of the segment PQ.

Q11. A drone is programmed to stay 50 m away from its charging station. What is the possible flight path of the drone?

Step 1: the locus of points 50 m from a fixed point is a circle of radius 50 m. Answer: the circle of radius 50 m centred at the charging station.

Q12. A security camera needs to be installed at a hallway intersection such that it is equidistant from both walls forming the intersection. Where should the camera be placed?

Step 1: the locus of points equidistant from two intersecting lines is the angle bisector.

Answer: the camera should be placed on the angle bisector of the two walls.

Unit 10: Practical Geometry of Triangles

EXERCISE 10.1 (book p.233)

Q1. Construct a triangle ABC such that mAB = 7.5 cm, mBC = 5.5 cm and angle B = 45 degrees.

Step 1: draw BC = 5.5 cm. Step 2: at B construct angle 45 degrees. Step 3: cut off BA = 7.5 cm on the ray. Step 4: join A with C. Answer: triangle ABC constructed by SAS.

Q2. Construct an isosceles triangle having each of its sides equal to 6.5 cm and the vertical angle of 60 degrees.

Step 1: draw one side = 6.5 cm. Step 2: at one end construct the vertical angle 60 degrees. Step 3: cut off the second side = 6.5 cm on the ray (with a 60 degree vertical angle the triangle is in fact equilateral). Step 4: join the ends. Answer: isosceles triangle with sides 6.5 cm and vertical angle 60 degrees constructed.

Q3. Construct triangle ABC such that mAB = 8.3 cm, angle A = 45 degrees and angle B = 75 degrees.

Step 1: draw AB = 8.3 cm. Step 2: at A construct angle 45 degrees and at B construct angle 75 degrees. Step 3: the rays meet at C. Answer: triangle ABC constructed by ASA (angle C = 60 degrees).

Q4. Construct triangle ABC such that mAB = 3.3 cm, mBC = 4.3 cm and mCA = 5.5 cm.

Step 1: draw AB = 3.3 cm. Step 2: with centre A and radius 5.5 cm, and centre B and radius 4.3 cm, draw arcs meeting at C. Step 3: join AC and BC. Answer: triangle ABC constructed by SSS.

Q5. Construct triangle ABC such that mAB = 2.5 cm, angle A = 40 degrees and angle B = 110 degrees.

Step 1: draw AB = 2.5 cm. Step 2: at A construct angle 40 degrees and at B construct angle 110 degrees. Step 3: the rays meet at C. Answer: triangle ABC constructed by ASA (angle C = 30 degrees).

Q6. Construct triangle ACB, if AC = 3 cm, BC = 4 cm and angle C = 90 degrees.

Step 1: draw AC = 3 cm. Step 2: at C construct the right angle. Step 3: cut off CB = 4 cm on the ray. Step 4: join A with B. Answer: right-angled triangle ACB constructed by SAS (AB = 5 cm by Pythagoras).

Q7. Construct triangle RST, if angle R = 60 degrees, RS = 7 cm and angle S = 105 degrees.

Step 1: draw RS = 7 cm. Step 2: at R construct angle 60 degrees and at S construct angle 105 degrees. Step 3: the rays meet at T. Answer: triangle RST constructed by ASA (angle T = 15 degrees).

Q8. Construct triangle XYZ such that XY = 4 cm, XZ = 5 cm and angle Y = 45 degrees.

Step 1: at Y construct angle 45 degrees and cut off YX = 4 cm on one ray. Step 2: with centre X and radius 5 cm draw an arc cutting the other ray at Z. Step 3: join X with Z. Answer: one triangle XYZ (SSA case; the second intersection of the arc falls on the extension of the ray beyond Y, so it does not give a second triangle).

EXERCISE 10.2 (book pp.236-237)

Q1. Show that the angle bisectors of the given triangles are concurrent: (i) mAB = 4.7 cm, mBC = 3.6 cm and mCA = 5.4 cm.

Step 1: construct triangle ABC by SSS. Step 2: draw the bisectors of angles B and C meeting at I. Step 3: draw the bisector of angle A; it also passes through I. Answer: the three angle bisectors are concurrent at the incentre I.

Q1. (ii) mAB = 3.8 cm, mBC = 4.5 cm and angle B = 60 degrees.

Step 1: construct triangle ABC by SAS. Step 2: draw the bisectors of angles B and C meeting at I. Step 3: draw the bisector of angle A; it also passes through I. Answer: the three angle bisectors are concurrent at the incentre I.

Q2. Draw altitudes of the triangle with given dimensions and show that they are concurrent: (i) mPQ = 5.9 cm, mQR = 4.6 cm and mPR = 5.8 cm.

Step 1: construct triangle PQR by SSS. Step 2: draw two altitudes meeting at O. Step 3: draw the third altitude; it also passes through O. Answer: the three altitudes are concurrent at the orthocentre O.

Q2. (ii) mPQ = 4.5 cm, angle P = 60 degrees and angle Q = 45 degrees.

Step 1: construct triangle PQR by ASA (angle R = 75 degrees). Step 2: draw two altitudes meeting at O. Step 3: draw the third altitude; it also passes through O. Answer: the three altitudes are concurrent at the orthocentre O.

Q3. Show that the perpendicular bisectors of the given triangles are concurrent: (i) mAB = 5.6 cm, angle A = 60 degrees, angle B = 60 degrees.

Step 1: construct triangle ABC by ASA (angle C = 60 degrees, so the triangle is equilateral). Step 2: draw the perpendicular bisectors of two sides meeting at O. Step 3: draw the perpendicular bisector of the third side; it also passes through O. Answer: the three perpendicular bisectors are concurrent at the circumcentre O.

Q3. (ii) mAB = 2.6 cm, mAC = 3.7 cm, angle A = 120 degrees.

Step 1: construct triangle ABC by SAS. Step 2: draw the perpendicular bisectors of two sides meeting at O (outside the triangle, since angle A is obtuse). Step 3: draw the perpendicular bisector of the third side; it also passes through O. Answer: the three perpendicular bisectors are concurrent at the circumcentre O.

Q4. Draw three medians of the following triangles and show that they are

concurrent: (i) mYZ = 4.4 cm, angle Y = 45 degrees and angle Z = 75 degrees.

Step 1: construct triangle XYZ by ASA (angle X = 60 degrees). Step 2: mark the midpoints of the sides and join each vertex to the midpoint of the opposite side. Step 3: the third median also passes through the meeting point G. Answer: the three medians are concurrent at the centroid G, dividing each other in the ratio 2 : 1.

Q4. (ii) mXY = 4.6 cm, mXZ = 3.6 cm and angle Y = 60 degrees.

Step 1: construct triangle XYZ (at Y construct angle 60 degrees, cut off YX = 4.6 cm, then with centre X and radius 3.6 cm draw an arc cutting the other ray at Z). Step 2: mark the midpoints of the sides and join each vertex to the midpoint of the opposite side. Step 3: the third median also passes through the meeting point G. Answer: the three medians are concurrent at the centroid G, dividing each other in the ratio 2 : 1.

REVIEW EXERCISE 10 (book p.237)

Q1. (i) A triangle having all the three sides congruent is called:

(a) Scalene

(b) right angled

(c) equilateral ✓

(d) isosceles

Q1. (ii) The medians of a triangle are:

(a) collinear

(b) congruent

(c) concurrent ✓

(d) parallel

Q1. (iii) The _____ altitudes of an equilateral triangle are congruent.

(a) two

(b) three ✓

(c) four

(d) none

Q1. (iv) The medians of a triangle divide it into _____ triangles.

(a) three

(b) four

(c) five

(d) six ✓

Q1. (v) If two _____ of a triangle are congruent, then the triangle will be isosceles.

(a) Altitudes

(b) Medians

(c) Right bisectors

(d) sides ✓

Q2. Construct triangle PQR if mPQ = 6 cm, angle P = 60 degrees, angle Q = 45 degrees. Draw median and altitude.

Step 1: construct triangle PQR by ASA (angle R = 75 degrees). Step 2: draw a median by joining a vertex to the midpoint of the opposite side. Step 3: draw an altitude by drawing the perpendicular from a vertex to the opposite side. Answer: triangle PQR constructed with a median and an altitude drawn.

Q3. Construct a triangle DEF, also draw angle bisectors, median and altitude, where mDE = 6 cm, angle D = 60 degrees and angle E = 60 degrees.

Step 1: construct triangle DEF by ASA (angle F = 60 degrees, so the triangle is equilateral). Step 2: draw the angle bisectors; they meet at the incentre. Step 3: draw a median and an altitude (in this equilateral triangle they coincide with the bisectors). Answer: triangle DEF constructed with angle bisectors, median and altitude drawn.

Q4. Construct a triangle XYZ, where mXY = 5.5 cm, angle X = 45 degrees and angle Y = 60 degrees. Also draw their perpendicular bisectors and verify the concurrency.

Step 1: construct triangle XYZ by ASA (angle Z = 75 degrees). Step 2: draw the perpendicular bisectors of two sides meeting at O. Step 3: draw the perpendicular bisector of the third side; it also passes through O. Answer: the three perpendicular bisectors are concurrent at the circumcentre O.

Q5. Construct a triangle FGH where FG = 6 cm, GH = 5 cm and FH = 4.6 cm. Draw their median and verify the concurrency.

Step 1: construct triangle FGH by SSS. Step 2: draw two medians meeting at G (the centroid). Step 3: draw the third median; it also passes through the same point. Answer: the three medians are concurrent at the centroid, dividing each other in the ratio 2 : 1.

Unit 11: Basic Statistics and Probability

EXERCISE 11.1 (book p.246)

Q1. The following is a record of weights of 40 students in pounds (lbs). Tabulate the data in the form of frequency distribution by taking 6 classes. Also find cumulative frequency. Data: 72, 96, 69, 78, 76, 112, 93, 107, 73, 61, 75, 91, 106, 84, 103, 109, 96, 88, 80, 86, 90, 105, 77, 90, 101, 91, 102, 92, 91, 82, 99, 63, 95, 118, 77, 72, 114, 101, 113, 105.

Step 1: minimum = 61, maximum = 118, range = 57, class interval = 57/6 = 9.5, taken as 10. Step 2: classes 60-69: 3; 70-79: 8; 80-89: 5; 90-99: 11; 100-109: 9; 110-119: 4; total 40. Step 3: cumulative frequencies: 3, 11, 16, 27, 36, 40. Answer: frequency distribution as above, with cumulative frequencies 3, 11, 16, 27, 36, 40.

Q2. The marks in Mathematics of 50 students are given as under: 68, 93, 74, 94, 66, 76, 59, 71, 53, 75, 71, 75, 88, 90, 85, 60, 71, 82, 73, 88, 82, 96, 65, 78, 62, 75, 65, 72, 60, 69, 83, 76, 81, 78, 76, 63, 97, 74, 85, 57, 67, 77, 68, 75, 88, 61, 73, 95, 78, 73. (i) Construct frequency distribution by taking 5 as class interval. (ii) Draw histogram obtained in part (i).

Step 1: minimum = 53, maximum = 97. Step 2: classes of width 5: 50-54: 1; 55-59: 2; 60-64: 5; 65-69: 7; 70-74: 9; 75-79: 11; 80-84: 4; 85-89: 5; 90-94: 3; 95-99: 3; total 50. Step 3: (ii) the histogram has class boundaries on the x-axis and bars of heights 1, 2, 5, 7, 9, 11, 4, 5, 3, 3. Answer: frequency distribution as above; histogram as described.

Q3. The following data give the index numbers of 45 commodities in a certain year: 120, 138, 113, 119, 111, 118, 123, 151, 134, 141, 115, 119, 124, 112, 145, 114, 106, 112, 110, 106, 111, 153, 100, 101, 108, 133, 100, 109, 110, 143, 109, 138, 113, 136, 100, 104, 144, 128, 105, 106, 147, 127, 140, 129, 108. (i) Construct a frequency distribution by taking the classes as 100-109, 110-119 etc.

Step 1: classes 100-109: 13; 110-119: 13; 120-129: 6; 130-139: 5; 140-149: 6; 150-159: 2; total 45. Answer: frequency distribution as above.

Q4. Construct histogram for the data given below: Marks 16-25: 4 students; 26-35: 8; 36-45: 15; 46-55: 7; 56-65: 6.

Step 1: the class intervals are all equal (10), so bar heights equal the frequencies. Step 2: class boundaries are 15.5-25.5, 25.5-35.5, 35.5-45.5, 45.5-55.5, 55.5-65.5. Answer: histogram with bars of heights 4, 8, 15, 7, 6 over the five classes.

EXERCISE 11.2 (book p.252)

Q1. Find arithmetic mean by (a) direct method (b) indirect method for the following data: (i) 15, 29, 25, 18, 45, 34, 39, 32, 41, 26.

Step 1: (a) direct: sum = 304, n = 10, mean = 304/10 = 30.4. Step 2: (b) indirect: take assumed mean a = 30; deviations sum to 4; mean = 30 + 4/10 = 30.4. Answer: mean = 30.4.

Q1. (ii) 145, 240, 275, 185, 195, 220, 295, 300.

Step 1: (a) direct: sum = 1855, n = 8, mean = 1855/8 = 231.875. Step 2: (b) indirect: take assumed mean a = 200; deviations sum to 255; mean = 200 + 255/8 = 231.875. Answer: mean = 231.875.

Q2. Find arithmetic mean of the following marks obtained by students of 10th class in a test: classes 25-29: 8; 30-34: 13; 35-39: 19; 40-44: 14; 45-49: 10; 50-54: 4.

Step 1: midpoints are 27, 32, 37, 42, 47, 52. Step 2: sum f = 68; sum fx = 216 + 416 + 703 + 588 + 470 + 208 = 2601. Step 3: mean = 2601/68 = 38.25. Answer: mean = 38.25.

Q3. Find mean, median and mode of the runs scored by 11 players: 45, 58, 61, 35, 29, 37, 58, 55, 48, 57, 42.

Step 1: mean = 525/11 = 47.73. Step 2: in ascending order: 29, 35, 37, 42, 45, 48, 55, 57, 58, 58, 61; median (6th value) = 48. Step 3: 58 occurs twice, more than any other value. Answer: mean = 47.73; median = 48; mode = 58.

Q4. A health survey recorded the weights (in pounds) of 60 adults in a factory and grouped them as follows: 65-84: 3; 85-104: 8; 105-124: 19; 125-144: 14; 145-164: 9; 165-184: 5. Find median and mode of the above data.

Step 1: total frequency = 58 (as grouped; the survey states 60 adults). Step 2: class boundaries: 64.5-84.5, 84.5-104.5, 104.5-124.5, 124.5-144.5, 144.5-164.5, 164.5-184.5; cumulative frequencies 3, 11, 30, 44, 53, 58. Step 3: median class is 105-124: median = 104.5 + (20/19)(29 - 11) = 123.45. Step 4: modal class is 105-124: mode = 104.5 + (19 - 8) x 20/((19 - 8) + (19 - 14)) = 118.25. Answer: median = 123.45; mode = 118.25.

Q5. A teacher records final exam marks of 100 students grouped as: 10-19: 5; 20-29: 25; 30-39: 40; 40-49: 20; 50-59: 10. Find the mean, median and mode of the above data.

Step 1: midpoints 14.5, 24.5, 34.5, 44.5, 54.5; sum fx = 72.5 + 612.5 + 1380 + 890 + 545 = 3500; mean = 3500/100 = 35. Step 2: cumulative frequencies 5, 30, 70, 90, 100; median class 30-39: median = 29.5 + (10/40)(50 - 30) = 34.5. Step 3: modal class 30-39: mode = 29.5 + (40 - 25) x 10/((40 - 25) + (40 - 20)) = 33.79. Answer: mean = 35; median = 34.5; mode = 33.79.

Unit 11: Basic Statistics and Probability (continued: Exercise 11.2 Q6-7 and Exercise 11.3)

EXERCISE 11.2 continued (book p.253)

Q6. A person purchased the following food items: Rice 10 kg at Rs.2500 per kg; Flour 13 kg at Rs.1950 per kg; Ghee 15 kg at Rs.2200 per kg; Sugar 4 kg at Rs.520 per kg; Mutton 3 kg at Rs.5500 per kg. What is the average cost of food items per kg?

Step 1: total quantity = 10 + 13 + 15 + 4 + 3 = 45 kg. Step 2: total cost = 25000 + 25350 + 33000 + 2080 + 16500 = 101930. Step 3: average cost = 101930/45 = 2265.11. Answer: Rs.2265.11 per kg (weighted mean).

Q7. Following table lists the size of families: size 2: 4; 3: 50; 4: 35; 5: 27; 6: 13; 7: 5; 8: 2. Calculate mean, median and mode. NOTE: the table heading says 127 families, but the frequencies add to 136; the calculations use the frequencies as printed.

Step 1: total families = 4 + 50 + 35 + 27 + 13 + 5 + 2 = 136. Step 2: mean = (8 + 150 + 140 + 135 + 78 + 35 + 16)/136 = 562/136 = 4.13. Step 3: cumulative frequencies 4, 54, 89, 116, 129, 134, 136; the 68th and 69th families are both in size 4. Step 4: size 3 has the highest frequency 50. Answer: mean = 4.13; median = 4; mode = 3.

EXERCISE 11.3 (book pp.255-256)

Q1. The mean of a set of 6 numbers is 2 and the mean of another set of 10 numbers is x. If the mean of all 16 numbers is 7, find the value of x.

Step 1: sum of the first set = 6 x 2 = 12; sum of the second set = 10x. Step 2: (12 + 10x)/16 = 7, so 12 + 10x = 112 and 10x = 100. Answer: x = 10.

Q2. The numbers 24, 22, 34, 28, 29, 24, 25, 29, a and b have a median of 27 and a mode of 29. Given that a < b, state the value of a and of b.

Step 1: with a = 26 and b = 29 the sorted list is 22, 24, 24, 25, 26, 28, 29, 29, 29, 34. Step 2: median = (26 + 28)/2 = 27. Step 3: 29 occurs three times, more than any other value. Answer: a = 26, b = 29.

Q3. 2000 people responded to the blood bank's appeal for blood donors. The blood type, i.e. A, B, AB or O, of each donor is recorded: A: h; B: 800; AB: 200; O: k. (a) (i) Write down an equation in terms of h and k. (ii) Given that 9h = k, find the value of h and of k. (b) Using your answers in (a) (ii), (i) draw a bar graph to illustrate the data.

Step 1: (a) (i) h + 800 + 200 + k = 2000, so h + k = 1000. Step 2: (a) (ii) with k = 9h, 10h

= 1000, so h = 100 and k = 900. Step 3: (b) (i) bar graph with bars A: 100, B: 800, AB: 200, O: 900. Answer: (a) (i) h + k = 1000; (a) (ii) h = 100, k = 900; (b) bar graph as described.

Q4. A box contains five cards numbered 1, 2, 3, 4 and 5. A card is drawn from the box, its number noted and then replaced. The process is repeated 100 times. The table shows the resulting frequency distribution: card 1: 21; 2: x; 3: y; 4: 18; 5: 17. (i) Show that x + y = 44. (ii) If the mean of the distribution is 2.9, show that 2x + 3y = 112. (iii) Hence, find the value of x and of y.

Step 1: (i) total frequency: 21 + x + y + 18 + 17 = 100, so x + y = 44. Step 2: (ii) total of values = 21 + 2x + 3y + 72 + 85 = 178 + 2x + 3y = 2.9 x 100 = 290, so 2x + 3y = 112. Step 3: (iii) subtracting twice (i) from (ii): y = 24, so x = 20. Answer: (i) x + y = 44; (ii) 2x + 3y = 112; (iii) x = 20, y = 24.

Q5. Three netball teams each played five games. Scores: Cheetah 65, 95, 32, 96, 88; Jaguar 50, 90, 65, 87, 87; Puma 90, 85, 46, 44, 80. (a) (i) By considering the mean score of each team, which team would you join? Justify by showing working. (ii) If instead you base your decision on the median score of each team, which team would you join? Justify by showing working. (b) Suppose you are the coach of Team Jaguar and are being interviewed about your team for the local newspaper. Would it be better to report the mean score or the median score of your team? Explain your answer.

Step 1: (a) (i) means: Cheetah 376/5 = 75.2; Jaguar 379/5 = 75.8; Puma 345/5 = 69.0. Jaguar has the highest mean. Step 2: (a) (ii) medians: Cheetah 88; Jaguar 87; Puma 80. Cheetah has the highest median. Step 3: (b) Jaguar's mean (75.8) is pulled down by the one low score of 50, while its median (87) reflects the team's usual level; reporting the median is better and more representative. Answer: (a) (i) Jaguar; (a) (ii) Cheetah; (b) the median, because the mean is dragged down by the single low score of 50.

Unit 11: Basic Statistics and Probability (continued: Exercises 11.4-11.5 and Review Exercise 11)

EXERCISE 11.4 (book p.263)

Q1. Two fair coins are tossed. Find the probability of getting (i) at most one head; (ii) same faces.

Step 1: sample space S = {HH, HT, TH, TT}, n(S) = 4. Step 2: (i) at most one head: {HT, TH, TT}, probability 3/4. Step 3: (ii) same faces: {HH, TT}, probability 2/4 = 1/2. Answer: (i) 3/4; (ii) 1/2.

Q2. Two dice, one red and the other green, are rolled simultaneously. The numbers of dots on the tops are added. Find the probability of getting a sum of: (i) 7; (ii) 9; (iii) 12.

Step 1: n(S) = 36. Step 2: (i) sum 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) = 6 outcomes, probability 6/36 = 1/6. Step 3: (ii) sum 9: (3,6), (4,5), (5,4), (6,3) = 4 outcomes, probability 4/36 = 1/9. Step 4: (iii) sum 12: (6,6) = 1 outcome, probability 1/36. Answer: (i) 1/6; (ii) 1/9; (iii) 1/36.

Q3. A bag contains 30 balls out of which 6 are green, 10 are red and the remaining are black. A ball is drawn out of the bag. Find the probability of getting: (i) the ball is green; (ii) the ball is black; (iii) the ball is not green.

Step 1: black balls = 30 - 6 - 10 = 14. Step 2: (i) P(green) = 6/30 = 1/5. Step 3: (ii) P(black) = 14/30 = 7/15. Step 4: (iii) P(not green) = 24/30 = 4/5. Answer: (i) 1/5; (ii) 7/15; (iii) 4/5.

Q4. A die is rolled. Find the probability by Venn diagram of getting: (i) an odd number; (ii) a number greater than 3; (iii) a number which is odd and greater than 3.

Step 1: S = {1, 2, 3, 4, 5, 6}, n(S) = 6. Step 2: (i) odd: {1, 3, 5}, probability 3/6 = 1/2. Step 3: (ii) greater than 3: {4, 5, 6}, probability 3/6 = 1/2. Step 4: (iii) odd and greater than 3: {5}, probability 1/6. Answer: (i) 1/2; (ii) 1/2; (iii) 1/6.

Q5. In a three child family, what is the probability (by using tree diagram) of: (i) three girls; (ii) exactly two boys; (iii) at least one girl.

Step 1: sample space has 8 outcomes: BBB, BBG, BGB, BGG, GBB, GBG, GGB, GGG. Step 2: (i) three girls: {GGG}, probability 1/8. Step 3: (ii) exactly two boys: {BBG, BGB, GBB}, probability 3/8. Step 4: (iii) at least one girl: all except BBB, probability 7/8. Answer: (i) 1/8; (ii) 3/8; (iii) 7/8.

Q6. In a single throw of two fair dice, find the probability that the product of the numbers on the dice is: (i) between 3 and 12 (both inclusive); (ii) divisible by 2.

Step 1: n(S) = 36. Step 2: (i) products from 3 to 12: counting the 36 products gives 20

outcomes, probability 20/36 = 5/9. Step 3: (ii) the product is odd only when both dice are odd: 3 x 3 = 9 outcomes; so even products = 36 - 9 = 27, probability 27/36 = 3/4. Answer: (i) 5/9; (ii) 3/4.

Q7. A marble is drawn at random from a box containing 30 red, 10 white, 20 orange and 15 blue marbles. Find the probability that it is: (i) orange or red; (ii) not blue or red; (iii) red, white or blue.

Step 1: total = 75. Step 2: (i) orange or red: (20 + 30)/75 = 50/75 = 2/3. Step 3: (ii) 'not blue or red' is read as neither blue nor red: white or orange = (10 + 20)/75 = 30/75 = 2/5 (the event 'not blue' alone would be 60/75 = 4/5). Step 4: (iii) red, white or blue: (30 + 10 + 15)/75 = 55/75 = 11/15. Answer: (i) 2/3; (ii) 2/5 (neither blue nor red); (iii) 11/15.

Q8. Two coins are tossed simultaneously. What is the probability of getting: (i) one head; (ii) two heads; (iii) same faces.

Step 1: S = {HH, HT, TH, TT}, n(S) = 4. Step 2: (i) one head: {HT, TH}, probability 2/4 = 1/2. Step 3: (ii) two heads: {HH}, probability 1/4. Step 4: (iii) same faces: {HH, TT}, probability 2/4 = 1/2. Answer: (i) 1/2; (ii) 1/4; (iii) 1/2.

EXERCISE 11.5 (book p.266)

Q1. Calculate relative frequencies and expected frequencies of the following data, where N = 400: Marks 31-40: 5; 41-50: 9; 51-60: 15; 61-70: 12; 71-80: 9. Also find (i) P(X >= 60); (ii) P(X <= 50).

Step 1: total = 50; relative frequencies: 5/50 = 0.10; 9/50 = 0.18; 15/50 = 0.30; 12/50 = 0.24; 9/50 = 0.18. Step 2: expected frequencies (N = 400): 40, 72, 120, 96, 72. Step 3: (i) P(X >= 60) = (12 + 9)/50 = 0.42 (classes wholly above 60). Step 4: (ii) P(X <= 50) = (5 + 9)/50 = 0.28. Answer: relative frequencies 0.10, 0.18, 0.30, 0.24, 0.18; expected frequencies 40, 72, 120, 96, 72; (i) 0.42; (ii) 0.28.

Q2. (i) Find the relative frequencies of the data: Colour Red: 35/100; Green: (blank); Yellow: 5/100. (ii) Find probability of not red.

Step 1: relative frequencies add to 1, so Green = 1 - 0.35 - 0.05 = 0.60 = 60/100. Step 2: (ii) P(not red) = 1 - 0.35 = 0.65. Answer: (i) Red 0.35, Green 0.60, Yellow 0.05; (ii) 0.65.

Q3. A tin contains some biscuits. The flavours of the biscuits and the amount of each flavor are shown in the table: Chocolate 14; Lemon 16; Ginger 28; Caramel 8. Find: (i) total biscuits; (ii) relative frequency of lemon biscuits.

Step 1: (i) total = 14 + 16 + 28 + 8 = 66. Step 2: (ii) relative frequency of lemon = 16/66 = 8/33 = 0.24 approx. Answer: (i) 66; (ii) 0.24 approx.

Q4. A biased coin is tossed 500 times and probability of tail is 0.45. Find the expected frequency of occurring head and tail.

Step 1: expected tails = 500 x 0.45 = 225. Step 2: P(head) = 1 - 0.45 = 0.55; expected heads = 500 x 0.55 = 275. Answer: heads 275; tails 225.

Q5. Find the expected frequencies of the following data, where N = 512: x: 0: 5; 1: 6; 2: 15; 3: 4; 4: 2.

Step 1: total = 32; relative frequencies: 5/32, 6/32, 15/32, 4/32, 2/32. Step 2: expected frequencies: 512 x 5/32 = 80; 512 x 6/32 = 96; 512 x 15/32 = 240; 512 x 4/32 = 64; 512 x 2/32 = 32. Answer: 80, 96, 240, 64, 32.

Q6. A spinner was spun 50 times, and it landed on red 12 times. What is the relative frequency of the spinner landing on red?

Step 1: relative frequency = 12/50 = 0.24. Answer: 0.24.

Q7. A survey of 200 people showed that 60 prefer chocolate ice cream. What is the expected number of people who would prefer chocolate ice cream out of a group of 500?

Step 1: relative frequency = 60/200 = 0.30. Step 2: expected number = 500 x 0.30 = 150. Answer: 150 people.

Q8. A factory machine produced 400 items out of which 16 were defective. What is the relative frequency of producing a defective item?

Step 1: relative frequency = 16/400 = 0.04. Answer: 0.04.

REVIEW EXERCISE 11 (book p.267)

Q1. (i) Histogram is a graph of:

(a) frequency distribution ✓

(b) probability

(c) ungrouped data

(d) lower class limit

Q1. (ii) In cumulative frequency polygon, the frequencies are plotted against:

(a) Midpoint

(b) Class limit

(c) Upper class boundaries ✓

(d) lower class limit

Q1. (iii) In a distribution, 15, 15, 15, 15, 15, the mean is:

(a) 75

(b) 15 ✓

(c) 30

(d) 45

Q1. (iv) The most occurring value in a set of data is called:

(a) mean

(b) median

(c) mode ✓

(d) harmonic mean

Q1. (v) Probability of an event lies between:

(a) -1 and 1

(b) -1 and 0

(c) 0 and 1 ✓

(d) -0.5 to 0.5

Q1. (vi) The probability of an impossible event is:

(a) -1

(b) 0 ✓

(c) 1

(d) 2

Q1. (vii) Probability of a certain event is:

(a) -2

(b) -1

(c) 0

(d) 1 ✓

Q2. Find the mean, median and mode of the marks of 10 students in a test of Mathematics (out of 50): 45, 32, 36, 48, 45, 24, 28, 35, 46, 40.

Step 1: mean = 379/10 = 37.9. Step 2: in ascending order: 24, 28, 32, 35, 36, 40, 45, 45, 46, 48; median = (36 + 40)/2 = 38. Step 3: 45 occurs twice, more than any other value. Answer: mean = 37.9; median = 38; mode = 45.

Q3. Find arithmetic mean, median and mode of the following data: classes 36-40: 9; 41-45: 16; 46-50: 24; 51-55: 15; 56-60: 10; 61-65: 8.

Step 1: midpoints 38, 43, 48, 53, 58, 63; sum f = 82; sum fx = 342 + 688 + 1152 + 795 + 580 + 504 = 4061; mean = 4061/82 = 49.52. Step 2: cumulative frequencies 9, 25, 49, 64, 74, 82; median class 46-50: median = 45.5 + (5/24)(41 - 25) = 48.83. Step 3: modal class 46-50: mode = 45.5 + (24 - 16) x 5/((24 - 16) + (24 - 15)) = 47.85. Answer: mean = 49.52; median = 48.83; mode = 47.85.

Q4. A coin is tossed 3 times. Find the probability of (i) at least two heads; (ii) at the most two tails.

Step 1: sample space has 8 outcomes. Step 2: (i) at least two heads: {HHH, HHT, HTH, THH}, probability 4/8 = 1/2. Step 3: (ii) at most two tails: all except TTT, probability 7/8. Answer: (i) 1/2; (ii) 7/8.

Q5. Find relative frequency of the data: Group 51-60: 3; 61-70: 8; 71-80: 15; 81-90: 9; 91-100: 4.

Step 1: total = 39. Step 2: relative frequencies: 3/39 = 0.08; 8/39 = 0.21; 15/39 = 0.38; 9/39 = 0.23; 4/39 = 0.10 (approx). Answer: 0.08, 0.21, 0.38, 0.23, 0.10.

Unit 1: Real Numbers (continued: Exercises 1.1-1.2, recovered by OCR cross-check)

EXERCISE 1.1 (book p.16)

Q1. Name the properties of real numbers: (i) a + sqrt(2) = sqrt(2) + a.

Step 1: the order of addition is changed. Answer: commutative property of addition.

Q1. (ii) 4 x sqrt(11) = sqrt(11) x 4.

Step 1: the order of multiplication is changed. Answer: commutative property of multiplication.

Q1. (iii) (a + 2) + sqrt(2) = a + (2 + sqrt(2)).

Step 1: the grouping of addition is changed. Answer: associative property of addition.

Q1. (iv) a - a = 0.

Step 1: a number minus itself is zero. Answer: additive inverse property.

Q1. (v) (sqrt(2)/3) x 1 = sqrt(2)/3.

Step 1: multiplying by 1 leaves the number unchanged. Answer: multiplicative identity property.

Q1. (vi) (6.2) x 1/(6.2) = 1.

Step 1: a number times its reciprocal is 1. Answer: multiplicative inverse property.

Q2. Which properties of the real numbers is used in the following expressions: (i) -6 < 5 implies 0 < 11.

Step 1: 6 is added to both sides. Answer: additive property of inequality.

Q2. (ii) 7 > 3 implies 1/3 > 1/7.

Step 1: reciprocals of positive numbers reverse the inequality. Answer: reciprocal property of inequality.

Q2. (iii) 8 < 10 implies -24 > -30.

Step 1: both sides are multiplied by -3, which reverses the inequality. Answer: multiplicative property of inequality with a negative number.

Q2. (iv) x + z = y + z implies x = y.

Step 1: z is cancelled from both sides. Answer: cancellation property with respect to addition.

Q2. (v) x = y implies x + z = y + z.

Step 1: the same number is added to both sides. Answer: additive property of equality.

Q2. (vi) 3x = 3y implies x = y.

Step 1: 3 is cancelled from both sides. Answer: cancellation property with respect to multiplication.

Q3. Using properties of real numbers, verify the following: (i) a/b + c/d = (ad +

bc)/bd.

Step 1: a/b = ad/bd and c/d = bc/bd. Step 2: adding gives (ad + bc)/bd. Answer: verified.

Q3. (ii) 3/4 - 1/2 = 1/4.

Step 1: 1/2 = 2/4. Step 2: 3/4 - 2/4 = 1/4. Answer: verified.

Q3. (iii) (4 + 36)/4 = 10.

Step 1: 4 + 36 = 40. Step 2: 40/4 = 10. Answer: verified.

Q3. (iv) 5(3/4 + 1/4) = 5.

Step 1: 3/4 + 1/4 = 1. Step 2: 5 x 1 = 5. Answer: verified.

Q4. Using properties of inequalities of real numbers, verify the following: (i) If 3 < 5 then -15 > -25.

Step 1: multiply both sides by -5; the inequality reverses. Step 2: 3 x (-5) = -15 and 5 x (-5) = -25, so -15 > -25. Answer: verified.

Q4. (ii) If 4 < 5 then 1/4 > 1/5.

Step 1: reciprocals of positive numbers reverse the inequality. Answer: verified.

Q4. (iii) 4/5 + 3 > 1/5 then 4/5 > 1/5.

Step 1: 4/5 + 3 = 19/5, and 19/5 > 1/5. Step 2: with the common denominator 5, 4 > 1, so 4/5 > 1/5. Answer: verified.

Q4. (iv) If 25 > 20 then -5 < -4.

Step 1: multiply both sides by -1/5; the inequality reverses. Step 2: 25 x (-1/5) = -5 and 20 x (-1/5) = -4, so -5 < -4. Answer: verified.

EXERCISE 1.2 (book pp.22-23)

Q1. Write the base and exponent in each of the following: (i) (a/b)^4.

Step 1: the whole fraction is raised to the power. Answer: base = a/b, exponent = 4.

Q1. (ii) (b/a)^(-3).

Answer: base = b/a, exponent = -3.

Q1. (iii) -30a^4.

Answer: base = a, exponent = 4 (with coefficient -30).

Q1. (iv) (-b^2/a)^2.

Answer: base = -b^2/a, exponent = 2.

Q1. (v) -(3/8)^101.

Answer: base = 3/8, exponent = 101 (with a negative sign outside).

Q2. Laws of indices to simplify: (i) (2a^2)^3 x (4ab^(-1))^2.

Step 1: (2a^2)^3 = 8a^6. Step 2: (4ab^(-1))^2 = 16a^2b^(-2). Step 3: product = 128a^8b^(-2). Answer: 128a^8/b^2.

Q2. (ii) 6x^5y^3 x x^2y^7.

Step 1: add exponents of like bases. Answer: 6x^7y^10.

Q2. (iii) h^7k^5 x h^4k^3.

Step 1: add exponents of like bases. Answer: h^11k^8.

Q2. (iv) (2^(-3)a^4b)^(-1) x (4^(-2)b^(-5)).

Step 1: (2^(-3)a^4b)^(-1) = 2^3a^(-4)b^(-1). Step 2: 4^(-2) = 2^(-4). Step 3: product = 2^(-1)a^(-4)b^(-6). Answer: 1/(2a^4b^6).

Q2. (v) (a^3b^2)^3 / (a^3b^4).

Step 1: numerator = a^9b^6. Step 2: subtract exponents: a^(9-3)b^(6-4). Answer: a^6b^2.

Q2. (vi) (a^3b^5)^3 x (3ab^3)^2.

Step 1: (a^3b^5)^3 = a^9b^15. Step 2: (3ab^3)^2 = 9a^2b^6. Step 3: product = 9a^11b^21. Answer: 9a^11b^21.

Q3. Simplify: (i) ((2xy^3)^4/(5xy)^2) x ((5x^4y^5)^3/(3x^2y^3)^2).

Step 1: first fraction = 16x^4y^12/(25x^2y^2) = (16/25)x^2y^10. Step 2: second fraction = 125x^12y^15/(9x^4y^6) = (125/9)x^8y^9. Step 3: product = (80/9)x^10y^19. Answer: (80/9)x^10y^19.

Q3. (ii) 21(3a + b)^11 / 3(3a + b)^5.

Step 1: 21/3 = 7. Step 2: (3a + b)^(11-5) = (3a + b)^6. Answer: 7(3a + b)^6.

Q3. (iii) (p^3q^2r)^(-4) / (p^2r^3q)^2.

Step 1: numerator = p^(-12)q^(-8)r^(-4). Step 2: denominator = p^4q^2r^6. Step 3: quotient = p^(-16)q^(-10)r^(-10). Answer: 1/(p^16q^10r^10).

Q3. (iv) (2^3)^4 / 3^6 x 3^(-3).

Step 1: (2^3)^4 = 2^12 = 4096. Step 2: 3^6 x 3^(-3) in the denominator gives 3^3 = 27. Answer: 4096/27.

Q3. (v) 4a^(-3)b^3 / (2a^2b^3)^2.

Step 1: denominator = 4a^4b^6. Step 2: quotient = a^(-7)b^(-3). Answer: 1/(a^7b^3).

Q3. (vi) (3^(-3))^6 x (9^(-2))^4.

Step 1: (3^(-3))^6 = 3^(-18). Step 2: (9^(-2))^4 = 3^(-16). Step 3: product = 3^(-34). Answer: 1/3^34.

Q3. (vii) ((4x^9y^2z^(-3))/(9x^(-2)y^(-3)))^(-3) / ((3x^(-4)y^(-5))/(8x^5y^3z^(-6)))^5.

Step 1: first bracket = (4/9)x^11y^5z^(-3); cubed negatively gives (729/64)x^(-33)y^(-15)z^9. Step 2: second bracket = (3/8)x^(-9)y^(-8)z^6; to the power 5 gives (243/32768)x^(-45)y^(-40)z^30. Step 3: dividing gives 1536x^12y^25z^(-21). Answer: 1536x^12y^25/z^21.

Q3. (viii) ((x^5y^(-5)z^3)/(y^3z^(-5)))^(-7) x ((x^(-6)y^2z^(-4))/(x^3y^(-5)z^7))^5.

Step 1: first bracket = x^5y^(-8)z^8; to the power -7 gives x^(-35)y^56z^(-56). Step 2: second bracket = x^(-9)y^7z^(-11); to the power 5 gives x^(-45)y^35z^(-55). Step 3: product = x^(-80)y^91z^(-111). Answer: y^91/(x^80z^111).

Q4. Write the radicand and index of the following: (i) cube root of (pq/z).

Answer: radicand = pq/z, index = 3.

Q4. (ii) q-th root of (c/d).

Answer: radicand = c/d, index = q.

Q4. (iii) 5th root of (8c/(b + c)).

Answer: radicand = 8c/(b + c), index = 5.

Q4. (iv) square root of (5/19).

Answer: radicand = 5/19, index = 2.

Q5. Express the following in radical form: (i) (8x^3y^3)^(-1/3).

Step 1: cube root of 8x^3y^3 is 2xy. Answer: 1/(2xy).

Q5. (ii) 9^(-3/4).

Step 1: 9^(3/4) = (3^2)^(3/4) = 3^(3/2) = 3sqrt(3). Answer: 1/(3sqrt(3)).

Q5. (iii) 216^(1/3).

Step 1: 6^3 = 216. Answer: 6.

Q5. (iv) (u^2/v^2)^(-1/5).

Step 1: invert and take the 5th root. Answer: 5th root of (v^2/u^2).

Q6. Express the following in exponential form: (i) cube root of (25x^2y^3).

Answer: (25x^2y^3)^(1/3).

Q6. (ii) 4th root of (3u^(1/2)y^6).

Answer: (3u^(1/2)y^6)^(1/4).

Q6. (iii) 1/(n-th root of x^q).

Answer: x^(-q/n).

Q6. (iv) p-th root of (36p/(qr)).

Answer: (36p/(qr))^(1/p).

Q7. Simplify the following: (i) 32^(-2/5).

Step 1: 32^(1/5) = 2, so 32^(2/5) = 4. Answer: 1/4.

Q7. (ii) (125/64)^(-1/3).

Step 1: invert: (64/125)^(1/3) = 4/5. Answer: 4/5.

Q7. (iii) (100u^2)^(-3/2).

Step 1: 100^(3/2) = 1000. Answer: 1/(1000u^3).

Q7. (iv) 16^(-1/4).

Step 1: 16^(1/4) = 2. Answer: 1/2.

Q7. (v) (x^(-2)y^4)^(1/3) x (x^4y^(-3)).

Step 1: first factor = x^(-2/3)y^(4/3). Step 2: multiply: x^(10/3)y^(-5/3). Answer: cube root of (x^10/y^5).

Q7. (vi) (625e^(-4)f^4)^(1/4).

Step 1: 625^(1/4) = 5. Answer: 5f/e.

Q7. (vii) x^(1/10) / c^(-1/5) x c^(-3/2).

Step 1: dividing by c^(-1/5) multiplies by c^(1/5). Step 2: c^(1/5 - 3/2) = c^(-13/10). Answer: x^(1/10)/c^(13/10).

Q7. (viii) sqrt(x^a/y^b) x sqrt(y^b/z^c) x sqrt(z^c/x^a).

Step 1: the product under one square root is (x^a/y^b)(y^b/z^c)(z^c/x^a) = 1. Answer: 1.

Q7. (ix) ((36^(-1/2) x 6^(3/2))/(216^(1/2) x 27^(1/3)))^(1/6).

Step 1: numerator = (1/6) x 6sqrt(6) = sqrt(6). Step 2: denominator = 6sqrt(6) x 3 = 18sqrt(6). Step 3: ratio = 1/18. Answer: (1/18)^(1/6).

Q8. Show that: (i) 4th root of (a^3/b^3) x 4th root of (b^3/c^3) x 4th root of (c^3/a^3) = 1.

Step 1: the product under one 4th root is (a^3/b^3)(b^3/c^3)(c^3/a^3) = 1. Step 2: 4th root of 1 = 1. Answer: shown.

Q8. (ii) (27^(1/2) x 125^(1/3))/(625^(-1/4) x 243^(1/3)) = 25/3. NOTE: as printed, this identity does NOT hold: the left side equals 25 x 3^(-1/6) = about 20.82, not 25/3 (verified numerically). The notes give the correct value of the left side and flag the printed right side as a book error.

Step 1: numerator = sqrt(27) x 5 = 15sqrt(3). Step 2: denominator = (1/5) x 3 x cube root of 9 = (3/5) x cube root of 9. Step 3: quotient = 25sqrt(3)/cube root of 9 = 25 x 3^(-1/6) = about 20.82. Answer: the left side equals about 20.82, NOT 25/3 as printed.